Special DPP on Straight line **DPP # 1**

1. Consider the points P(2, – 4); Q (4, – 2) and R = (7, 1). The points P, Q, R :
   (A) form an equilateral triangle
   (B) form a right angled triangle
   (C) form an isosceles triangle which is not equilateral
   (D) are collinear

   माना कि बिन्दु P(2, – 4) ; Q (4, – 2) तथा R = (7, 1) दिये गये है तो P, Q, R बनाते है :
   (A) एक समबाहु त्रिभुज (B) एक समकोणीय त्रिभुज
   (C) एक समद्विबाहु त्रिभुज जो कि समबाहु नही है (D) समरेखीय है

Ans. D

Sol. PQ = $2\sqrt{2}$; QR = $3\sqrt{2}$; PR = $5\sqrt{2}$ $\Rightarrow$ PQ + QR = PR $\Rightarrow$ (D)

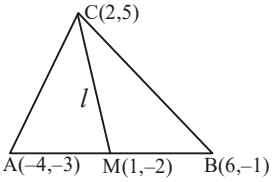
2. Let ΔABC have vertices A(–4, –3), B(6, – 1) and C(2, 5). Find the length of the median from C to AB?

   माना ΔABC के शीर्ष A (–4, –3), B (6, – 1) तथा C (2, 5) है। C से AB पर मध्यिका की लम्बाई ज्ञात कीजिये?

   (A) $\sqrt{50}$ (B) $\sqrt{53}$ (C) $\sqrt{89}$ (D) $\sqrt{104}$

Ans. A

Sol. $l = \sqrt{(2-1)^2 + 7^2} = \sqrt{50}$ Ans.



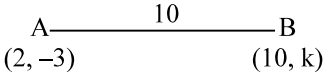
3. The length of a line segment AB is 10 units. If the coordinates of one extremity are (2, – 3) and the abscissa of the other extremity is 10 then the sum of all possible values of the ordinate of the other extremity is :

   रेखा खण्ड AB की लम्बाई 10 इकाई है। यदि उसके एक छोर का निर्देशांक, (2, – 3) है तथा अन्य छोर का भुज 10 है तो उसके अन्य छोर की कोटि के सभी मानो का योग है :

   (A) 3 (B) – 4 (C) 12 (D) – 6

Ans. D

Sol. $64 + (k + 3)^2 = 100$



$(k+3)^2 = 36$

$k + 3 = 6$ or -6 $\Rightarrow$ $k = +3$ or -9

4. If $P(1,2)$, $Q(4,6)$, $R(5,7)$ & $S(a,b)$ are the vertices of a parallelogram PQRS, then :

यदि $P(1,2)$, $Q(4,6)$, $R(5,7)$ एवं $S(a,b)$ समान्तर चतुर्भुज PQRS के शीर्ष है तो–

(A) $a = 2, b = 4$ (B) $a = 3, b = 4$ (C) $a = 2, b = 3$ (D) $a = 3, b = 5$

Ans. C

Sol. $\because$ diagonals of a parallelogram bisects each other

$\therefore \quad \frac{4+a}{2} = 3 \Rightarrow a = 2$ and $\frac{6+b}{2} = \frac{9}{2} \Rightarrow b = 3$

5. The four points whose co–ordinates are (2, 1), (1, 4), (4, 5), (5, 2) form :

(A) a rectangle which is not a square (B) a trapezium which is not a parallelogram

(C) a square (D) a rhombus which is not a square

चार बिन्दु जिनके निर्देशांक (2, 1), (1, 4), (4, 5), (5, 2) है, बनाते है :

(A) एक आयत जो वर्ग नही है (B) एक समलम्ब चतुर्भुज जो समानान्तर चतुर्भुज नही है

(C) एक वर्ग (D) एक समचतुर्भुज जो कि वर्ग नही है

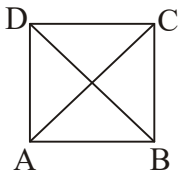
Ans. C

Sol. A (2, 1), B (1, 4), C(4, 5), D(5, 2)

$AB = BC = CD = DA = \sqrt{10}$

$AC = BD = \sqrt{20}$

$\angle A = \angle B = \angle C = \angle D = 90^\circ$



6. If A and B are the points $(-3, 4)$ and (2, 1), then the co–ordinates of the point C on AB produced such that $AC = 2BC$ are :

यदि A तथा B बिन्दु $(-3, 4)$ तथा (2, 1) है, तथा A B को बढ़ाने पर C बिन्दु इस प्रकार प्राप्त होते है कि $AC = 2BC$ है तो C के निर्देशांक हैं :

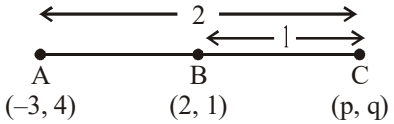
(A) (2, 4) (B) (3, 7) (C) (7, –2) (D) $\left(-\frac{1}{2}, \frac{5}{2}\right)$

Ans. C

Sol. Let co-ordinates of point C be (p, q)

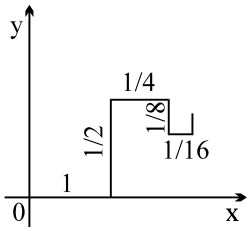
C divides AB externally in the ratio 2 : 1.

$$p = \frac{2\times 2-(-3)(1)}{2-1} = 7 \quad \text{and} \quad q = \frac{2\times 1-(1)(4)}{1} = -2$$

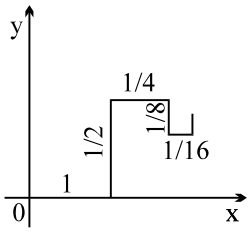


Hence, co-ordinates of point C(7, –2). Ans.

7. A particle begins at the origin and moves successively in the following manner as shown,
1 unit to the right, 1/2 unit up, 1/4 unit to the right, 1/8 unit down, 1/16 unit to the right etc.
The length of each move is half the length of the previous move and movement continues in the 'zigzag' manner indefinitely. The co-ordinates of the point to which the 'zigzag' converges is :



एक कण मूल बिन्दु से प्रारम्भ होकर निम्न प्रकार से क्रमागत गमन कर रहा है
1 इकाई दायी ओर, 1/2 इकाई उपर, 1/4 इकाई दायी ओर, 1/8 इकाई नीचे, 1/16 दायी ओर इत्यादि
प्रत्येक चाल की लम्बाई पिछली चाल लम्बाई की आधी है तथा यह चलन अनन्त तक जा रहा है। तो वह बिन्दु ज्ञात कीजिये जहां पर यह पथ अभिमुख हो जायें :

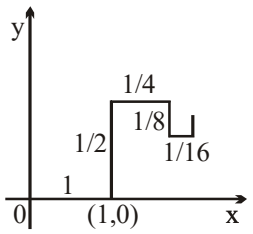


(A) (4/3, 2/3) (B) (4/3, 2/5) (C) (3/2, 2/3) (D) (2, 2/5)

Ans. B

Sol. x-coordinate : $1+\frac{1}{4}+\frac{1}{16}+.....\infty = \frac{1}{1-\frac{1}{4}} = \frac{4}{3}$

y-coordinate : $\frac{1}{2}-\frac{1}{8}+\frac{1}{32}+.....\infty=\frac{\frac{1}{2}}{1+\frac{1}{4}}=\frac{2}{5}$

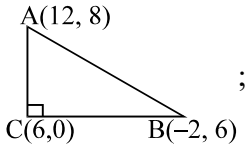


Hence $\left(\frac{4}{3},\frac{2}{5}\right)$. Ans.

8. Coordinates of the vertices of a triangle ABC are (12, 8), (–2, 6) and (6, 0) then the correct statement is :
(A) triangle is right but not isosceles.
(B) triangle is isosceles but not right.
(C) triangle is obtuse.
(D) the product of the abscissae of the centroid, orthocentre and circumcentre is 160.

यदि त्रिभुज ABC के शीर्षो के निर्देशांक (12, 8), (–2, 6) तथा (6, 0) है तो सही कथन बताइये :
(A) त्रिभुज समकोणीय है लेकिन समद्विबाहु नही
(B) त्रिभुज समद्विबाहु है लेकिन समकोणीय नही
(C) त्रिभुज अधिक कोणीय त्रिभुज है
(D) केन्द्रक, लम्बकेन्द्र, परिकेन्द्र के भुजो का गुणन 160 है

Ans. D

Sol. 

; ABC is right isosceles triangle

AB = $10\sqrt{2}$

AC = BC = 10

centroid is $\left(\frac{16}{3},\frac{14}{3}\right)$; orthocentre is (6, 0) and circumcentre is (5, 7)

9. Suppose ABC is a triangle with 3 acute angle A, B and C. The point whose coordinates are (cos B – sin A, sin B – cos A) can be in the :
(A) first and 2nd quadrant (B) second the 3rd quadrant

(C) third and 4th quadrant (D) second quadrant only

माना ABC न्यूनकोणीय त्रिभुज है। बिन्दु जिसके निर्देशांक (cos B – sin A, sin B – cos A) है, हो सकता है :

(A) प्रथम तथा द्वितीय चतुर्थांश में (B) द्वितीय तथा तृतीय चतुर्थांश में

(C) तृतीय तथा चतुर्थ चतुर्थांश में (D) केवल द्वितीय चतुर्थांश में

Ans. D

Sol. Since ABC are acute angle

$\therefore$ $A + B > \pi/2$; $A > \frac{\pi}{2} - B$; $\sin A - \cos B > 0 \Rightarrow \cos B - \sin A < 0$ ....(1)

Again, $B > \frac{\pi}{2} - A$; $\sin B > \cos A$; $\sin B - \cos A > 0$ ....(2)

from (1) and (2)

x-coordinate is –ve and y-coordinate is +ve

$\Rightarrow$ line in 2nd quadrant only Ans.

10. Consider the points P(2, – 4) ; Q (4, – 2) and R = (7, 1). If the points S is the harmonic conjugate of Q w.r.t. P and R then PQ, PR and PS are in :

माना बिन्दु P(2, – 4) ; Q (4, – 2) तथा R = (7, 1) दिये गये है। यदि S बिन्दु Q का P तथा R के सापेक्ष हरात्मक संयुग्मी है, तो PQ, PR तथा PS है :

(A) A.P. (B) G.P. (C) H.P. (D) A.G.P.

Ans. C

Sol. (a) Refer to the basic definition.

(b) Let $S(\alpha, \beta)$ be the harmonic conjugates of the point R(5, 1). Suppose R divides PQ in the ratio $\lambda : 1$ internally then S divides PQ in the ratio $\lambda : 1$ externally then

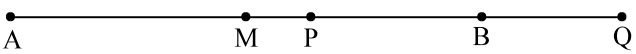
$5 = \frac{6\lambda + 2}{\lambda + 1} \Rightarrow 5\lambda + 5 = 6\lambda + 2 \Rightarrow \lambda = 3$

Also, $1 = \frac{-2\lambda + 10}{\lambda + 1} \Rightarrow \lambda + 1 = -2\lambda + 10 \Rightarrow 3\lambda = 9 \Rightarrow \lambda = 3$

Now, $\alpha = \frac{3 \times 6 - 1 \times 2}{3 - 1} = 8$ and $\beta = \frac{3 \times (-2) - 1 \times 10}{3 - 1} = -8$

Hence harmonic conjugate of R (5, 1) is S(8, – 8)]

11. A line segment AB is divided internally and externally in the same ratio at P and Q respectievly and M is the mid point of AB.



Statement-1 : MP, MB, MQ are in G.P.

Statement-2 : AP, AB and AQ are in H.P.

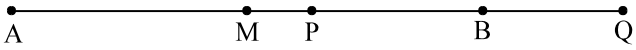
(A) Statement-1 is true, statement-2 is true; statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true; statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

रेखाखण्ड AB को बिन्दु P तथा Q समान अनुपात में आन्तरिक एवं बाह्य विभाजित करता है यदि M, रेखा AB का मध्य बिन्दु है तो :



कथन–1 : MP, MB, MQ गु.श्रे. में होगा।

कथन–2 : AP, AB, AQ ह.श्रे. में होगा।

(A) कथन–1 सत्य है, कथन–2 सत्य है, कथन–2 कथन–1 का सही स्पष्टीकरण है

(B) कथन–1 सत्य है, कथन–2 सत्य है, कथन–2 कथन–1 का सही स्पष्टीकरण नही है

(C) कथन–1 सत्य है, कथन–2 असत्य है

(D) कथन–1 असत्य है, कथन– 2 सत्य है

Ans. A

Sol. $\frac{2}{AB} = \frac{1}{AP} + \frac{1}{AQ}$

$\frac{2}{2(MB)} = \frac{1}{MB+MP} + \frac{1}{MQ+MB}$ $\Rightarrow \frac{1}{MB} - \frac{1}{MB+MP} = \frac{1}{MQ+MB}$

$\Rightarrow$ MP(MQ + MB) = MB(MB + MP) $\Rightarrow$ (MP)(MQ) = $(MB)^2$

INTEGER TYPE

12. Find the distances between the following pairs of points

(a) $\left(t_1^2, 2t_1\right)$ and $\left(t_2^2, 2t_2\right)$ if t_1 and t_2 are the roots of $x^2 - 2\sqrt{3}x + 2 = 0$.

(b) (a cos θ, a sin θ) and (a cos ϕ, a sin ϕ)

निम्नलिखित बिन्दुओं के मध्य दूरी ज्ञात कीजिये।

(a) $\left(t_1^2, 2t_1\right)$ तथा $\left(t_2^2, 2t_2\right)$ यदि t_1 तथा t_2 समीकरण $x^2 - 2\sqrt{3}x + 2 = 0$ के मूल है

(b) (a cos θ, a sin θ) तथा (a cos ϕ, a sin ϕ)

Ans. (a) 8 ; (b) $2a\sin\left|\frac{\theta-\phi}{2}\right|$

Sol. (a) $\sqrt{\left(t_1^2 - t_2^2\right)^2 + 4\left(t_1 - t_2\right)^2}$

$\Rightarrow (t_1 - t_2)\sqrt{(t_1 + t_2)^2 + 4}$

$\Rightarrow \left|\frac{\sqrt{D}}{a}\right|\sqrt{\left(2\sqrt{3}\right)^2 + 4} = \frac{\sqrt{12-8}}{1}\sqrt{12+4} = 2 \times 4 = 8$

(b) $|a|\sqrt{(\cos\theta - \cos\phi)^2 + (\sin\theta - \sin\phi)^2}$

$|a|\sqrt{2 - 2\cos(\theta - \phi)}$

$|a|\sqrt{4\sin^2\left(\frac{\theta - \phi}{2}\right)}$

$2|a|\sin\left|\frac{\theta - \phi}{2}\right|$

13. The points (2, 1), (5, 4) and (1, 4) are three of the corners of a parallelogram. Find the coordinates of the remaining corner which is opposite to (2, 1).

बिन्दुओं (2, 1), (5, 4) तथा (1, 4) एक समान्तर चतुर्भुज के तीन कोने है। चौथे शीर्ष के निर्देशांक ज्ञात कीजिये जो (2, 1) के विपरीत है।

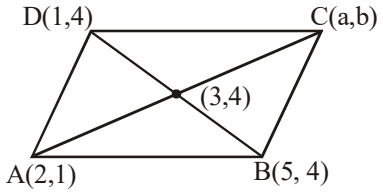
Ans. (4, 7)

Sol. $\because$ Diagonals of parallelogram bisect each other

$\therefore \quad \frac{a+2}{2} = 3$ and $\frac{b+1}{2} = 4$

a = 4 and b = 7

c (4, 7). Ans.



14. The line joining the points (1, – 2) and (– 3, 4) is trisected, find the coordinates of the points of the trisection.

बिन्दुओं (1, – 2) तथा (– 3, 4) को मिलाने वाली रेखा के समत्रिभाजक बिन्दुओं के निर्देशांक ज्ञात करें।

Ans. $\left(\frac{-1}{3}, 0\right)$; $\left(\frac{-5}{3}, 2\right)$

Sol. Let point P divides AB in the ratio 1 : 2 (internally)

$$P\left(\frac{-3\times1 + 2\times1}{3}, \frac{1\times4 + 2\times-2}{3}\right)$$

$$P\left(\frac{-1}{3}, 0\right)$$

Q divides AB is the ratio 2 : 1 (internally)

$$Q\left(\frac{2\times -3+1\times 1}{3}, \frac{2\times 4+(-1)(-2)}{3}\right)$$

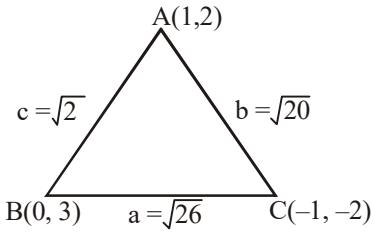
$$Q\left(\frac{-5}{3}, 2\right) \text{ Ans.}$$

15. Find the lengths of the medians of the triangle whose vertices are (1, 2), (0, 3) and (–1, – 2).

शीर्षो (1, 2), (0, 3) तथा (–1, – 2) द्वारा बनाये गये त्रिभुज की मध्यिकाओं की लम्बाई ज्ञात करें।

Ans. $\frac{3\sqrt{2}}{2}, 3, \frac{3\sqrt{10}}{2}$

Sol. Method - 1 :

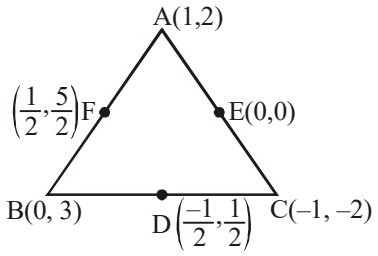


$$l_A = \frac{1}{2}\sqrt{2b^2+2c^2-a^2} = \frac{1}{2}\sqrt{40+4-26} = \frac{3\sqrt{2}}{2}$$

$$l_B = \frac{1}{2}\sqrt{2a^2+2c^2-b^2} = \frac{1}{2}\sqrt{52+4-20} = 3$$

$$l_C = \frac{1}{2}\sqrt{2b^2+2a^2-c^2} = \frac{1}{2}\sqrt{40+52-2} = \frac{3\sqrt{10}}{2}. \text{ Ans.]}$$

Method - 2 :



Using Distance formula,

$$AD = \sqrt{\frac{9}{4}+\frac{9}{4}} = \frac{3}{\sqrt{2}} = \frac{3\sqrt{2}}{2}$$

$BE = 3$

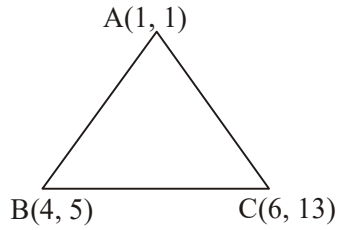
$$CF = \sqrt{\frac{9}{4}+\frac{81}{4}} = \frac{\sqrt{90}}{2} = \frac{3\sqrt{10}}{2}.$$ Ans.

16. The vertices of a triangle are A(1, 1), B(4, 5) and C(6, 13). Find cos A.

एक त्रिभुज का शीर्ष A(1, 1), B(4, 5) तथा C(6, 13) है। cos A ज्ञात करें।

Ans. $\frac{63}{65}$

Sol. $c = AB = \sqrt{9+16} = 5$



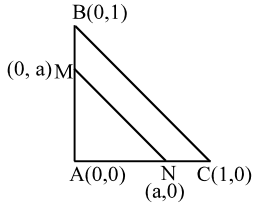
$a = BC = \sqrt{4+64} = \sqrt{68} = 2\sqrt{17}$

$b = CA = \sqrt{25+144} = \sqrt{169} = 13$

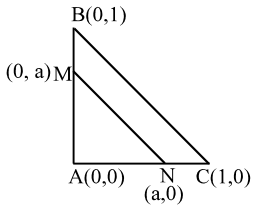
$$\cos A = \frac{b^2+c^2-a^2}{2bc} = \frac{169+25-68}{2\times13\times5} = \frac{126}{130}$$

$$\cos A = \frac{63}{65}$$

17. ΔABC lies in the plane wit A = (0, 0), B = (0, 1) and C = (1, 0). Points M and N are chosen on AB and AC, respectively, such that MN is parallel to BC and MN divides the area of ΔABC in half. Find the coordinates of M.



ΔABC में, A = (0, 0), B = (0, 1) तथा C = (1, 0) है। बिन्दु M तथा N भुजा AB तथा AC पर क्रमशः इस प्रकार चुने गये है कि MN भुजा BC के समानान्तर है तथा वह ΔABC को दो समान क्षेत्रफलों में विभाजित करती है। M के निर्देशांक ज्ञात करें।



Ans. $\left(0, \frac{1}{\sqrt{2}}\right)$

Sol. Since MN is parallel to BC, we have $|AN| = |AM|$ denotes this values of x.

The area of ΔABC is $\frac{1}{2}$. Since MN halves this area, the area of ΔAMN is $\frac{1}{4}$. So we have

$$\frac{x \cdot x}{2} = \frac{1}{4} \qquad \Rightarrow x^2 = \frac{1}{2} \qquad \Rightarrow x = \frac{1}{\sqrt{2}}$$

Therefore M is $\frac{1}{\sqrt{2}}$ units above the x-axis. That is, $M = \left(0, \frac{1}{\sqrt{2}}\right)$

18. (a) Prove that the coordinates x and y of the middle point of the line segment joining the point (2, 3) to the point (3, 4) satisfy the equation $x - y + 1 = 0$.

(b) If the points (x, y) be equidistant from the points (6, – 1) and (2, 3), prove that $x - y = 3$.

(a) सिद्ध कीजिये कि : बिन्दु (2, 3) तथा (3, 4) को मिलाने वाली रेखाखंड के मध्य बिन्दु के निर्देशांक x तथा y समीकरण $x - y + 1 = 0$ को सतुष्ट करते है।

(b) यदि बिन्दु (x, y) बिन्दुओं (6, – 1) तथा (2, 3) समान दूरी पर है, सिद्ध करें $x - y = 3$ है।

Ans.

Sol. (a) (2, 3) $\left(\frac{5}{2}, \frac{7}{2}\right)$ (3, 4)

$x - y + 1 = 0$

$$\frac{5}{2} - \frac{7}{2} + 1 = -1 + 1 = 0$$

(b) Let $P \equiv (x, y)$, $A \equiv (6, -1)$ and $B \equiv (2, 3)$

By the given condition, $PA = PB$

$$\Rightarrow \sqrt{(x-6)^2 + (y+1)^2} = \sqrt{(x-2)^2 + (y-3)^2}$$

or $(x - 6)^2 + (y + 1)^2 = (x - 2)^2 + (y - 3)^2$

or $x^2 - 12x + 36 + y^2 + 2y + 1 = x^2 - 4x + 4 + y^2 - 6y + 9$

or $8x - 8y = 24$

or $x - y = 3$ Hence proved.

19. Find the harmonic conjugates of the point R(5, 1) with respect to the points P(2, 10) and Q(6, – 2).

बिन्दु R(5, 1) का बिन्दुओं P(2, 10) तथा Q(6, – 2) के सापेक्ष हरात्मक संयुग्मी ज्ञात करें।

Ans. (8, – 8)

Sol. Let $S(\alpha, \beta)$ be the harmonic conjugates of the point R(5, 1). Suppose R divides PQ in the ratio $\lambda : 1$ internally then S divides PQ in the ratio $\lambda : 1$ externally then

$$5 = \frac{6\lambda + 2}{\lambda + 1} \Rightarrow 5\lambda + 5 = 6\lambda + 2 \Rightarrow \lambda = 3$$

Also, $\Rightarrow \lambda + 1 = -2\lambda + 10 \Rightarrow 3\lambda = 9 \Rightarrow \lambda = 3$

Now, $\alpha = \frac{3\times 6 - 1\times 2}{3-1} = 8$ and $\beta = \frac{3\times(-2) - 1\times 10}{3-1} = -8$

Hence harmonic conjugate of R (5, 1) is S(8, – 8)

20. If $P(t^2, 2t)$, $Q\left(\frac{1}{t^2}, \frac{-2}{t}\right)$ and S(1, 0) be any three points, find the value of $\left(\frac{1}{SP} + \frac{1}{SQ}\right)$.

यदि $P(t^2, 2t)$, $Q\left(\frac{1}{t^2}, \frac{-2}{t}\right)$ तथा S(1, 0) कोई तीन बिन्दु है, $\left(\frac{1}{SP} + \frac{1}{SQ}\right)$ का मान ज्ञात करें।

Ans. 1

Sol. $P\left(t^2, 2t\right)$, $Q\left(\frac{1}{t^2}, \frac{-2}{t}\right)$, S(1, 0)

$$SP = \sqrt{\left(t^2 - 1\right)^2 + 4t^2} = \sqrt{\left(t^2 + 1\right)^2} = t^2 + 1$$

$$SQ = \sqrt{\left(1 - \frac{1}{t^2}\right)^2 + \frac{4}{t^2}} = \frac{t^2 + 1}{t^2}$$

$$\frac{1}{SP} + \frac{1}{SQ} = \frac{1}{t^2 + 1} + \frac{t^2}{t^2 + 1} = \frac{1 + t^2}{1 + t^2} = 1$$

21. If a and b are real numbers between 0 and 1 such that the point (a, 1), (1, b) and (0, 0) form an equilateral triangle, find a and b.

यदि a तथा b वास्तविक संख्याऐं 0 से 1 के मध्य इस प्रकार है कि बिन्दु (a, 1), (1, b) तथा (0, 0) एक समबाहु त्रिभुज बनाते है। a तथा b ज्ञात करें।

Ans. $a = b = 2 - \sqrt{3}$

Sol. ΔABC is an equilateral triangle

$AB = BC = AC$

$(AB)^2 = (BC)^2 = (AC)^2$

$$AB = \sqrt{(1-a)^2 + (b-1)^2}$$

$$BC = \sqrt{1+b^2}$$

$$AC = \sqrt{a^2+1}$$

Also, $(1 - a)^2 + (b - 1)^2 = 1 + b^2 = 1 + a^2$

$\Rightarrow a^2 = b^2$

Also, $(1 - a)^2 + (b - 1)^2$

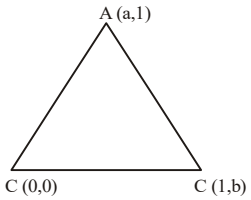
$= 1 + a^2 - 2a + 1 + b^2 - 2b$

$\Rightarrow 1 + a^2 - 2a + 1 + b^2 - 2b = 1 + b^2$

$\Rightarrow a^2 - 2a - 2b + 2 = 1$

$\Rightarrow b^2 - 2b - 2b + 1 = 0$

$\Rightarrow b^2 - 4b + 1 = 0$



$$\Rightarrow b = \frac{4 \pm \sqrt{16-4}}{2} = \frac{4 \pm \sqrt{12}}{2}$$

$$\Rightarrow b = 2 \pm \sqrt{3}$$

$$b = 2 - \sqrt{3}$$

$$a = 2 - \sqrt{3}$$

22. If O be the origin, and if the coordinates of any two points P_1 and P_2 be respectively (x_1, y_1) and (x_2, y_2), prove that $OP_1 \cdot OP_2 \cos P_1OP_2 = x_1x_2 + y_1y_2$.

यदि O मूल बिन्दु है, तथा यदि P_1 तथा P_2 क्रमशः (x_1, y_1) तथा (x_2, y_2) है तो सिद्ध कीजिये $OP_1 \cdot OP_2 \cos P_1OP_2 = x_1x_2 + y_1y_2$.

Ans.

Sol. **Method I** : Given : $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$

Now, $OP_1 = \sqrt{(x_1)^2 + (y_1)^2}$

$\Rightarrow OP_2 = \sqrt{x_2^2 + y_2^2}$

Now, $\cos\theta = \dfrac{(OP_1)^2 + (OP_2)^2 - (P_1P_2)^2}{2(OP_1)(OP_2)}$

$\Rightarrow (OP_1)(OP_2)\cos P_1OP_2 = \dfrac{(OP_1)^2 + (OP_2)^2 - (P_1P_2)^2}{2}$

$\Rightarrow (OP_1)(OP_2)\cos P_1OP_2 = \dfrac{x_1^2 + y_1^2 + x_2^2 + y_2^2 - \left((x_2 - x_1)^2 + (y_2 - y_1)^2\right)}{2}$

$\Rightarrow (OP_1)(OP_2)\cos P_1OP_2 = \dfrac{x_1^2 + x_2^2 + y_1^2 + y_2^2 - x_1^2 - x_2^2 + 2x_1x_2 - y_1^2 - y_2^2 + 2y_1y_2}{2}$

$\Rightarrow (OP_1)(OP_2)\cos P_1OP_2 = \dfrac{2(x_1x_2 + y_1y_2)}{2}$

$\Rightarrow (OP_1)\,(OP_2)\cos P_1OP_2 = x_1x_2 + y_1y_2$

Method I : Let D be the area of the triangle P_1OP_2

$$\therefore \Delta = \frac{1}{2}\begin{vmatrix} 0 & 0 & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = \frac{1}{2}(x_1y_2 - x_2y_1) \qquad ...(1)$$

We know that the area of triangle $= \dfrac{1}{2} \times \text{base} \times \text{height}$

$$\therefore \Delta = \frac{1}{2} \times OP_2 \times OP_1 \sin(P_1OP_2) \qquad ...(2)$$

from (1) & (2)

$OP_2 \times OP_1 \sin(P_1OP_2) = (x_1y_2 - x_2y_1)$

$OP_2^{\,2} \times OP_1^{\,2} \sin^2(P_1OP_2) = (x_1y_2 - x_2y_1)^2 \qquad ...(3)$

Also, $OP_1^2 \times OP_2^2 = \left\{(x_1-0)^2+(y_1-0)^2\right\}^2 \times \left\{(x_2-0)^2+(y_2-0)^2\right\}^2 = \left(x_1^2+y_1^2\right)\left(x_2^2+y_2^2\right)$ ...(4)

Now (4) – (3) gives

$OP_1^2 \times OP_2^2 - OP_2^2 \times OP_1^2 \sin^2(P_1OP_2) = (x_1^2 + y_1^2)(x_2^2 + y_2^2) - (x_1y_2 - x_2y_1)^2$

$\Rightarrow OP_1^2 \times OP_2^2 \left(1-\sin^2\left(P_1OP_2\right)\right) = x_1^2x_2^2 + x_1^2y_2^2 + x_2^2y_1^2 + y_1^2y_2^2 - x_1^2y_2^2 - x_2^2y_1^2 + 2x_1x_2y_1y_2$

$\Rightarrow OP_1^2 \times OP_2^2 \times \cos^2\left(P_1OP_2\right) = x_1^2x_2^2 + y_1^2y_2^2 + 2x_1x_2y_1y_2$

$\Rightarrow OP_1^2 \times OP_2^2 \times \cos^2\left(P_1OP_2\right) = \left(x_1x_2 + y_1y_2\right)^2$

$\Rightarrow OP_1 \times OP_2 \cos\left(P_1OP_2\right) = x_1x_2 + y_1y_2$

23. Match the following column :

Column-I	**Column-II**
(A) The points (2, – 2) , (8, 4), (5, 7) and (–1, 1) taken in order constitute the vertices of a	(P) square
(B) The points (0, – 1), (2, 1), (0, 3) and (– 2, 1) taken in order are the vertices of a	(Q) rectangle
(C) The points (3, – 5), (– 5, – 4), (7, 10), (15, 9) taken in order are the vertices of a	(R) trapezium
(D) The points (1, 0), (–1, 0), (–3, 4) and (3, 4) taken in order are the vertices of a	(S) parallelogram
	(T) cyclic quadrilateral

निम्न कॉलम को सुमेलित कीजिए :

Column-I	**Column-II**
(A) बिन्दु (2, – 2) , (8, 4), (5, 7) तथा (–1, 1) क्रमावार रूप में लेने पर किसके शीर्ष है	(P) वर्ग
(B) बिन्दु (0, – 1), (2, 1), (0, 3) तथा (– 2, 1) क्रमावार रूप में लेने पर किसके शीर्ष है	(Q) आयत
(C) बिन्दु (3, – 5), (– 5, – 4), (7, 10) तथा (15, 9) क्रमावार रूप में लेने पर किसके शीर्ष है	(R) समलम्ब
(D) बिन्दु (1, 0), (–1, 0), (–3, 4) तथा (3, 4) क्रमावार रूप में लेने पर किसके शीर्ष है	(S) समान्तर चतुर्भुज
	(T) चक्रीय चतुर्भुज

Ans. (A) Q, S, T; (B) P, Q, S, T ; (C) S (D) RT

Sol.

	A	B	C	D
(A)	(2, –2)	(8,4)	(5,7)	(–1,1)

$AB = \sqrt{36+36} = 6\sqrt{2}$

$M = \left(\frac{7}{2}, \frac{5}{2}\right)$

$BC = \sqrt{9+9} = 3\sqrt{2}$

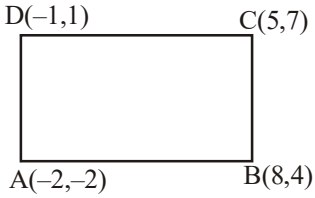
$AC = \sqrt{9+81} = 3\sqrt{10}$

$CD = \sqrt{36+36} = 6\sqrt{2}$

$BD = \sqrt{81+9} = 3\sqrt{10}$

$AD = \sqrt{9+9} = 3\sqrt{2}$

Rectangle



(B) $AB = 2\sqrt{2}$

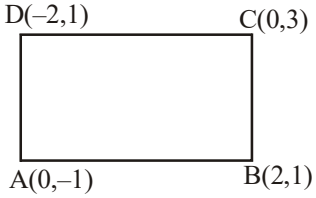
$BC = 2\sqrt{2}$

$CD = 2\sqrt{2}$

$AD = 2\sqrt{2}$

Also, AC = 4 and BD = 4

Hence quadratic is a square.

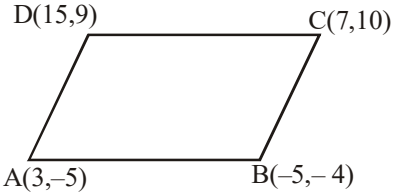


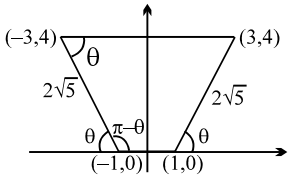
(C) $AB = \sqrt{65}$ $CD = \sqrt{65}$

$BC = \sqrt{340}$ $AD = \sqrt{340}$

$AC = \sqrt{241}$ and $BD = \sqrt{569}$

∵ opposite sides are equal but diag. are not equal hence quadratic is a parallelogram.



(D) 

; Note that opposite angles are supplementary

24. Match the following column :

Column-I	**Column-II**
(A) The points (2, – 2) , (– 2, 1) and (5, 2)	(P) are the vertices of aright angled triangle
(B) The points (1, – 2), (– 3, 0) and (5, 6)	(Q) are the vertices of a right angled isosceles triangle
(C) The points (3, 7), (6, 5) and (15, – 1)	(R) are the vertices ofan equilateral triangle
(D) The points (2, 2), (– 2, – 2) & $\left(-2\sqrt{3}, 2\sqrt{3}\right)$	(S) do not form a triangle

निम्न कॉलम को सुमेलित कीजिए :

Column-I	**Column-II**
(A) बिन्दु (2, – 2) , (– 2, 1) तथा (5, 2)	(P) एक समकोणीय त्रिभुज के शीर्ष है
(B) बिन्दु (1, – 2), (– 3, 0) तथा (5, 6)	(Q) एक समकोणीय समद्विबाहु त्रिभुज के शीर्ष है
(C) बिन्दु (3, 7), (6, 5) तथा (15, – 1)	(R) एक समबाहु त्रिभुज के शीर्ष है
(D) बिन्दु (2, 2), (– 2, – 2) तथा $\left(-2\sqrt{3}, 2\sqrt{3}\right)$	(S) कोई त्रिभुज नहीं बनाते हैं

Ans. (A) P, Q ; (B) P ; (C) S; (D) R

Sol. (A) $\rightarrow$ P, Q

A(2, –2), B(–2,1) & C(5,2)

$AB = \sqrt{16+9} = 5$

$BC = \sqrt{49+1} = \sqrt{50} = 5\sqrt{2}$

$AC = \sqrt{9+16} = 5$

$AB^2 + AC^2 = BC^2$

(B) $\rightarrow$ P

A(1,–2), B(–3,0), C(5,6)

$AB = \sqrt{16+4} = \sqrt{20} = 2\sqrt{5}$

$BC = \sqrt{64+36} = \sqrt{100} = 10$

$AC = \sqrt{16+64} = \sqrt{80} = 4\sqrt{5}$

$AB^2 + AC^2 = BC^2$

(C) $\rightarrow$ S

A(3,7), B(6,5) & C(15,–1)

$AB = \sqrt{9+4} = \sqrt{13}$

$BC = \sqrt{81+36} = \sqrt{117}$

$AC = \sqrt{144+64} = \sqrt{208}$

$AB + BC < AC$

$(D) \rightarrow R$

$A(2,2), B(-2,-2), C\left(-2\sqrt{3}, 2\sqrt{3}\right)$

$AB = \sqrt{16+16} = \sqrt{32}$

$BC = \sqrt{4\left(\sqrt{3}-1\right)^2 + 4\left(\sqrt{3}+1\right)^2} = \sqrt{8[3+1]} = \sqrt{32}$

$AC = \sqrt{32}$

1. The orthocentre of the triangle ABC is 'B' and the circumcentre is 'S' (a, b) . If A is the origin then the co-ordinates of C are :

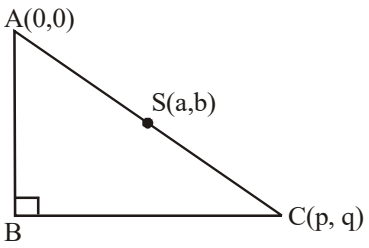
(A) (2a, 2b) (B) $\left(\frac{a}{2}, \frac{b}{2}\right)$ (C) $\left(\sqrt{a^2 + b^2}, 0\right)$ (D) none

त्रिभुज ABC का लम्बकेन्द्र 'B' है तथा परिकेन्द्र 'S' (a, b) है। यदि A मूल बिन्दु है तो C के निर्देशांक है :

(A) (2a, 2b) (B) $\left(\frac{a}{2}, \frac{b}{2}\right)$ (C) $\left(\sqrt{a^2 + b^2}, 0\right)$ (D) इनमें से कोई नहीं

Ans. A

Sol. Let co-orinates of C be (p, q) $\frac{p+0}{2} = a$ and $\frac{q+0}{2} = b$



p = 2a and q = 2b

c (2a, 2b). Ans.

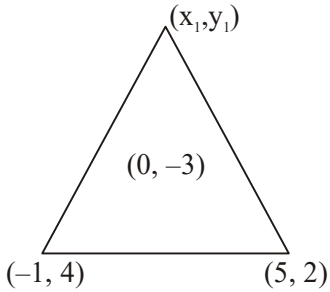
2. The medians of a triangle meet at (0, –3) and its two vertices are at (–1, 4) and (5, 2). Then the third vertex is at:

एक त्रिभुज की मध्यिकाऐं (0, –3) पर मिलती है तथा इसके दो शीर्ष (–1, 4) तथा (5, 2) है तो तीसरा शीर्ष है :

(A) (4, 15) (B) (–4, –15) (C) (–4, 15) (D) (4, –15)

Ans. B

Sol.



$\frac{x_1 - 1 + 5}{3} = 0$ $\qquad$ $\frac{y_1 + 4 + 2}{3} = -3$

$x_1 = -4$ $\qquad$ $y_1 = -9 - 6$

$(-4, -15)$ $\qquad$ $y_1 = -15$

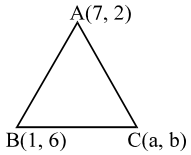
3. If the two vertices of a triangle are (7, 2) and (1, 6) and its centroid is (4, 6) then the coordinate of the third vertex are (a, b). The value of (a + b), is :

एक त्रिभुज के दो शीर्ष (7, 2) तथा (1, 6) है तथा इसका केन्द्रक (4, 6) है, तो तीसरे सिरे के निर्देशांक (a, b) है। (a + b) का मान है :

(A) 13 $\qquad$ (B) 14 $\qquad$ (C) 15 $\qquad$ (D) 16

Ans. B

Sol. $\frac{7 + 1 + a}{3} = 4$ and $\frac{2 + 6 + b}{3} = 6$



$a = 4$ and $b = 10$

$\Rightarrow$ $a + b = 14$ Ans.

4. If in triangle ABC, $A \equiv (1, 10)$, circumcentre $\equiv \left(-\frac{1}{3}, \frac{2}{3}\right)$ and orthocentre $\equiv \left(\frac{11}{3}, \frac{4}{3}\right)$ then the co-ordinates of mid-point of side opposite to A is :

यदि त्रिभुज ABC में, $A \equiv (1, 10)$, परिकेन्द्र $\equiv \left(-\frac{1}{3}, \frac{2}{3}\right)$ तथा लम्बकेन्द्र $\equiv \left(\frac{11}{3}, \frac{4}{3}\right)$ तो A के विपरीत भुजा का मध्य बिन्दु है

(A) $(1, -11/3)$ $\qquad$ (B) $(1, 5)$ $\qquad$ (C) $(1, -3)$ $\qquad$ (D) $(1, 6)$

Ans. A

Sol. Correct option is (A).

We know that,

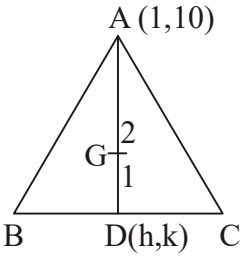
The orthocentre, centroid and circumcentre of any triangle are collinear. And the centroid divides the distance from orthocentre to circumcentre in the ratio 2 : 1.

Let the centroid be G(x, y), its coordinates can be found using the section formula.

$$\therefore (x, y) \equiv \left(\frac{\frac{-2}{3} + \frac{11}{3}}{3}, \frac{\frac{4}{3} + \frac{4}{3}}{3} \right) \equiv \left(1, \frac{8}{9}\right)$$

Also, the centroid (G) divides the medians (AD) in the ratio 2 : 1.

Let the coordinates of D be (h, k)



$\therefore 1 = \frac{2h+1}{3}$ and $\frac{8}{9} = \frac{2k+10}{3}$

h = 1 and 8 = 6k + 30

and $k = \frac{-11}{3}$

$\therefore D(h,k) = \left(1, \frac{-11}{3}\right)$

5. ABC is an isosceles triangle . If the co-ordinates of the base are $(1, 3)$ and $(-2, 7)$, then co-ordinates of vertex A can be :

ABC एक समद्विबाहु त्रिभुज है। यदि आधार के निर्देशांक (1, 3) एवं (– 2, 7) है तो A को निर्देशांक हो सकते है :

(A) $\left(-\frac{1}{2}, 5\right)$ (B) $\left(-\frac{1}{8}, 5\right)$ (C) $\left(\frac{5}{6}, -5\right)$ (D) $\left(-7, \frac{1}{8}\right)$

Ans. D

Sol. The vertex A(x,y) is equidistant from B and C because ABC is an isosceles triangle.

Therefore,

$(x - 1)^2 + (y - 3)^2 = (x + 2)^2 + (y - 7)^2$

$\Rightarrow 6x - 8y + 43 = 0$

Thus, any point lying on this line can be the vertex A except the mid-point $\left(-\frac{1}{2}, 5\right)$ of BC.

Note : A lies perpendicular bisector of BC.

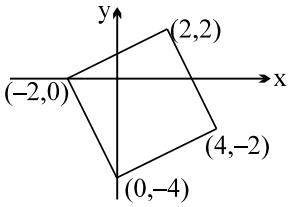
6. Find the area of the quadrilateral ABCD with vertices A(–2, 0), B(0, –4), C(4, –2), and D(2, 2).

चतुर्भुज ABCD का क्षेत्रफल ज्ञात करें जहाँ A(–2, 0), B(0, –4), C(4, –2), तथा D(2, 2) शीर्ष है।

(A) 12 sq. units (B) 16 sq. units (C) 20 sq. units (D) 32 sq. units

Ans. C

Sol. It is a square of side $\sqrt{20}$ as shown $\Rightarrow A = 20$



7. The line x = c cuts the triangle with corners (0, 0); (1, 1) and (9, 1) into two regions. For the area of the two regions to be the same c must be equal to :

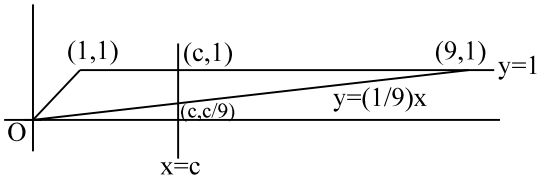
(A) 5/2 (B) 3 (C) 7/2 (D) 3 or 15

एक रेखा x = c त्रिभुज (0, 0); (1, 1) तथा (9, 1) को दो समान क्षेत्रफलों वाले भाग में विभाजित करती है तो c का मान होना चाहिये:

(A) 5/2 (B) 3 (C) 7/2 (D) 3 या 15

Ans. B

Sol. $2 \cdot \frac{1}{2} \cdot \begin{vmatrix} 9 & 1 & 1 \\ c & 1 & 1 \\ c & c/9 & 1 \end{vmatrix} = \frac{1}{2} \begin{vmatrix} 9 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{vmatrix} = \frac{1}{2} \cdot 8 = 4$



$\Rightarrow$ c = 3 Ans.

or c = 15 which is not possible

8. A triangle has two of its vertices at (0, 1) and (2, 2) in the Cartesian plane. Its third vertex lies on the x-axis. If the area of the triangle is 2 square units then the sum of the possible abscissae of the third vertex, is :

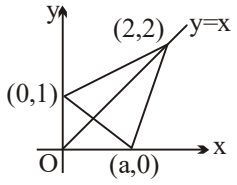
(A) – 4 (B) 0 (C) 5 (D) 6

एक त्रिभुज जिसके दो शीर्ष (0, 1) तथा (2, 2) कार्तिय तल में है । इसका तीसरा शीर्ष x अक्ष पर स्थित है। यदि त्रिभुज का क्षेत्रफल 2 वर्ग इकाई है तो इसके तीसरे शीर्ष के सम्भव भुजों के मानों का योग होगा :

(A) – 4 (B) 0 (C) 5 (D) 6

Ans. A

Sol. $\Delta = \frac{1}{2} \begin{vmatrix} 0 & 1 & 1 \\ 2 & 2 & 1 \\ a & 0 & 1 \end{vmatrix} = 2; \qquad \therefore \begin{vmatrix} 0 & 0 & 1 \\ 2 & 1 & 1 \\ a & -1 & 1 \end{vmatrix} = 4$



$1\ |-2-a| = 4 \qquad \Rightarrow |a+2| = 4$

$a = 2$ or $a = -6$

sum of the abscissa = – 4 Ans.

9. Given a ΔABC whose vertices are $A(x_1, y_1)$; $B(x_2, y_2)$; $C(x_3, y_3)$.

Let there exists a point P(a, b) such that $6a = 2x_1 + x_2 + 3x_3$; $6b = 2y_1 + y_2 + 3y_3$

Statement-1 : Area of triangle PBC must be less than the area of ΔABC

Statement-2 : P lies inside the triangle ABC

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

त्रिभुज ΔABC में $A(x_1, y_1)$; $B(x_2, y_2)$; $C(x_3, y_3)$.

माना कि P(a, b) ऐसा बिन्दु है कि $6a = 2x_1 + x_2 + 3x_3$; $6b = 2y_1 + y_2 + 3y_3$

कथन-1 : त्रिभुज PBC का क्षेत्रफल < ΔABC का क्षेत्रफल

कथन-2 : P त्रिभुज ABC के भीतर है।

(A) कथन–1 सत्य है, कथन–2 सत्य है, कथन–2 कथन–1 का सही स्पष्टीकरण है

(B) कथन–1 सत्य है, कथन–2 सत्य है, कथन–2 कथन–1 का सही स्पष्टीकरण नही है

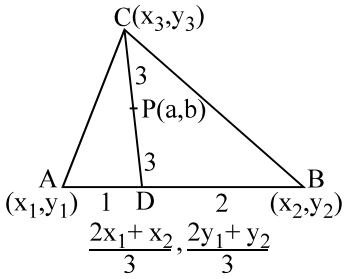
(C) कथन–1 सत्य है, कथन–2 असत्य है

(D) कथन–1 असत्य है, कथन– 2 सत्य है

Ans. A

Sol. $P = \dfrac{2x_1 + x_2 + 3x_3}{6}, \dfrac{2y_1 + y_2 + 3y_3}{6}$

hence P lies inside the triangle



Let D divides AB in 1 : 2 internally and P divides CD in 3 : 3 internally.

∴ area of ΔPBC < area of ΔABC

10. Two vertices of a triangle are at the points (3, – 1), (– 2, 3) and the centroid is at the origin. Find the coordinates of the remaining vertex.

त्रिभुज के दो शीर्ष (3, – 1), (– 2, 3) है तथा केन्द्रक मूल बिन्दु है अन्य शीर्ष ज्ञात करें।

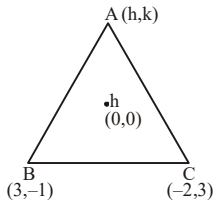
Ans. (– 1, – 2)

Sol. Centroid

$$\frac{3-2+h}{3}=0 \qquad \frac{-1+3+k}{3}=0$$

$$\frac{h+1}{3}=0 \qquad \frac{k+2}{3}=0$$

$h=-1 \qquad k=-2$



$\therefore$ Third vertices of $\Delta = (-1, -2)$

11. Find the incentre of the triangle whose vertices are (2, 3), (–2, – 5), (– 4, 6).

शीर्ष (2, 3), (–2, – 5), (– 4, 6) दारा बनाए त्रिभुज का अन्तः केन्द्र ज्ञात करें।

Ans. (–1, 2)

Sol. Use $x=\frac{ax_1+bx_2+cx_3}{a+b+c}$ and $y=\frac{ay_1+by_2+cy_3}{a+b+c}$. Note that the triangle is a right triangle whose sides are $3\sqrt{5}$, $4\sqrt{5}$ and $5\sqrt{5}$

12. If x_1, x_2, x_3 are the roots of the equation $x^3-3px^2+3qx-1=0$ then find the centroid of the Δ the co-ordinates of whose vertices are $(x_1, 1/x_1)$ $(x_2, 1/x_2)$ and $(x_3, 1/x_3)$.

यदि x_1, x_2, x_3 समीकरण $x^3-3px^2+3qx-1=0$ के मूल है तो त्रिभुज जिसके शीर्षो के निर्देशांक है $(x_1, 1/x_1)$ $(x_2, 1/x_2)$ तथा $(x_3, 1/x_3)$ का केन्द्रक ज्ञात करो।

Ans. (p, q)

Sol. x_1, x_2 and x_3 are the roots of equation.

$x^3-2px^2+3qx-1=0$

$\Rightarrow x_1+x_2+x_3=3p$

$x_1x_2+x_2+x_3+x_3x_1=3q$

$x_1x_2x_3=1$

Now, centroid of triangle with vertices (m_1, n_1) (m_2, n_2) (m_3, n_3) is

$$\left(\frac{m_1+m_2+m_3}{3}, \frac{n_1+n_2+n_3}{3}\right)$$

$\therefore$ vertices of centroid

$$\left(\frac{x_1+x_2+x_3}{3}, \frac{\frac{1}{x_1}+\frac{1}{x_2}+\frac{1}{x_3}}{3}\right)$$

$$\Rightarrow \left(\frac{3p}{3}, \frac{x_1x_2+x_2x_3+x_1x_3}{3x_1x_2x_3}\right)$$

$$\Rightarrow \left[P, \frac{3q}{3(1)}\right]$$

(p, q)

13. Find the circumcentre and circumradius of the triangle whose vertices are (1, 1), (2, – 1) and (3, 2).

शीर्षो (1, 1), (2, – 1) एवं (3, 2) द्वारा बनाये गये त्रिभुज का परिकेन्द्र तथा परित्रिज्या ज्ञात करें।

Ans. $\left(\frac{5}{2}, \frac{1}{2}\right)$; $\left(\frac{5}{2}\right)^{1/2}$

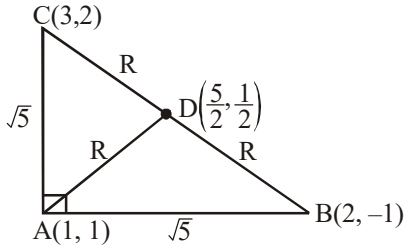
Sol. $AC = \sqrt{5}$

$BC = \sqrt{10}$

$AB = \sqrt{5}$

Δ is triangle angled at A.

$R = DC = BD = AD = \sqrt{\frac{5}{2}}$. Ans.



14. If a circle passes through the points (9, 3), (7, – 1) and (1, – 1), then find its :

(i) centre (ii) radius

यदि वृत (9, 3), (7, – 1) तथा (1, – 1) से गुजरता है तो :

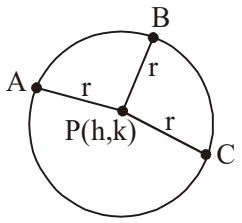
(i) केन्द्र ज्ञात करे (ii) इसकी त्रिज्या ज्ञात करें।

Ans. (i) (4, 3) (ii) 5

Sol. If a circle passes through the points

$A(9, 3), B(7, -1), C(1, -1)$

Let centre of the circle is $p(h, k)$.



$PA = PB$

$PA^2 = PB^2$

$(x - 9)^2 = (9 - 3)^2 = (x - 7)^2 + (y + 1)^2$

$x^2 + 81 - 18x + y^2 + 9 - 6y = x^2 + 49 - 14x + y^2 + 1 + 2y$

$4x + 8y = 40$

$x + 2y = 10$ ...(1)

$PB^2 = PC^2$

$(x - 7)^2 + (y + 1)^2 = (x - 1)^2 + (y + 1)^2$

$x^2 + 49 - 14x = x^2 - 2x + 1$

$12x = 48$

$x = 4, y = 3$

centre $\equiv (4, 3)$

$r = \sqrt{25 + 0} = \sqrt{25} = PA = PB = PC$

15. The vertices of a triangle are $(1, a), (2, b)$ and $(c^2, -3)$:

(i) Prove that its centroid can not lie on the y-axis.

(ii) Find the condition that the centroid may lie on the x-axis for any value of $a, b, c \in R$.

एक त्रिभुज के शीर्ष $(1, a), (2, b)$ तथा $(c^2, -3)$ है :

(i) सिद्ध कीजिए कि केन्द्रक y-अक्ष पर स्थित नही होगा

(ii) वह शर्त ज्ञात करें जब $a, b, c \in R$ के लिये केन्द्रक x अक्ष पर स्थित है

Ans. (ii) $a + b = 3$

Sol. Centroid of the triangle is

$$G \equiv \left(\frac{1+2+c^2}{3}, \frac{a+b-3}{3}\right) \quad \text{i.e.} \quad \left(\frac{3+c^2}{3}, \frac{a+b-3}{3}\right)$$

(i) $\because$ G will lie on y-axis then $\frac{3+c^2}{3} = 0 \Rightarrow c^2 = -3;\ c = \pm i\sqrt{3}$

or $\because$ both balues of c are imaginary

Hence G can not lie on y-axis.

(ii) $\because$ G will lie on x-axis then $\frac{a+b-3}{3} = 0$ Þ $a + b - 3 = 0 \Rightarrow a + b = 3$ Ans.

16. If $\left(\frac{3}{2}, 0\right)$, $\left(\frac{3}{2}, 6\right)$ and $(-1, 6)$ are mid points of the sides of a triangle, then find

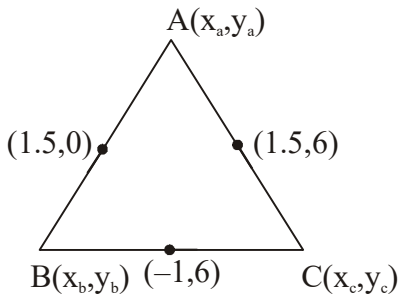
(i) centroid of the triangle (ii) incentre of the triangle.

यदि $\left(\frac{3}{2}, 0\right)$, $\left(\frac{3}{2}, 6\right)$ तथा $(-1, 6)$ एक त्रिभुज की भुजाओं के मध्य बिन्दु है तो

(i) त्रिभुज का केन्द्रक ज्ञात करें (ii) त्रिभुज का अन्तःकेन्द्र ज्ञात करें

Ans. (i) $\left(\frac{2}{3}, 4\right)$ (ii) $(1, 2)$

Sol.



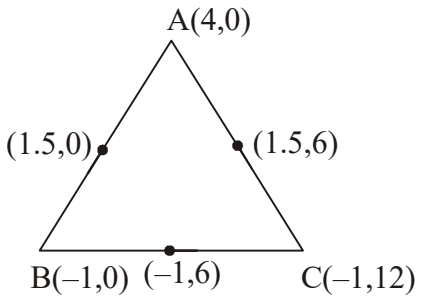
Find the values of vertices A, B, C

$X_a = 1.5 + 1.5 - (-1) = 4$ $\quad Y_a = 0 + 6 - 6 = 0$

$X_b = 1.5 + (-1) - 1.5 = -1$ $\quad Y_b = 0 + 6 - 6 = 0$

$X_c = 1.5 + (-1) - 1.5 = -1$ $\quad Y_c = 6 + 6 - 0 = 12$

The random triangle after calculating the vertices values is



Draw the random triangle on the coordinate axis with the vertices values.

From the above triangle ABC,

(x_1, y_1) are the coordinates of Point A and 'a' is the side of triangle opposite to the point A.

(x_2, y_2) are the coordinates of Point B and 'b' is the side of triangle opposite to the point B.

(x_3, y_3) are the coordinates of point C and 'c' is the side of triangle opposite to the point C.

AB = 5, BC = 12, AC = ?

By Pythagoras Theorem,

$AC^2 = AB^2 + BC^2$

$AC^2 = 5^2 + 12^2 = 169$

$\therefore$ AC = 13

Therefore, $(x_1, y_1) = (4,0)$ and a = 12

$(x_2, y_2) = (-1,0)$ and b = 13

$(x_3, y_3) = (-1, 12)$ and c = 5

We know that the coordinates of the incenter of the triangle ABC is,

$$\left(I_x, I_y\right) = \left(\frac{ax_1 + bx_2 + cx_3}{a+b+c}, \frac{ay_1 + by_2 + cy_3}{a+b+c}\right)$$

$$\left(I_x, I_y\right) = \left(\frac{12\times 4 + 13\times(-1) + 5\times(-1)}{12+13+5}, \frac{12\times 0 + 13\times 0 + 5\times 12}{12+13+5}\right)$$

$$\left(I_x, I_y\right) = \left(\frac{30}{30}, \frac{60}{30}\right)$$

$(I_x, I_y) = (1,2)$

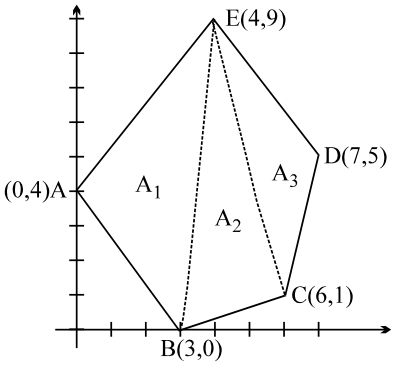
17. Find the area of the pentagon whose vertices taken in order are (0, 4), (3, 0), (6, 1), (7, 5) and (4, 9).

क्रमागत शीर्षो (0, 4), (3, 0), (6, 1), (7, 5) और (4, 9) द्वारा बनाये गये पंचभुज का क्षेत्रफल ज्ञात कीजिये।

Ans. 36.5

Sol. $A_1 = \frac{1}{2}\begin{vmatrix} 0 & 4 & 1 \\ 3 & 0 & 1 \\ 4 & 9 & 1 \end{vmatrix} = \frac{1}{2}|-4(-1)+1(27)| = \frac{31}{2}$

$A_2 = \frac{1}{2}\begin{vmatrix} 3 & 0 & 1 \\ 4 & 9 & 1 \\ 6 & 1 & 1 \end{vmatrix} = \frac{1}{2}|3\cdot(9-1)+1\cdot(4-54)| = \frac{1}{2}|24-50| = 13$



$A_3 = \frac{1}{2}\begin{vmatrix} 4 & 9 & 1 \\ 7 & 5 & 1 \\ 6 & 1 & 1 \end{vmatrix} = \frac{1}{2}|4\times4-9(1)+1(7-30)|$

$= \frac{1}{2}|16-9-23| = \frac{16}{2} = 8$

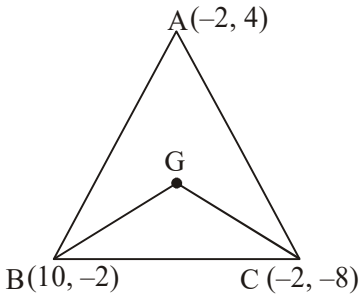
$\therefore$ Area of pentagon $= \frac{31}{2} + 13 + 8 = \frac{73}{2} = 36.5$ sq. units

18. Consider the triangle with vertices A(–2, 4), B(10, – 2), C(–2, – 8). If G is the centroid of the triangle, find the area of the triangle BGC.

शीर्षो A(–2, 4), B(10, – 2), C(–2, – 8) द्वारा बनाये गये त्रिभुज का केन्द्रक G है। त्रिभुज BGC का क्षेत्रफल ज्ञात कीजिये।

Ans. 24

Sol.



$\frac{0}{3} = \frac{1}{3}\times\frac{1}{2}\begin{vmatrix} -2 & 10 & -2 & -2 \\ 4 & -2 & -8 & 4 \end{vmatrix}$

$= \frac{1}{6}[-40-80-8-16]$

$$= \frac{144}{6} = 24$$

1. If x_1, y_1 are the roots of $x^2 + 8x - 20 = 0$, x_2, y_2 are the roots of $4x^2 + 32x - 57 = 0$ and x_3, y_3 are the roots of $9x^2 + 72x - 112 = 0$, then the points, (x_1, y_1), (x_2, y_2) and (x_3, y_3) :

(A) are collinear (B) form an equilateral triangle

(C) form a right angled isosceles triangle (D) are concyclic

यदि x_1, y_1 समीकरण $x^2 + 8x - 20 = 0$ व x_2, y_2 समीकरण $4x^2 + 32x - 57 = 0$ एवं x_3, y_3 समीकरण $9x^2 + 72x - 112 = 0$ के मूल हो, तो बिन्दु (x_1, y_1), (x_2, y_2) तथा (x_3, y_3) :

(A) संरेखीय (B) समबाहु त्रिभुज का शीर्ष होगा

(C) समकोणीय समद्विबाहु त्रिभुज होगा। (D) वृत्त पर स्थित होगा

Ans. A

Sol. Given $x_1 + y_1 = -8 \Rightarrow y_1 = -8 - x_1$; $x_2 + y_2 = -8 \Rightarrow y_2 = -8 - x_2$

and $x_3 + y_3 = -8 \Rightarrow y_3 = -8 - x_3$

now check

$\begin{vmatrix} x_1 & -8-x_1 & 1 \\ x_2 & -8-x_2 & 1 \\ x_3 & -8-x_3 & 1 \end{vmatrix}$ value of this determinant comes out to be zero

2. Let A(5, 12), B(–13 cos θ, 13 sin θ) and C(13 sin θ, – 13 cos θ) are angular points of ΔABC where $\theta \in R$. The locus of orthocentre of ΔABC is :

यदि A(5, 12), B(–13 cos θ, 13 sin θ) तथा C(13 sin θ, – 13 cos θ) ΔABC के शीर्ष हो जहाँ $\theta \in R$ तो त्रिभुज के लम्बकेन्द्र का बिन्दुपथ होगा :

(A) $x - y + 7 = 0$ (B) $x - y - 7 = 0$ (C) $x + y - 7 = 0$ (D) $x + y + 7 = 0$

Ans. A

Sol. Clearly circumcentre is (0, 0)

Now centroid G is $\left(\frac{5-13\cos\theta+13\sin\theta}{3}, \frac{12+13\sin\theta-13\cos\theta}{3}\right)$

G divides HO internally 2 : 1

$\therefore \frac{5-13\cos\theta+13\sin\theta}{3} = \frac{2(0)+1(h)}{3}$ and $\frac{12+13\sin\theta-13\cos\theta}{3} = \frac{2(0)+1(k)}{3}$

H (orthocentre) — 2 : 1 — G (centroid) — O (circumcentre)

$\Rightarrow \sin\theta - \cos\theta = \frac{h-5}{13}$ ..... (1)

and $\sin\theta - \cos\theta = \frac{k-12}{13}$ ..... (2)

$\therefore$ From (1) and (2), we get

$\frac{h-5}{13} = \frac{k-12}{13}$

$\Rightarrow h - k + 7 = 0$

$\therefore$ Locus of orthocentre H(h, k) is

$x - y + 7 = 0$ Ans.

3. A point P (x, y) moves so that the sum of the distances from P to the coordinate axes is equal to the distance from P to the point A(1, 1). The equation of the locus of P in the first quadrant is :

एक बिन्दु P (x, y) इस प्रकार गमन करता है कि उसकी निर्देशांक अक्षों से दूरियों का योग का मान बिन्दु P की बिन्दु A(1, 1) से दूरी के समान है। P के बिन्दु पथ का प्रथम चतुर्थांश में समीकरण होगा :

(A) $(x + 1)(y + 1) = 1$ (B) $(x + 1)(y + 1) = 2$ (C) $(x - 1)(y - 1) = 1$ (D) $(x - 1)(y - 1) = 2$

Ans. B

Sol. $(x + y)^2 = (x - 1)^2 + (y - 1)^2$

$2xy = 2 - 2x - 2y$

$x + y + xy = 1$

$x + y + xy + 1 = 2 \quad \Rightarrow (x + 1)(y + 1) = 2$ Ans.

4. Let A(2, –3) and B(–2, 1) be vertices of a Δ ABC. If the centroid of Δ ABC moves on the line $2x + 3y = 1$, then the locus of the vertex C is

माना A(2, –3) तथा B(–2, 1), Δ ABC के दो शीर्ष है। यदि त्रिभुज का केन्द्रक रेखा $2x + 3y = 1$ पर गमन कर रहा है, तो C का बिन्दुपथ होगा :

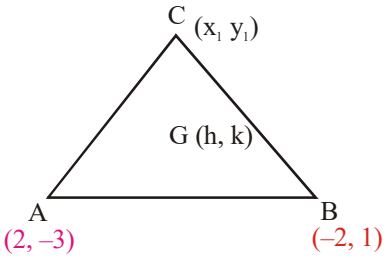
(A) $2x + 3y = 9$ (B) $2x - 3y = 7$ (C) $3x + 2y = 5$ (D) $3x - 2y = 3$

Ans. A

Sol. Let $C = (x_1, y_1)$

The centroid is $\left(\frac{2-2+x_1}{3}, \frac{-3+1+y_1}{3}\right)$ i.e. $\left(\frac{x_1}{3}, \frac{y_1-2}{3}\right)$

It lies on the line $2x + 3y = 1$,



$$\Rightarrow \frac{2}{3}(x_1) + 3\frac{(y_1 - 2)}{3} = 1$$

Hence the locus vertex of C is $2x + 3y = 9$

5. A stick of length 10 units rests against the floor and a wall of a room . If the stick begins to slide on the floor then the locus of its middle point is :

(A) $x^2 + y^2 = 2.5$ (B) $x^2 + y^2 = 25$ (C) $x^2 + y^2 = 100$ (D) none

एक 10 इकाई लम्बाई की छड़ कमरे की दिवार तथा तल पर स्थिर है। यदि छड़ तल पर फिसलना चालू होती है तो उसके मध्य बिन्दु का बिन्दुपथ क्या होगा :

(A) $x^2 + y^2 = 2.5$ (B) $x^2 + y^2 = 25$ (C) $x^2 + y^2 = 100$ (D) कोई नहीं

Ans. B

Sol. given $(a^2 + b^2) = 100$ ---(1)

p (h, k) be the middle pt then

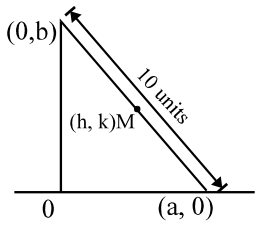
$h = a/2$, $k = b/2$

$\Rightarrow a = 2h$ $b = 2k$

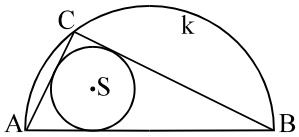
from (1) $4h^2 + 4k^2 = 100$

$h^2 + k^2 = 25$

locus $x^2 + y^2 = 25$



6. AB is the diameter of a semicircle k, C is an arbitrary point on the semicircle (other than A or B) and S is the centre of the circle inscribed into triangle ABC, then measure of :



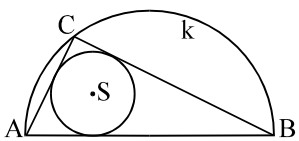
(A) angle ASB changes as C moves on k

(B) angle ASB is the same for all positions of C but it cannot be determined without knowing the radius

(C) angle ASB = 135° for all C

(D) angle ASB = 150° for all C

AB, अर्द्ध वृत k का व्यास है, C अर्द्धवृत पर स्थित कोई बिन्दु है (A अथवा B के अलावा) तथा S एक वृत का केन्द्र है जो त्रिभुज ABC में स्थित है, तो :



(A) कोण ASB का माप C का k पर गमन करने के साथ बदलता है

(B) कोण ASB का माप C की सभी स्थितियों के लिये समान होता है लेकिन त्रिज्या ज्ञात किये बिना इसका मान नही ज्ञात किया जा

सकता है

(C) कोण ASB = 135°, सभी C के लिये

(D) कोण ASB = 150°, सभी C के लिये

Ans. C

Sol. General result: angle subtended by AB at its incentre $= 90 + \frac{C}{2} = 90° + 45° = 135°$

7. Given the points A (0, 4) and B (0, – 4), the equation of the locus of the point P (x, y) such that $|AP - BP| = 6$ is :

दिया गया है कि बिन्दु A (0, 4) तथा B (0, – 4) है। बिन्दु P (x, y) का बिन्दुपथ ज्ञात कीजिये कि $|AP - BP| = 6$ हो :

(A) $9x^2 - 7y^2 + 63 = 0$ (B) $9x^2 - 7y^2 - 63 = 0$ (C) $7x^2 - 9y^2 + 63 = 0$ (D) $7x^2 - 9y^2 - 63 = 0$

Ans. A

Sol. Given that,

$|AP - BP| = 6$

$\Rightarrow AP - BP = \pm 6$

The point

$A(x_1, y_1) = (0,4)$

$B(x_2, y_2) = (4,0)$

Using distance formula,

$$\sqrt{x^2 + (y-4)^2} = \sqrt{x^2 + (y+4)^2} \pm 6$$

On squaring both side and we get,

$\Rightarrow 16y^2 + 81 + 72y = 9(x^2 + (y + 4)^2)$

$\Rightarrow 16y^2 + 81 + 72y = 9(x^2 + y^2 + 16 + 8y)$

$\Rightarrow 16y^2 + 81 + 72y = 9x^2 + 9y^2 + 144 + 72y$

$\Rightarrow 16y^2 + 81 = 9x^2 + 9y^2 + 144$

$\Rightarrow 16y^2 + 81 - 9y^2 = 9x^2 + 144$

$\Rightarrow 7y^2 + 81 = 9x^2 + 144$

$\Rightarrow 7y^2 - 9x^2 = 144 - 81$

$\Rightarrow 7y^2 - 9x^2 = 63$

$$\Rightarrow \frac{7y^2}{63} - \frac{9x^2}{63} = 1$$

$$\Rightarrow \frac{y^2}{9} - \frac{x^2}{7} = 1$$

8. Each member of the family of parabolas $y = ax^2 + 2x + 3$ has a maximum or a minimum point depending upon the value of a. The equation to the locus of the maxima or minima for all possible values of 'a' is :

(A) a straight line with slope 1 and y intercept 3

(B) a straight line with slope 2 and y intercept 2

(C) a straight line with slope 1 and x intercept 3

(D) a straight line with slope 2 and y intercept 3

परवलय $y = ax^2 + 2x + 3$ के समूह का प्रत्येक सदस्य का अधिकतम या न्यूनतम बिन्दु a पर निर्भर करता है। सभी सम्भावित a के मानो के लिये इसकी उच्चिष्ठ या निम्निष्ठ बिन्दुओं का बिन्दुपथ :

(A) एक सरल रेखा है जिसका ढाल 1 तथा y अन्तःखण्ड 3 है

(B) एक सरल रेखा है जिसका ढाल 2 तथा y अन्तःखण्ड 2 है

(C) एक सरल रेखा है जिसका ढाल 1 तथा x अन्तःखण्ड 3 है

(D) एक सरल रेखा है जिसका ढाल 2 तथा y अन्तःखण्ड 3 है

Ans. A

Sol. maxima / minima occurs at $= \left(-\frac{1}{a}, 3 - \frac{1}{a}\right)$ $\left(f(x) = ax^2 + bx + c \text{ has a maxima or minima if } x = -\frac{b}{2a}\right)$

$h = -\frac{1}{a}$ and $k = 3 - \frac{1}{a}$; elliminating 'a' we get [11th, 26-02-2009, P-2]

$k = 3 + x$

locus is $y = x + 3$ $\Rightarrow$ (A)

9. A variable straight line passes through a fixed point (a, b) intersecting the co–ordinates axes at A and B . If 'O' is the origin then the locus of the centroid of the triangle OAB is :

(A) $bx + ay - 3xy = 0$ (B) $bx + ay - 2xy = 0$ (C) $ax + by - 3xy = 0$ (D) none

एक चर सरल रेखा जो नियत बिन्दु (a, b) से गुजरती है तथा निर्देशांक अक्षो को A तथा B पर प्रतिच्छेद करती है। यदि 'O' मूल बिन्दु है तो त्रिभुज OAB के केन्द्र का बिन्दुपथ है :

(A) $bx + ay - 3xy = 0$ (B) $bx + ay - 2xy = 0$ (C) $ax + by - 3xy = 0$ (D) कोई नहीं

Ans. A

Sol. Let slope of line be 'm'.

Equation of line with slope m and passing through (a, b) is $(y-b) = m(x-a)$

Now, coordinate of point $A = \left(\frac{a-b}{m}, 0\right)$

Let (h, k) be the centroid of ΔAOB.

$$h = \frac{a - \frac{b}{m}}{3} \Rightarrow m = \frac{b}{a-3h}$$

$$k = \frac{b-ma}{3}$$

Put $m = \frac{b}{a-3h}$

$$\Rightarrow 3k = b - \frac{ab}{a-3h}$$

$$\Rightarrow ak + bh - 3hk = 0$$

Now, replace h with x and k with y.

$$\Rightarrow bx + ay - 3xy = 0$$

10. A and B are any two points on the positive x and y axis respectively satisfying 2(OA) + 3(OB) = 10. If P is the middle point of AB then the locus of P is :

Aतथा B दो बिन्दु है जो क्रमशः धनात्मक x एवं y अक्ष पर स्थित है जो 2(OA) + 3(OB) = 10 को संतुष्ट करते है। यदि AB का मध्य बिन्दु P है तो P का बिन्दुपथ है :

(A) $2x + 3y = 5$ (B) $2x + 3y = 10$ (C) $3x + 2y = 5$ (D) $3x + 2y = 10$

Ans. A

Sol. $2(OA) + 3(OB) = 10$

$2x + 3y = 10$

$$2 \times 2\left(\frac{x}{2}\right) + 3 \times 2\left(\frac{y}{2}\right) = 10$$

$$\Rightarrow \frac{4x}{2} + \frac{6y}{2} = 10$$

$\Rightarrow 4x + 6y = 10$

$\Rightarrow 2x + 3y = 5$

11. If the area of the triangle formed by the points (1, 2), (2, 3) and (x, 4) is 40 square units, find x.

यदि बिन्दुओं (1, 2), (2, 3) तथा (x, 4) द्वारा बनाये गये त्रिभुज का क्षेत्रफल 40 वर्ग इकाई है, x का मान ज्ञात कीजिये।

Ans. 83, – 77

Sol. $40 = \frac{1}{2}\left|1(3-4)+2(4-2)+x(2-3)\right|$

$40 \times 2 = |1(-1) + 2(2) + x(-1)|$

$80 = |-1 + 4 - x|$

$80 = |3 - x|$

$3 - x = \pm 80$

Equate 3 – x to 80 and –80 separately and solve for x to obtain the required result.

$3 - x = 80$

$x = -80 + 3$

$= -77$

$3 - x = -80$

$x = 3 + 80$

$= 83$

Thus, x = –77, 83.

12. Find the area of the of the quadrilateral whose vertices are A(1, 1), B(3, 4), C(5, – 2) and D(4, – 7).

शीर्षो A(1, 1), B(3, 4), C(5, – 2) तथा D(4, – 7) द्वारा बनाये गये चतुर्भुज का क्षेत्रफल ज्ञात कीजिये।

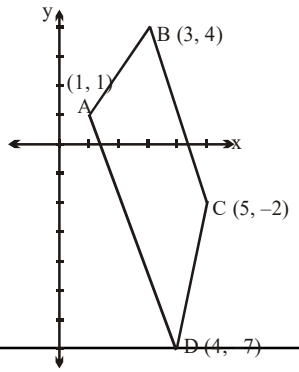
Ans. 20.5

Sol. Quadrilateral ADCB = ΔADC + ΔACB

$\Delta ADC = \frac{1}{2}[1(-7+2) + 4(-2-1) + 5(1+7)] = \frac{23}{2}$

$\Delta ACB = \frac{1}{2}[1(-2-4) + 5(4-1) + 3(1+2)] = 9$

$\therefore$ Area of the quadrilateral ADCB = $20\frac{1}{2}$ sq. units



13. Let Δ_1 denotes the area of the triangle formed by the vertices $\left(am_1^2, 2am_1\right)$, $\left(am_2^2, 2am_2\right)$, $\left(am_3^2, 2am_3\right)$ and Δ_2 denotes the area of the triangle formed by the vertices $\left(am_1m_2, a(m_1+m_2)\right)$, $\left(am_2m_3, a(m_2+m_3)\right)$ and $\left(am_3m_1, a(m_3+m_1)\right)$. Find $\frac{\Delta_1}{\Delta_2}$.

माना Δ_1 एक त्रिभुज का क्षेत्रफल है जिसके शीर्ष $\left(am_1^2, 2am_1\right)$, $\left(am_2^2, 2am_2\right)$, $\left(am_3^2, 2am_3\right)$ तथा Δ_2 एक त्रिभुज का क्षेत्रफल है जिसके शीर्ष $\left(am_1m_2, a(m_1+m_2)\right)$, $\left(am_2m_3, a(m_2+m_3)\right)$ तथा $\left(am_3m_1, a(m_3+m_1)\right)$ है। $\frac{\Delta_1}{\Delta_2}$ का मान ज्ञात कीजिये।

Ans. 2

Sol. $\Delta_1 = a^2(m_2-m_1)(m_3-m_1)(m_1-m_2)$

$\Delta_2 = \frac{1}{2}a^2(m_2-m_1)(m_3-m_1)(m_1-m_2)$

Hence $\frac{\Delta_1}{\Delta_2} = 2$ Ans.

14. Find the relation between x and y when the point (x, y) lies on the straight line joining the points (2, –3) and (1, 4).

x और y के मध्य सम्बन्ध ज्ञात कीजिये जिसके लिये बिन्दु (x, y) बिन्दुओं (2, –3) व (1, 4) को मिलाने वाली रेखा पर स्थित है।

Ans. $7x + y = 11$

Sol. Determinant = 0

15. Find the locus of a point which is equidistant from the points (3, 4) and (5, – 2).

उस बिन्दु का बिन्दुपथ ज्ञात कीजिये जो बिन्दुओं (3, 4) तथा (5, – 2) से समान दूरी पर स्थित है।

Ans. $x - 3y = 1$

Sol. A(3,4), B(5,–2), P(x, y)

$PA = PB \Rightarrow (x-3)^2 + (y-4)^2 = (x-2)^2 + (y+2)^2$

16. If the distances of a point P from the points (2, 1) and (1, 2) are in the ratio 2 : 1, find the locus of the point P.

यदि बिन्दु P की बिन्दुओं (2, 1) तथा (1, 2) से दूरियो का अनुपात 2 : 1 है, तो बिन्दु P का बिन्दुपथ ज्ञात कीजिये।

Ans. $3x^2 + 3y^2 - 4x - 14y + 15 = 0$

Sol. Let $P(x_1, y_1)$ be any point on the locus.

Let A(2,1) and B(1,2) be the given point.

Given that PA : PB = 2 : 1

i.e., $\frac{PA}{PB} = \frac{2}{1}$

$PA = 2PB$

$PA^2 = 4PB^2$

$(x_1 - 2)^2 + (y_1 - 1)^2 = 4\,[(x_1 - 1)^2 + (y_1 - 2)^2]$

$x_1^2 - 4x_1 + 4 + y_1^2 - 2y_1 + 1$

$= 4(x_1^2 - 2x_1 + 1 + y_1^2 - 4y_1 + 4]$

$x_1^2 + y_1^2 - 4x_1 - 2y_1 + 5$

$= 4x_1^2 - 8x_1 + 4y_1^2 - 16y_1 + 20$

$-3x_1^2 - 3y_1^2 + 4x_1 + 14y_1 - 15 = 0$

$\therefore\ 3x_1^2 + 3y_1^2 - 4x_1 - 14y_1 + 15 = 0$

$\therefore$ The locus of the point (x_1, y_1) is $3x^2 + 3y^2 - 4x - 14y + 15 = 0$

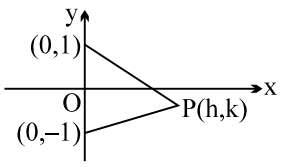
17. A(0, 1) and B(0, – 1) are 2 points if a variable point P moves such that sum of its distance from A and B is 4. Then the locus of P is the equation of the form of $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. Find the value of $(a^2 + b^2)$.

A(0, 1) तथा B(0, – 1) दो बिन्दु है यदि बिन्दु P इस प्रकार गमन करता है कि इसकी A एवं B से दूरियो को योग 4 है। तो बिन्दु P का बिन्दुपथ $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ के रूप में लिखा जाता है। $(a^2 + b^2)$ का मान ज्ञात कीजिये।

Ans. 7

Sol. $\sqrt{h^2 + (k-1)^2} + \sqrt{h^2 + (k+1)^2} = 4$ [11th, 14-10-2007]

$h^2 + (k - 1)^2 = 16 + h^2 + (k + 1)^2 - 8\sqrt{h^2 + (k+1)^2}$



$16 + 4k = 8\sqrt{h^2 + (k+1)^2} \Rightarrow$ $4 + k = 2\sqrt{h^2 + (k+1)^2}$

$16 + k^2 + 8k = 4h^2 + 4(k + 1)^2$

$4h^2 + 3k^2 = 12$

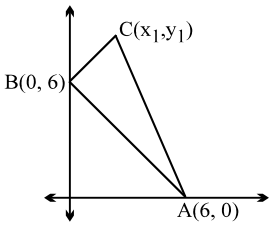
$\frac{h^2}{3} + \frac{k^2}{4} = 1 \Rightarrow \frac{x^2}{3} + \frac{y^2}{4} = 1 \Rightarrow a^2 = 3$ and $b^2 = 4 \Rightarrow 3 + 4 = 7$ Ans

18. The ends of the hypotenuse of a right angled triangle are (6, 0) and (0, 6). Find the locus of the third vertex and interpret the locii geometrically.

एक समकोणीय त्रिभुज के कर्ण के शीर्ष (6, 0) तथा (0, 6) है। तीसरे शीर्ष का बिन्दु पथ ज्ञात कीजिये तथा ज्यामितीय रूप से अनुमान लगाइये।

Ans. $x^2 + y^2 - 6x - 6y = 0$

Sol. Let $C(x_1, y_1)$ be the moving point (third vertex) whose locus is required and A(6, 0) and B(0, 6) be the given vertices.



By hypothesis

$(AC)^2 + (BC)^2 = (AB)^2$ $(\because \angle ACB = 90°)$

$\Rightarrow (x_1 - 6)^2 + (y_1 - 0)^2 + (x_1 - 0)^2 + (y_1 - 6)^2 = 6^2 + 6^2$

$\Rightarrow 2x_1^2 + 2y_1^2 - 12x_1 - 12y_1 = 0$ or $x_1^2 + y_1^2 - 6x_1 - 6y_1 = 0$

Changing (x_1, y_1) by (x, y) then $x^2 + y^2 - 6x - 6y = 0$

which is the required locus of third vertex C.

1. The position vectors of vertices of ΔABC are (1, –2), (– 7, 6) and $\left(\frac{11}{5}, \frac{2}{5}\right)$ respectively.

The measure of the interior angle A of the ΔABC, is :

(A) acute and lies in (75°, 90°) (B) acute and lies in (60°, 75°)

(C) acute and lies in (45°, 60°) (D) obtuse and lies in (120°, 150°)

यदि त्रिभुज ABC के शीर्ष की स्थिति सदिश यदि (1, –2), (– 7, 6) एवं $\left(\frac{11}{5}, \frac{2}{5}\right)$ है, तो आन्तरिक कोण A होगा :

(A) न्यूनकोण जो कि (75°, 90°) के बीच है (B) न्यून कोण जो कि (60°, 75°) के बीच है

(C) न्यूनकोण जो कि (45°, 60°) के बीच है (D) अधिक कोण (120°, 150°) के बीच है

Ans. B

Sol. Method 1

use cosine rule

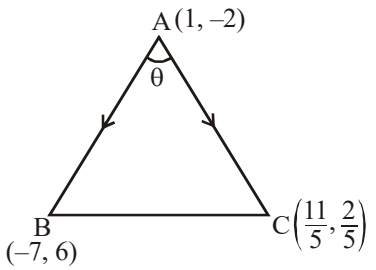
$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

Method 2

Method 3

We have $\overrightarrow{AB} = \vec{V}_1 = -8\hat{i} + 8\hat{j} = 8(-\hat{i} + \hat{j})$ [13th, 13-12-2009,P1]

$$\overrightarrow{AC} = \vec{V}_2 = \left(\frac{11}{5} - 1\right)\hat{i} + \left(\frac{2}{5} + 2\right)\hat{j} = \frac{6}{5}\hat{i} + \frac{12}{5}\hat{j} = \frac{6}{5}(\hat{i} + 2\hat{j})$$



$$\therefore \quad \cos\theta = \frac{(-\hat{i}+\hat{j})\cdot(\hat{i}+2\hat{j})}{(\sqrt{2})(\sqrt{5})} = \frac{-1+2}{\sqrt{10}} = \frac{1}{\sqrt{10}}$$

$\Rightarrow \tan\theta \Rightarrow 3 \Rightarrow$ (B)

2. The line through point (m, – 9) and (7, m) has slope *m*. The y-intercept of this line, is :

बिन्दुओं (m, – 9) तथा (7, m) को मिलाने वाली रेखा का ढाल *m* है। रेखा का y-अन्तःखण्ड है :

(A) – 18 (B) – 6 (C) 6 (D) 18

Ans. A

Sol. $m = \frac{m+9}{7-m} \Rightarrow 7m - m^2 = m + 9 \Rightarrow m^2 - 6m + 9 = 0 \Rightarrow m = 3$

$y + 9 = 3(x - 3)$

at $x = 0$ $y = -18$ Ans.

3. A line passes through (2, 2) and cuts a triangle of area 9 square units from the first quadrant. The sum of all possible values for the slope of such a line, is :

एक रेखा जो (2, 2) से गुजरती है तथा प्रथम चतुर्थांश में 9 वर्ग इकाई का एक त्रिभुज काटती है। रेखा की सभी सम्भावित प्रवणताओं का योग ज्ञात कीजिए :

(A) – 2.5 (B) – 2 (C) – 1.5 (D) – 1

Ans. A

Sol. Let the slope of a line be = m, and it pass through point (2,2) then its equation is $y - 2 = m(x - 2)$

$y = mx - 2m + 2$

Now, area of triangle in such a case is half the product of x-intercept and y-intercept.

x-intercept is given by putting $y = 0$, i.e.,

$mx - 2m + 2 = 0$ or $x = \frac{2m-2}{m}$

and y intercept is given by putting $x = 0$, i.e.,

$y = -2m + 2$

or $18 = -4m^2 + 8m - 4m$

$\because$ Area of triangle $= \frac{1}{2} \times \text{base} \times \text{height}$

$$2A = (-2m + 2) \times \frac{(2m - 2)}{m} = \frac{-4m + 8m - 4}{m}$$

$2Am = -4m^2 + 8m - 4$

Put $A = 9$ unit, we get

$18m = -4m^2 + 8m - 4$

$4m^2 + 10m + 4 = 0$

we know that in a quadratic equation $ax^2 + bx + c = 0$

sum of roots is $-\frac{b}{a}$

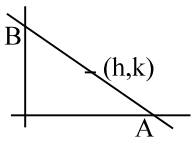
sum of slopes is $-\frac{10}{4} = -2.5$

4. A variable straight line passes through the points of intersection of the lines, $x + 2y = 1$ and $2x - y = 1$ and meets the co–ordinate axes in A and B . The locus of the middle point of AB is :
(A) $x + 3y - 10xy = 0$ (B) $x - 3y + 10xy = 0$ (C) $x + 3y + 10xy = 0$ (D) none

एक चर सरल रेखा जो रेखाओं $x + 2y = 1$ तथा $2x - y = 1$ के प्रतिच्छेद बिन्दु से गुजरती है तथा निर्देशांक अक्षो को A तथा B पर मिलती है। AB के मध्य बिन्दु का बिन्दुपथ है :
(A) $x + 3y - 10xy = 0$ (B) $x - 3y + 10xy = 0$ (C) $x + 3y + 10xy = 0$ (D) कोई नहीं

Ans. A

Sol. $\frac{x}{2h} + \frac{y}{2k} = 1$. (It passes through $\frac{3}{5}, \frac{1}{5}$)



5. The equations of L_1 and L_2 are $y = mx$ and $y = nx$, respectively. Suppose L_1 makes twice as large of an angle with the horizontal (measured counterclockwise from the positive x-axis) as does L_2 and that L_1 has 4 times the slope of L_2. If L_1 is not horizontal, then the value of the product (mn) equals :

L_1 तथा L_2 के समीकरण क्रमशः $y = mx$ तथा $y = nx$ है। माना L_1 का क्षैतिज से कोण (धनात्मक x-अक्ष से वामावृत) L_2 के क्षेतिज के कोण का दुगना है तथा L_1 का ढाल L_2 से चार गुना है। यदि L_1 क्षैतिज नही है, तो (mn) गुणनफल है :

(A) $\frac{\sqrt{2}}{2}$ (B) $-\frac{\sqrt{2}}{2}$ (C) 2 (D) –2

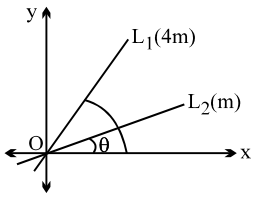
Ans. C

Sol. Let $m = \tan 2\theta$ and $n = \tan \theta$

and $m = 4n$ and $m \neq 0$, $\therefore$ $\tan 2\theta \neq 0$; $\therefore$ $\theta \neq 0$

$\tan 2\theta = 4 \tan \theta$

$$\frac{2\tan\theta}{1-\tan^2\theta} = 4\tan\theta$$



$\therefore$ $1 = 2 - 2\tan^2\theta$

$2\tan^2\theta = 1 \quad \Rightarrow \quad \tan^2\theta = 1/2$

$$\therefore \quad mn = \tan 2\theta \cdot \tan\theta = \frac{2\tan^2\theta}{1-\tan^2\theta} = \frac{1}{1-(1/2)} = 2 \text{ Ans.}$$

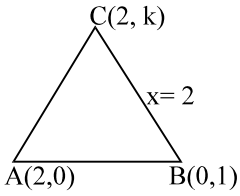
6. The extremities of the base of an isosceles triangle ABC are the points A (2, 0) and B (0, 1). If the equation of the side AC is x = 2 then the slope of the side BC is :

समद्विबाहु त्रिभुज ABC के आधार A (2, 0) तथा B (0, 1) है। यदि समीकरण x = 2 भुजा AC है तो BC का ढाल है :

(A) $\frac{3}{4}$ (B) $\frac{4}{3}$ (C) $\frac{3}{2}$ (D) $\sqrt{3}$

Ans. A

Sol. AB = BC



$$\therefore \quad \sqrt{k^2} = \sqrt{4+(k-1)^2}$$

$k^2 = 4 + k^2 - 2k + 1$

$$\therefore \quad k = \frac{5}{2}$$

$$\therefore \quad \text{slope of BC} = \frac{5/2-1}{2} = \frac{5-2}{4} = \frac{3}{4} \text{ Ans.}$$

7. A line with gradient 2 intersects a line with gradient 6 at the point (40, 30). The distance between x-intercepts of

these lines, is :

एक 2 प्रवणता की रेखा, एक अन्य 6 प्रवणता की रेखा को (40, 30) पर प्रतिच्छेद करती है। दोनों के x-अन्तःखण्डो के मध्य दूरी है :

(A) 6 (B) 8 (C) 10 (D) 12

Ans. C

Sol. Let C_1 and C_2 be the x-intercept of lines with slope 2 and 6 respectively

$y - 0 = 2(x - c_1)$

$y = 2x - 2C_1$ ....(1)

|||ly $y = 6x - 6C_2$ ....(2)

both (1) and (2) satisfy $x = 40$ and $y = 30$

$30 = 80 - 2C_1 \Rightarrow C_1 = 25$

and $30 = 240 - 6C_2$

$\Rightarrow 6 \cdot C_2 = 210 \Rightarrow C_2 = 35$

hence $C_2 - C_1 = 10$ Ans.

8. Locus of a point which is equidistant from the point (3, 4) and (5, – 2) is a straight line whose x-intercept is :

एक बिन्दु का बिन्दुपथ जो बिन्दुओं (3, 4) तथा (5, – 2) से समान दूरी पर है, एक सरल रेखा है जिसका x-अन्तःखण्ड है :

(A) 1/3 (B) 2/3 (C) 1 (D) – 1/3

Ans. C

Sol. Locus is $x - 3y = 1 \Rightarrow$ x-intercept is 1

9. The diagonals of a parallelogram PQRS are along the lines $x + 3y = 4$ and $6x - 2y = 7$. Then PQRS must be a :

(A) rectangle (B) square (C) cyclic quadrilateral (D) rhombus

एक समान्तर चतुर्भुज PQRS के विकर्ण रेखाओं $x + 3y = 4$ तथा $6x - 2y = 7$ के सम्पाति है। तो PQRS होना चाहिये :

(A) एक आयत (B) एक वर्ग (C) एक चक्रीय चतुर्भुज (D) एक समचतुर्भुज

Ans. D

Sol. lines are perpendicular $\Rightarrow$ (D)

10. The sides of a triangle ABC lie on the lines $3x + 4y = 0$; $4x + 3y = 0$ and $x = 3$. Let (h, k) be the centre of the circle inscribed in ΔABC. The value of $(h + k)$ equals :

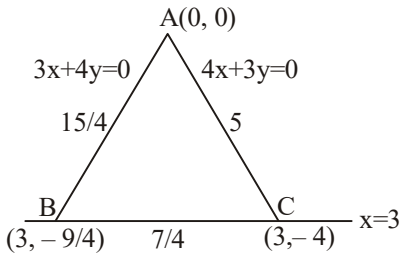
एक त्रिभुजABC की भुजाऐं रेखाओं $3x + 4y = 0$; $4x + 3y = 0$ तथा $x = 3$ द्वारा दी जाती है। माना (h, k) त्रिभुज ABC का का अतःकेन्द्र है, तो $(h + k)$ का मान होगा।

(A) 0 (B) 1/4 (C) – 1/4 (D) 1/2

Ans. A

Sol. $h+k=\dfrac{a(x_1+y_1)+b(x_2+y_2)+c(x_3+y_3)}{a+b+c}$

$$=\frac{\frac{7}{4}(0)+5\left(3-\frac{9}{4}\right)+\frac{15}{4}(3-4)}{5+\frac{15}{4}+\frac{7}{4}}=\frac{\frac{15}{4}-\frac{15}{4}}{5+\frac{22}{4}}=0$$



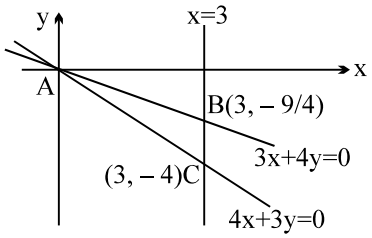
Alternately: Equation of angle bisector of angle A

$$\frac{3x+4y}{5}=\pm\frac{4x+3y}{5} \quad \Rightarrow \quad x=\pm y$$

equation of internal bisector is $x=-y$

since h and k lie on the line $x=-y$

$\Rightarrow$ $h+k=0$ Ans.



11. If m and b are real numbers and $mb > 0$, then the line whose equation is $y = mx + b$ cannot contain the point :

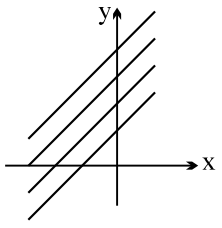
यदि m तथा b वास्तविक संख्याऐं है तथा $mb > 0$ है, तो रेखा जिसका समीकरण $y = mx + b$ है पर कौनसा बिन्दु स्थित नही होगा :

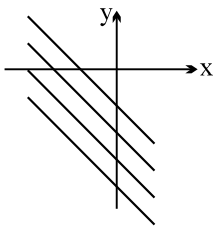
(A) (0, 2009) (B) (2009, 0) (C) (0, – 2009) (D) (20, – 100)

Ans. B

Sol. $mb > 0 \Rightarrow$ $m > 0$ and $b > 0$ or $m < 0$ and $b < 0$

hence possible lines are as shown $\Rightarrow$ x-intercept can not be + ve





$m > 0, b > 0$ $m < 0, b < 0$

In both the cases x intercept cannot be + ve $\Rightarrow$ (B)

12. If the vertices P and Q of a triangle PQR are given by (2, 5) and (4, –11) respectively, and the point R moves along the line N: $9x + 7y + 4 = 0$, then the locus of the centroid of the triangle PQR is a straight line parallel to :

त्रिभुज PQR के शीर्ष P (2, 5) तथा Q (4, –11) है तथा बिन्दु R रेखा N: $9x + 7y + 4 = 0$, पर गमन करता है, तो त्रिभुज PQR के केन्द्रक का बिन्दुपथ एक रेखा है तो वह किसके समानान्तर होगी :

(A) PQ (B) QR (C) RP (D) N

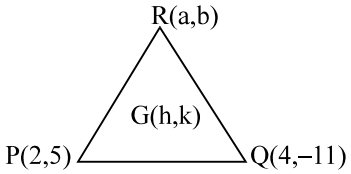
Ans. D

Sol. R (a, b) lies on $9x + 7y + 4 = 0$

$\Rightarrow R\left(a, \frac{(4+9a)}{7}\right)$, centroid of Δ PQR = (h, k)

$$h = \left(\frac{2+4+a}{3}\right) = \frac{6+a}{3} \qquad ....(1)$$

$$k = \frac{5-11-\frac{(4+9a)}{7}}{3} = \frac{-46-9a}{7\times 3} \qquad ....(2)$$



from (1) and (2) we get

equating x $\qquad 3h - 6 = \frac{-(21k-46)}{9} \Rightarrow 27h + 21k - 54 + 46 = 0$

or locus is $9x + 7y - 8/3 = 0$

this line is || to N

13. The co-ordinates of the orthocentre of the triangle bounded by the lines, $4x - 7y + 10 = 0$; $x + y = 5$ and $7x + 4y = 15$ is :

रेखाओं $4x - 7y + 10 = 0$; $x + y = 5$ तथा $7x + 4y = 15$ द्वारा बनाए गए त्रिभुज का लम्बकेन्द्र है :

(A) (2, 1) (B) (– 1, 2) (C) (1, 2) (D) (1, – 2)

Ans. C

Sol. Given lines

$4x - 7y + 10 = 0$ ...(1)

$x + y - 5 = 0$ ...(2)

$7x + 4y - 15 = 0$ ...(3)

On solving eq (1) & (3)

$$7x + 4\left(\frac{4x+10}{7}\right) - 15 = 0$$

$49x + 16x + 40 - 105 = 0$

$65x = 65 \Rightarrow x = 1$

from equation (1)

$4 - 7y = -10$

$-7y = -14$

$y = 2$

Point of intersection is (1,2)

Here equation (1) and (3) are perpendicular so the lines formed right angle triangle.

Hence the orthocentre of Right angle triangle is the point at whaich 90° angle is formed.

Hence (1,2) is orthocentre.

14. Find the area of the triangle formed by the straight lines whose equations are $2y + x - 5 = 0$, $y + 2x - 7 = 0$ and $x - y + 1 = 0$.

रेखाओं $2y + x - 5 = 0$, $y + 2x - 7 = 0$ एवं $x - y + 1 = 0$ द्वारा बनाए गए त्रिभुज का क्षेत्रफल ज्ञात करें।

Ans. 3/2

Sol. vertex are (1,2), (3,1), (2,3) and $\Delta = \frac{3}{2}$

15. (a) Find the equation of the straight line which passes through the point (1, 2) and is such that the given point bisects the part intercepted between the axes.
(b) Find the equations to the straight lines which join the origin and the points of trisection of the portion of the line $x + 3y - 12 = 0$ intercepted between the axes of co-ordinates.

(a) बिन्दु (1, 2) से गमन करने वाली रेखा का समीकरण ज्ञात करो जहाँ इस बिन्दु पर अक्षों के मध्य काटा गया अन्तः खण्ड समद्विभाजित होता है।

(b) उन रेखाओं का समीकरण ज्ञात करें, जो रेखा $x + 3y - 12 = 0$ द्वारा अक्षों के मध्य काटे गए अन्तःखण्ड के समत्रिभाजक बिन्दुओं को मूलबिन्दु से जोड़ती है।

Ans. (a) $2x + y = 4$ (b) $6y = x$ & $3y = 2x$

Sol. (a) Given : P(1,2)

We know that the equation of a line making intercepts a and b with x-axis and y-axis respectively, is given by

$$\Rightarrow \frac{x}{a} + \frac{y}{b} = 1$$

Here we have

$$\Rightarrow \frac{a+0}{2} = 1 \text{ and } \frac{b+0}{2} = 2$$

which give $a = 2$ and $b = 4$

Therefore, the required equation of the line is

$\Rightarrow \frac{x}{2}+\frac{y}{4}=1$

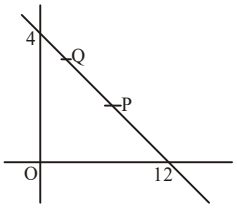
$\Rightarrow \frac{2x+y}{4}=1$

$\Rightarrow 2x+y=4$

$\Rightarrow 2x+y-4=0$

(b) $P\left(8,\frac{4}{3}\right)$

$Q\left(4,\frac{8}{3}\right)$



16. Find the equation of the straight line passing through (3, 4) and the intersecting point of the two lines $5x - y = 9$ and $x + 6y = 8$.

रेखाओं $5x - y = 9$ तथा $x + 6y = 8$ के प्रतिच्छेदन बिन्दु से तथा (3, 4) से गुजरने वाली रेखा का समीकरण ज्ञात कीजिए।

Ans. $3x - y - 5 = 0$

Sol. Solving the equations $5x - y = 9$ and $x + 6y = 8$, we get the intersecting point of the two lines as (2, 1).
Then we have to find the straight line joining the points (2, 1) and (3, 4).

Its equation is $\frac{y-1}{1-4}=\frac{x-2}{2-3}$

i.e., $3x - y - 5 = 0$. Ans.

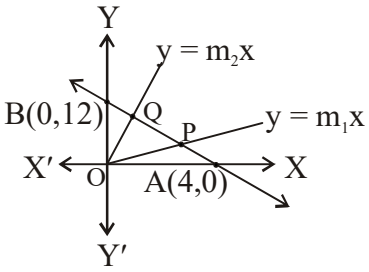
17. (a) Find the equations to the straight lines which go through the origin and trisect the portion of the straight line $3x + y = 12$ which is intercepted between the axes of coordinates.
(b) Find the equation to the straight line which passes through the point $(-5, 4)$ and is such that the portion of it between the axes is divided by the point in the ratio of 1 : 2.

(a) रेखा $3x + y = 12$ द्वारा अक्षों के मध्य काटे गये अन्तःखण्ड को समत्रिभाषित करने वाले बिन्दुओं को मूल बिन्दु से मिलने वाली रेखाओं का समीकरण ज्ञात करें।
(b) रेखा का समीकरण ज्ञात करें, जिनका अक्षों के मध्य काटा गया अन्तः खण्ड बिन्दु $(-5, 4)$ पर 1 : 2 में विभाजित होती है।

Ans. (a) $y = 6x, 2y = 3x$ (b) $5y = 8x + 60$

Sol. (a) $P \equiv \left(\frac{2\times4+1\times0}{2+1}, \frac{2\times0+1\times12}{2+1}\right) \equiv \left(\frac{8}{3}, 4\right)$

$Q \equiv \left(\frac{1\times4+2\times0}{2+1}, \frac{1\times0+2\times12}{2+1}\right) \equiv \left(\frac{4}{3}, 8\right)$



Clearly P and Q lie on $y = m_1x$ and $y = m_2x$.

$\therefore 4 = m_1 \times \frac{8}{3}$ and $8 = m_2 \times \frac{4}{3}$

$\Rightarrow m_1 = \frac{3}{2}$ and $m_2 = 6$

Hence the required lines are

$y = \frac{3}{2}x \Rightarrow 2y = 3x$ and $y = 6x$

(b) Let the required straight line be $\frac{x}{a} + \frac{y}{b} = 1$. This meets the axes in the points whose coordinates are (a, 0) and (0, b).

The coordinates of the point dividing the line joining these points in the ratio 1 : 2, are (Art. 22).

$\frac{2\cdot a+1\cdot 0}{2+1}$ and $\frac{2\cdot 0+1\cdot b}{2+1}$, i.e. $\frac{2a}{3}$ and $\frac{b}{3}$.

If this be the point (– 5, 4) we have

$-5 = \frac{2a}{3}$ and $4 = \frac{b}{3}$,

so that $a = -\frac{15}{2}$ and $b = 12$.

The required straight line is therefore

$$\frac{x}{-\frac{15}{2}} + \frac{y}{12} = 1,$$

i.e., $5y - 8x = 60$. Ans.

18. Find the equations of the sides and of the medians of the triangle formed by joining the points (2, 4), (4, 6) and (– 6, – 10).

बिन्दुओं (2, 4), (4, 6) एवं (– 6, – 10) द्वारा बनाये त्रिभुज की भुजाओं एवं मध्यिकाओं का समीकरण ज्ञात करें।

Ans. $x - y + 2 = 0, 7x - 4y + 2 = 0, 8x - 5y - 2 = 0, y = 2x, 3x = 2y, 5x = 3y$

Sol. Since median divideds the hint segment into two parts.

Let AD be the median on aC and EC be the median on AB.

Coordinates of D, E and F are

$$D = \equiv \left(\frac{4-6}{2}, \frac{6-10}{2}\right), E \equiv \left(\frac{2+4}{2}, \frac{4+6}{2}\right)$$

$$F \equiv \left(\frac{2+(-6)}{2}, \frac{4+(-10)}{2}\right)$$

$D \equiv (-1,-2), E \equiv (3,5), F \equiv (-2,-3)$

$$\text{Equation of line : } (y - y_1) = \left(\frac{y_2 - y_1}{x_2 - x_1}\right)(x - x_1)$$

Equation of AD :

A (2, 4) D(–1, –2)

x_1, y_1 x_2, y_2

$$(y-4) = \left(\frac{-2-4}{-1-2}\right)(x-2)$$

$$(y-4) = \left(\frac{-6}{-3}\right)(x-2)$$

$\Rightarrow y - 4 = 2(x - 2)$

$\Rightarrow y - 4 = 2x - 4$

$AD \Rightarrow 2x - y = 0$

Equation of BF :

B(4,6), F(–2,–3)

$$(y-6) = \left(\frac{-3-6}{-2-4}\right)(x-4)$$

$$\Rightarrow (y-6) = \frac{-9}{-6}(x-4)$$

$\Rightarrow 2(y - 6) = 3(x - 4) \Rightarrow 2y - 12 = 3x - 12$

$BF \Rightarrow 3x - 2y = 0$

Equation of EC :

(3,5), C(–6,–10)

$$(y-5)=\left(\frac{-10-5}{-6-3}\right)(x-3)$$

$\Rightarrow 3y - 15 = 5x - 15$

$$\Rightarrow (y-5)=\frac{-15}{-9}(x-3)$$

EC $\Rightarrow 5x - 3y = 0$

19. Find the equation to the straight line which passes through the point (5, 6) and has intercepts on the axes :
(i) equal in magnitude and both positive.
(ii) equal in magnitude but opposite in sign.

सरल रेखा का समीकरण ज्ञात करें जो बिन्दु (5, 6) से गुजरती है तथा अक्षों पर अन्तःखण्ड निम्न प्रकार से काटती है :
(i) परिमाण में समान तथा दोनों धनात्मक
(ii) परिमाण में समान लेकिन विपरीत चिन्ह के

Ans. (i) $x + y = 11$ (ii) $y - x = 1$

Sol. (i) intercepts are equal and positive

$\Rightarrow a = b = k$

The equation of straight line is

$$\frac{x}{a}+\frac{y}{b}=1 \qquad ...(i)$$

Since this line passes through (5,6) and a = b = k, we get :

$$\frac{5}{k}+\frac{6}{k}=1$$

$k = 1$

$$\therefore \frac{x}{11}+\frac{y}{11}=1$$

$\Rightarrow x + y = 11$

(ii) intercepts are equal but opposite in sign

Let a = k, b = –k

putting in (i), we get

$$\frac{5}{k}+\frac{6}{-k}=1$$

$$\frac{5}{k}-\frac{6}{k}=1$$

$\Rightarrow k = -1$

Thus from (i)

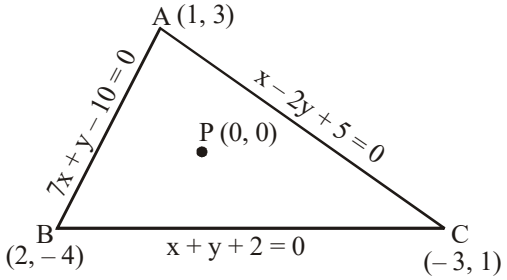
$x - y = -1$

20. A triangle ABC is formed by three lines $x + y + 2 = 0$, $x - 2y + 5 = 0$ and $7x + y - 10 = 0$. P is a point inside the triangle ABC such that areas of the triangles PAB, PBC and PCA are equal. If the co-ordinates of the point P are (a, b) and the area of the triangle ABC is δ, then find $(a + b + \delta)$.

तीन सरल रेखाओं $x + y + 2 = 0$, $x - 2y + 5 = 0$ एवं $7x + y - 10 = 0$ द्वारा एक त्रिभुज ABC बनाया जाता है। त्रिभुज ABC में एक बिन्दु P इस प्रकार है कि त्रिभुज PAB, PBC और PCA के क्षेत्रफल बराबर है। यदि बिन्दु P के निर्देशाक (a, b) है और त्रिभुज ABC का क्षेत्रफल δ हैं, तब $(a + b + \delta)$ ज्ञात कीजिए।

Ans. 15

Sol.



P is the centroid (0, 0). Ans.

21. Match the following column :

Column-I	**Column-II** (Equation of Straight Line)
(A) Which cuts-off an intercept 4 on the x-axis and passes through the point (2, – 3).	(P) $2x + y + 1 = 0$
(B) Which cuts-off equal intercepts on the co-ordinate axes and passes through (2, 5)	(Q) $x + y = 7$
(C) Which makes an angle of 135° with the axis of x and which cuts the axis of y at a distance - 8 from the origin and.	(R) $3x - 2y = 12$
(D) Through the point (4, 1) and making with the axes in the first quadrant	(S) $x + 4y = 8$

a triangle whose area is 8.

(T) $x + y + 8 = 0$

निम्न कॉलम को सुमेलित कीजिए :

Column-I	**Column-II** (सरल रेखा का समीकरण)
(A) जो x अक्ष पर 4 का अन्तःखण्ड काटती है तथा बिन्दु (2, – 3) से गुजरती हैं	(P) $2x + y + 1 = 0$
(B) जो निर्देशांक अक्षो पर समान अन्तःखण्ड काटती है एवं (2, 5) से गुजरती है	(Q) $x + y = 7$
(C) जो x अक्ष के साथ 135° कोण बनाती है तथा y अक्ष को मूलबिन्दु से - 8 दूरी पर काटती है	(R) $3x – 2y = 12$
(D) (4, 1) से गुजरती है तथा अक्षों के साथ प्रथम चतुर्थांश में 8 इकाई का त्रिभुज बनाती है	(S) $x + 4y = 8$
	(T) $x + y + 8 = 0$

Ans. (A) R, (B) Q (C) T (D) S

Sol. (A) (4,0), (2, –3)

$$(y-0)=\left(\frac{-3-0}{2-4}\right)(x-4)$$

$3x – 2y – 12 = 0$

(B) $\frac{x}{a}+\frac{y}{a}=1$

(2,5)

$a = 7$

$x + y = 7$

(C) $m = – 1$

$y = –x – 8$

(D) $ab = 16$

$\frac{x}{a}+\frac{ay}{16}=1$

(4,1)

$a = 8$

$x + 4y = 8$

1. If the x intercept of the line $y = mx + 2$ is greater than 1/2 then the gradient of the line lies in the interval :

यदि रेखा $y = mx + 2$ का x अन्तःखण्ड 1/2 से अधिक है तो रेखा की प्रवणता किस अंतराल में स्थित है :

(A) $(-1, 0)$ (B) $(-1/4, 0)$ (C) $(-\infty, -4)$ (D) $(-4, 0)$

Ans. D

Sol. x-intercept is the point where the line cuts the x-axis. Therefore, its y-coordinate is 0 :

$$0 = mx + 2 \Rightarrow x = \frac{-2}{m}$$

Given :

$$\frac{-2}{m} > \frac{1}{2}$$

$$\Rightarrow \frac{2}{m} + \frac{1}{2} < 0$$

$$\Rightarrow \frac{4+m}{2m} < 0$$

Critical points : 0, –4

Using Wavy-curve method, required interval : (–4,0)

2. Let the co-ordinates of the two points A and B be (1, 2) and (7, 5) respectively . The line AB is rotated through 45º in anti clockwise direction about the point of trisection of AB which is nearer to B . The equation of the line in new position is :

(A) $2x - y - 6 = 0$ (B) $x - y - 1 = 0$ (C) $3x - y - 11 = 0$ (D) none of these

माना A तथा B दो बिन्दुओं के निर्देशांक क्रमशः (1, 2) तथा (7, 5) है। यदि बिन्दु B के निकट समत्रिभाजक बिन्दु के सापेक्ष रेखा AB को 45º से वामावृत घुमा दिया जाता है तो नई स्थिति में रेखा का समीकरण है।

(A) $2x - y - 6 = 0$ (B) $x - y - 1 = 0$ (C) $3x - y - 11 = 0$ (D) इनमें से कोई नहीं

Ans. C

Sol. point of trisection is (5, 4) . After rotation the slope of the line

$$\tan\theta_1 = \frac{\tan45°+\tan\theta_2}{1-\tan45°\tan\theta_2} = \frac{1+\frac{1}{2}}{1-\frac{1}{2}} = 3 \Rightarrow C$$

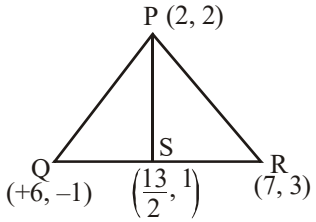
3. Let PS be the median of the triangle with vertices, P (2, 2) , Q (6, - 1) and R (7, 3) . The equation of the line passing through (1, – 1) and parallel to PS is :

माना PS शीर्षो P (2, 2) , Q (6, - 1) तथा R (7, 3) द्वारा बनाए त्रिभुज मध्यिका है तो उस रेखा का समीकरण जो (1, – 1) से गुजरती है तथा PS के समान्तर है :

(A) $2x - 9y - 7 = 0$ (B) $2x - 9y - 11 = 0$ (C) $2x + 9y - 11 = 0$ (D) $2x + 9y + 7 = 0$

Ans. D

Sol. $m_{PS} = \frac{1}{2-\frac{13}{2}} = \frac{-2}{9}$



$\therefore\ y + 1 = \frac{-2}{9}(x - 1)$

$9y + 9 = -2x + 2$

$2x + 9y + 7 = 0$

$\therefore$ (D) Ans.

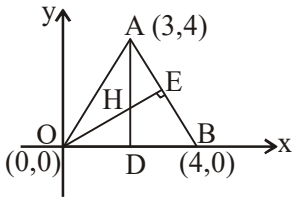
4. Co-ordinates of the orthocentre of the triangle whose vertices are A(0, 0) , B(3, 4) and C(4, 0) is :

शीर्षो A(0, 0) , B(3, 4) एवं C(4, 0) द्वारा बनाए त्रिभुज का लम्बकेन्द्र है :

(A) (3, 1) (B) (3, 4) (C) (3, 3) (D) $\left(3, \frac{3}{4}\right)$

Ans. D

Sol. We know that orthocentre is the meeting point of altitudes of a Δ.



Equation of altitude AD

$\Rightarrow$ AD line is parallel to Y-axis through (3,4)

$\Rightarrow x = 3$ ...(1)

Similarly equation of OE is

$$y = -\frac{3-4}{4-0}x$$

$$\Rightarrow y = \frac{x}{4} \quad ...(2)$$

Solving (1) and (2), we get orthocentre as $\left(3, \frac{3}{4}\right)$

5. Let the lines $(y - 2) = m_1(x - 5)$ and $(y + 4) = m_2(x - 3)$ intersect at right angles at P (where m_1 and m_2 are parameters). If locus of P is $x^2 + y^2 + gx + fy + 7 = 0$, then $(f - g)$ equals :

माना कि सरल रेखाऐं $(y - 2) = m_1(x - 5)$ और $(y + 4) = m_2(x - 3)$, 90° पर एक दूसरे को बिन्दु P पर काटती है यदि बिन्दु P का बिन्दुपथ $x^2 + y^2 + gx + fy + 7 = 0$ है तो $(f - g)$ का मान होगा : (जहाँ m_1 तथा m_2 प्राचल है)

(A) 1 (B) 2 (C) 8 (D) 10

Ans. D

Sol. Clearly locus of point of intersection of lines is $(x-5)(x-3)+(y-2)(y+4)=0$

$\Rightarrow x^2 + y^2 - 8x + 2y + 7 = 0$

Hence $(f - g) = 2 - (-8) = 10$ Ans.

6. If each of the points (a, 6) and (3, b) lies on the line joining the points (3, 2) and (5, 1) then the point (a, b) lies on the line :

यदि (a, 6) तथा (3, b) बिन्दु (3, 2) तथा (5, 1) को मिलाने वाली रेखा पर हो, तो (a, b) इनमें से किस रेखा पर होगा :

(A) $3x + 4y + 7 = 0$ (B) $2x + 3y - 7 = 0$ (C) $4x - 3y - 7 = 0$ (D) $3x - 2y - 7 = 0$

Ans. A

Sol. Line joining (3,2) & (5,1) is

$$y - 2 = \frac{1-2}{5-3}(x-3)$$

2y Â 4 = 3 – x

$2y + x = 7$ ...(1)

then

$12 + a = 7$ .... As (a, 6) lies on equation (1)

$a = -5$

also $2b + 3 = 7$ .... As (3,b) lies on equation (1)

$b = 2$

then $3x + 4y + 7 = 0$

will satisfy (a, b) or (–5,2)

7. Equation of a straight line passing through the origin and making with x-axis an angle twice the size of the angle made by the line $y = 0.2\,x$ with the x – axis, is :

(A) $y = 0.4\,x$ (B) $y = (5/12)\,x$ (C) $6y - 5x = 0$ (D) none of these

मूल बिन्दु से जाने वाली तथा x-अक्ष से रेखा $y = 0.2\,x$ द्वारा बनाये गये कोण का दुगना कोण बनाने वाली रेखा का समीकरण होगा :

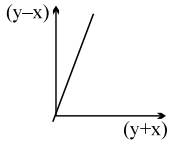
(A) $y = 0.4\,x$ (B) $y = (5/12)\,x$ (C) $6y - 5x = 0$ (D) इनमें से कोई नहीं

Ans. B

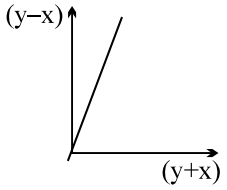
Sol. here $\tan\theta = \frac{1}{5}$ $\therefore$ $\tan 2\theta = \frac{2\left(\frac{1}{5}\right)}{1-\left(\frac{1}{5}\right)^2} = \frac{5}{12}$

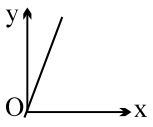
$\therefore$ required line $y = \frac{5x}{12}$

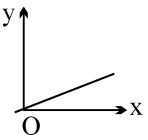
8. The graph of (y – x) against (y + x) is as shown. Which one of the following shows the graph of y against x?

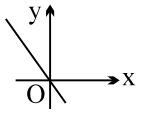


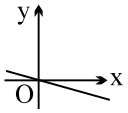
(y – x) के सापेक्ष (y + x) का आरेख दिया गया हो, तो y का x के सापेक्ष वक्र कौन सा होगा?



(A) 

(B) 

(C) 

(D) 

Ans. C

Sol. $\frac{y-x}{y+x} = k\ (k > 1); \qquad y - x = k(y + x)$

$y(1 - k) = x(1 + k)$

$y = \left(\frac{1+k}{1-k}\right)x$ where $\frac{1+k}{1-k} < -1$ $\Rightarrow$ (C) Ans.

9. The number of possible straight lines , passing through (2, 3) and forming a triangle with coordinate axes, whose area is 12 sq. units , is :

(A) one (B) two (C) three (D) four

(2, 3) से गुजरने वाली तथा निर्देशांक अक्ष से 12 मात्रक का क्षेत्रफल बनाने वाली कुल सरल रेखाओं की संख्या होगी :

(A) एक (B) दो (C) तीन (D) चार

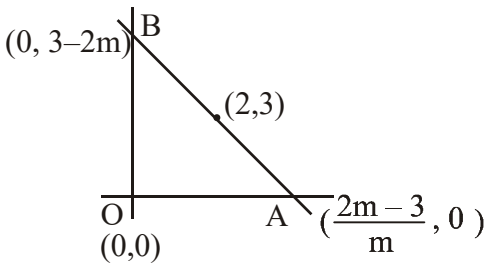
Ans. C

Sol. equation of any line through (2, 3) is $y - 3 = m(x - 2)$

$y = mx - 2m + 3$

with the help of the fig. area of Δ OAB = $\pm$ 12

ie. $\frac{1}{2}\left(\frac{2m-3}{m}\right)(3-2m) = \pm 12$



taking + sign me get $(2m+3)^2 = 0$

this gives one value of m = -3/2

taking negative sign we get

$4m^2 - 36m + 9 = 0 \quad (D > 0)$

quadratic in m gives 2 values of m

$\Rightarrow$ 3 st. lines are possible.

10. Statement-1 : Incentre of the triangle formed by the lines whose sides are $3x + 4y = 0$; $5x - 12y = 0$ and $y - 15 = 0$ is the point P whose co-ordinates are (1, 8).

Statement-2 : Point P is equidistant from the 3 lines forming the triangle.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

कथन–1 : यदि त्रिभुज की भुजाऐं $3x + 4y = 0$; $5x - 12y = 0$ एवं $y - 15 = 0$ है जिसका अन्तः केन्द्र P (1, 8) होगा।

कथन–2 : बिन्दु P तीनों रेखाओं से समान दूरी पर है।

(A) कथन–1 सत्य है, कथन–2 सत्य है, कथन–2 कथन–1 का सही स्पष्टीकरण है।

(B) कथन–1 सत्य है, कथन–2 सत्य है, कथन–2 कथन–1 का सही स्पष्टीकरण नही है।

(C) कथन–1 सत्य है, कथन–2 असत्य है।

(D) कथन–1 असत्य है, कथन– 2 सत्य है।

Ans. B

Sol. Step-1 : Find the vertices of triangle

Given the sides of triangle are

$AC \equiv 3x - 4y = 0$ ...(i)

$AB \equiv 5x + 12y = 0$ ...(ii)

$BC \equiv y - 15 = 0$ ...(iii)

Finding vertices of the triangle by solving equations (i), (ii) & (iii)

From (ii) & (ii), we get

$x = 0, y = 0$

$\therefore A(0,0)$

from (ii) & (iii), we get

$x = -36$

$\Rightarrow y = 15$

$\therefore B(-36,15)$

from (i) & (iii) we get

$y = 15$

$\Rightarrow x = 20$

$\therefore C(20,15)$

Therefore the vertices of the triangle are

A(0,0), B(–36,15) & C(20,15)

Step 2 : The length of sides

The distance formula is given by

$$D=\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Now calculating the sides of triangle by using the distance formula

$$AB = \sqrt{(-36-0)^2 + (15-0)^2}$$

$= 39$

$= a$

$$BC = \sqrt{(20+36)^2 + (15-15)^2}$$

$= 56$

$= b$

$$AC = \sqrt{(20-0)^2 + (15+0)^2}$$

$= 25$

$= c$

Step 3 : Find the incentre of triangle

Now, the incentre of the triangle is

$$= \left(\frac{x_1a + x_2b + x_3c}{a+b+c}, \frac{y_1a + y_2b + y_3c}{a+b+c} \right)$$

$$= \left(\frac{0\times 56 - 36\times 25 + 20\times 39}{120}, \frac{0\times 56 + 15\times 25 + 15\times 39}{120} \right)$$

$$= \left(\frac{-900 + 780}{120}, \frac{375 + 585}{120} \right)$$

$$= \left(\frac{-120}{120}, \frac{960}{120} \right)$$

$= (-1, 8)$

Paragraph for Question nos. 11 to 13

Let $M\left(2, \frac{13}{8}\right)$ is the circumcentre of ΔPQR whose sides PQ and PR are represented by the straight lines $4x - 3y = 0$ and $4x + y = 16$ respectively.

माना $M\left(2,\frac{13}{8}\right)$, ΔPQR का परिकेन्द्र है जिसकी भुजाऐं PQ तथा PR सरल रेखाओं $4x - 3y = 0$ तथा $4x + y = 16$ द्वारा दी जाती है।

11. The orthocentre of ΔPQR is :

ΔPQR का लम्ब केन्द्र है :

(A) $\left(\frac{7}{3},\frac{4}{3}\right)$ (B) $\left(\frac{4}{3},\frac{7}{3}\right)$ (C) $\left(3,\frac{3}{4}\right)$ (D) $\left(\frac{3}{4},3\right)$

Ans. C

Sol. On solving $4x - 3y = 0$ and $4x + y = 16$, we get P(3, 4).

The equation of line perpendicular to PQ is $3x + 4y = k$

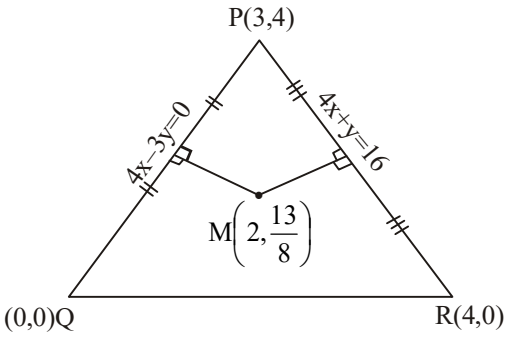
As above line passes through $M\left(2,\frac{13}{8}\right)$,

So $3(2) + 4\left(\frac{13}{8}\right) = k \Rightarrow k = \frac{25}{2}$

$\therefore$ Equation of AM is $3x + 4y = \frac{25}{2}$

Similarly, the equation of line perpendicular to PR is $x - 4y = \lambda$

As above line passes through $M\left(2,\frac{13}{8}\right)$,



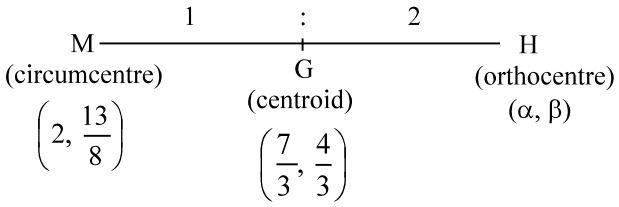
So $2 - 4\left(\frac{13}{8}\right) = \lambda \quad \Rightarrow \lambda = -\frac{9}{2}$

$\therefore$ Equation of CM is $x - 4y = -\frac{9}{2}$

Now, on solving $4x - 3y = 0$ and $3x + 4y = \frac{25}{2}$, we get $A\left(\frac{3}{2},2\right)$

Also on solving $4x + y = 16$ and $x - 4y = -\frac{9}{2}$, we get $C\left(\frac{7}{2},2\right)$

As A is the midpoint of PQ, so $Q \equiv (0, 0)$ and C is the midpoint of PR, so $R \equiv (4, 0)$



$\therefore$ By application of section formula, we get

$$\frac{1(\alpha)+2(2)}{1+2} = \frac{7}{3}$$

$\Rightarrow \alpha = 7 - 4 = 3$

And $\dfrac{1(\beta)+2\left(\frac{13}{8}\right)}{1+2} = \dfrac{4}{3} \Rightarrow \beta = 4 - \dfrac{13}{4} = \dfrac{16-13}{4} = \dfrac{3}{4}$

Hence orthocentre of ΔPQR is $\left(3, \frac{3}{4}\right)$

12. If A, B and C are the midpoint of the sides PQ, QR and PR of ΔPQR respectively, then the area of ΔABC equals:

यदि A, B तथा C भुजाओं PQ, QR तथा PR के क्रमशः मध्यबिन्दु है तो ΔABC का क्षेत्रफल होगा :

(A) 1 (B) 2 (C) 3 (D) 4

Ans. B

Sol. On solving $4x - 3y = 0$ and $4x + y = 16$, we get P(3, 4).

The equation of line perpendicular to PQ is $3x + 4y = k$

As above line passes through $M\left(2, \frac{13}{8}\right)$,

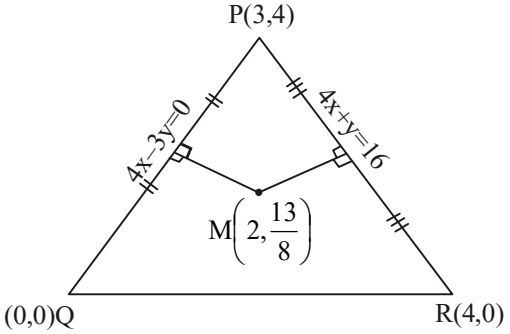
So $3(2) + 4\left(\frac{13}{8}\right) = k \Rightarrow k = \frac{25}{2}$

$\therefore$ Equation of AM is $3x + 4y = \frac{25}{2}$

Similarly, the equation of line perpendicular to PR is $x - 4y = \lambda$

As above line passes through $M\left(2, \frac{13}{8}\right)$,

So $2 - 4\left(\frac{13}{8}\right) = \lambda \quad \Rightarrow \lambda = -\frac{9}{2}$



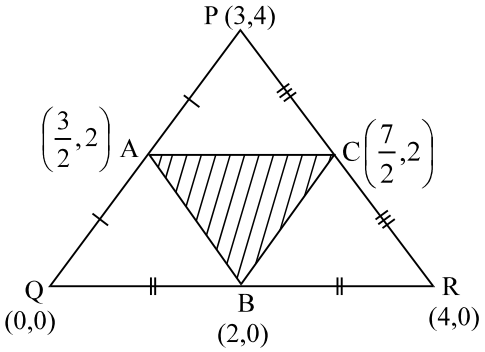
$\therefore$ Equation of CM is $x - 4y = -\frac{9}{2}$

Now, on solving $4x - 3y = 0$ and $3x + 4y = \frac{25}{2}$, we get $A\left(\frac{3}{2}, 2\right)$

Also on solving $4x + y = 16$ and $x - 4y = -\frac{9}{2}$, we get $C\left(\frac{7}{2}, 2\right)$

As A is the midpoint of PQ, so $Q \equiv (0, 0)$ and C is the midpoint of PR, so $R \equiv (4, 0)$

We have area $(\Delta ABC) = \frac{1}{2}\left|\begin{vmatrix} \frac{3}{2} & 2 & 1 \\ 2 & 0 & 1 \\ \frac{7}{2} & 2 & 1 \end{vmatrix}\right| = 2$ square units.



Aliter : area $(\Delta ABC) = \frac{1}{4}$ area $(\Delta PQR) = \frac{1}{4} \times \left(\frac{1}{2} \times 4 \times 4\right) = 2$ sq.units.

13. If PB be the median of the ΔPQR, then the equation of the straight line passing through N(–2, 3) and perpendicular to PB is :

यदि PB, ΔPQR की एक मध्यिका है, तो उस सरल रेखा का समीकरण ज्ञात कीजिये जो N(–2, 3) से गुजरती है तथा PB के लम्बवत् है :

(A) $4x + y + 5 = 0$ (B) $x - 4y + 14 = 0$ (C) $4x - y + 11 = 0$ (D) $x + 4y - 10 = 0$

Ans. D

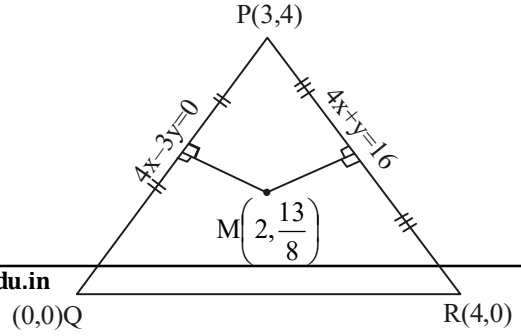
Sol. On solving $4x - 3y = 0$ and $4x + y = 16$, we get P(3, 4).
The equation of line perpendicular to PQ is $3x + 4y = k$

As above line passes through $M\left(2, \frac{13}{8}\right)$,

So $3(2) + 4\left(\frac{13}{8}\right) = k \Rightarrow k = \frac{25}{2}$

$\therefore$ Equation of AM is $3x + 4y = \frac{25}{2}$

Similarly, the equation of line perpendicular to PR is $x - 4y = \lambda$



As above line passes through $M\left(2, \frac{13}{8}\right)$,

So $2 - 4\left(\frac{13}{8}\right) = \lambda \qquad \Rightarrow \lambda = -\frac{9}{2}$

$\therefore$ Equation of CM is $x - 4y = -\frac{9}{2}$

Now, on solving $4x - 3y = 0$ and $3x + 4y = \frac{25}{2}$, we get $A\left(\frac{3}{2}, 2\right)$

Also on solving $4x + y = 16$ and $x - 4y = -\frac{9}{2}$, we get $C\left(\frac{7}{2}, 2\right)$

As A is the midpoint of PQ, so $Q \equiv (0, 0)$ and C is the midpoint of PR, so $R \equiv (4, 0)$

Slope of PB = $\frac{4-0}{3-2} = \frac{4}{1}$

$\therefore$ Equation of straight line passing through N(–2, 3) and perpendicular to PB is

$(y - 3) = \frac{-1}{4}(x + 2) \Rightarrow 4y - 12 = -x - 2$

Hence $x + 4y = 10$

Paragraph for Question nos. 14 to 16

Consider a Δ ABC whose sides BC, CA and AB are represented by the straight lines $x - 2y + 5 = 0$, $x + y + 2 = 0$ and $8x - y - 20 = 0$ respectively.

त्रिभुज ABC की भुजा BC, CA तथा AB का समीकरण क्रमशः $x - 2y + 5 = 0$, $x + y + 2 = 0$ तथा $8x - y - 20 = 0$ है।

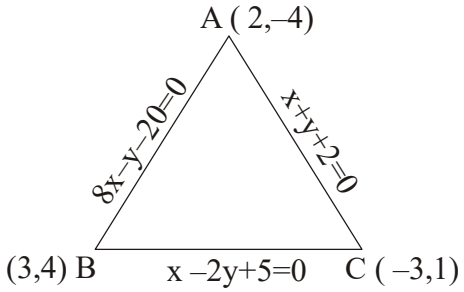
14. The area of Δ ABC equals :

Δ ABC का क्षेत्रफल होगा :

(A) $\frac{41}{2}$ (B) $\frac{43}{2}$ (C) $\frac{45}{2}$ (D) $\frac{47}{2}$

Ans. C

Sol.



$$\text{Area of } \Delta ABC = \frac{1}{2}\left|\begin{vmatrix} 2 & -4 & 1 \\ 3 & 4 & 1 \\ -3 & 1 & 1 \end{vmatrix}\right| = \frac{45}{2}$$

15. If AD be the median of the Δ ABC then the equation of the straight line passing through (2, –1) and parallel to AD is :

यदि ABC की माध्यिका AD हो, तो रेखा AD के समान्तर तथा बिन्दु (2, –1) से गुजरने वाली रेखा का समीकरण होगा :

(A) $4x - 3y - 11 = 0$ (B) $13x - 4y - 30 = 0$ (C) $4x + 13y + 5 = 0$ (D) $13x + 4y - 22 = 0$

Ans. D

Sol. As D is midpoint of BC, so $D \equiv (0, \frac{5}{2})$

$$\text{Slope of median AD} = -\frac{13}{4}$$

$\therefore$ The equation of straight line passing through (2, –1) and parallel to AD is

$$(y+1) = -\frac{13}{4}(x-2) \Rightarrow 13x + 4y - 22 = 0$$

16. The orthocentre of the Δ ABC is :

Δ ABC का लम्बकेन्द्र होगा :

(A) (–3, 1) (B) $\left(-\frac{1}{3}, \frac{2}{3}\right)$ (C) (–2, 4) (D) $\left(-\frac{2}{3}, \frac{4}{3}\right)$

Ans. B

Sol. The equation of altitude through angular point A to side BC is

$2x + y = 0$ .....(1)

And the equation of altitude through angular point C to side AB is

$x + 8y - 5 = 0$ .....(2)

$\therefore$ On solving (1) and (2), we get $x = -\frac{1}{3}$, $y = \frac{2}{3}$

Hence orthocentre of ΔABC is $\left(-\frac{1}{3}, \frac{2}{3}\right)$

17. Consider 3 lines

L_1: $5x - y + 4 = 0$

L_2 : $3x - y + 5 = 0$

L_3 : $x + y + 8 = 0$

If these lines enclose a triangle ABC and sum of the squares of the tangent of the interior angles can be expressed in the form p/q where p and q are relatively prime numbers, compute the value of $(p + q)$.

तीन रेखाऐं

L_1: $5x - y + 4 = 0$

L_2 : $3x - y + 5 = 0$

L_3 : $x + y + 8 = 0$

एक त्रिभुज ABC बनाती है। यदि सारे आन्तरिक कोणों के स्पर्शज्या के वर्ग का योग यदि p/q है जहाँ p एवं q का महत्तम समापवर्तक 1 है तो $(p + q)$ का मान क्या होगा।

Ans. 465

Sol. Arranging the lines in descending order

$m_1 = 5$; $m_2 = 3$ and $m_3 = -1$

$\therefore \quad \tan A = \frac{m_1 - m_2}{1 + m_1 m_2} = \frac{2}{1+15} = \frac{1}{8}$

$\tan B = \frac{m_2 - m_3}{1 + m_2 m_3} = \frac{3+1}{1-3} = -2$

$\tan C = \frac{m_3 - m_1}{1 + m_3 m_1} = \frac{-1-5}{1-5} = \frac{3}{2}$

$\sum \tan^2 A = \frac{1}{64} + 4 + \frac{9}{4} = \frac{1+256+144}{64} = \frac{401}{64} \Rightarrow p + q = 465$ Ans.

18. Consider the line $Ax + By + C = 0$.

Match the nature of intercept of the line given in column-I with their corresponding conditions in column-II. The mapping is one to one only.

Column-I	**Column-II**
(A) x intercept is finite and y intercept is infinite	(P) $A = 0,\ B, C \neq 0$

(B) x intercept is infinite and y intercept is finite — (Q) $C = 0,\ A, B \neq 0$

(C) both x and y intercepts are zero — (R) $A, B = 0$ and $C \neq 0$

(D) both x and y intercepts are infinite — (S) $B = 0,\ A, C \neq 0$

माना रेखा $Ax + By + C = 0$ है।

स्तम्भ–I से (I) में स्थित रेखाओं के अंन्तः खण्डों की प्रकृति का स्तम्भ–II में स्थित शर्तो से मिलान कीजिये। मिलान एकैकी है।

Column-I	**Column-II**
(A) x अन्तःखण्ड परिमित है तथा y अन्तःखण्ड अनन्त	(P) $A = 0,\ B, C \neq 0$
(B) x अन्तः खण्ड अनन्त है तथा y अन्तःखण्ड परिमित	(Q) $C = 0,\ A, B \neq 0$
(C) x तथा y अन्तःखण्ड दोनों शून्य है	(R) $A, B = 0$ and $C \neq 0$
(D) x तथा y अन्तःखण्ड दोनों अनन्त है	(S) $B = 0,\ A, C \neq 0$

Ans. (A) S; (B) P; (C) Q; (D) R

Sol. (A) $\rightarrow$ S

$\frac{x}{a} + \frac{y}{\infty} = 1$ $\quad A \neq 0, B = 0, C \neq 0$

(B) $\rightarrow$ P

$\frac{x}{\infty} - \frac{y}{b} = 1$ $\quad A = 0, B \neq 0, C \neq 0$

(C) $\rightarrow$ Q

$\frac{x}{0} + \frac{y}{0} = 0$ $\quad A_y\ B = 0, C = 0$

(D) $\rightarrow$ R

$\frac{x}{\infty} + \frac{y}{\infty} = 1$ $\quad A = 0, B = 0, C \neq 0$

19. Match the following column :

Column-I	**Column-II**
(A) Four lines $x + 3y - 10 = 0$, $x + 3y - 20 = 0$ $3x - y + 5 = 0$ and $3x - y - 5 = 0$ form a figure which is	(P) a quadrilateral which is neither a parallelogram nor a trapezium nor a kite
(B) The point A(1, 2), B(2, – 3), C(– 1, – 5) and D(–2, 4) in order are the vertices of	(Q) a parallelogram
(C) The lines $7x + 3y - 33 = 0$, $3x - 7y + 19 = 0$ $3x - 7y - 10 = 0$ and $7x + 3y - 4 = 0$ form a figure which is	(R) a rectangle of area 10 sq. units

(D) Four lines $4y - 3x - 7 = 0$, $3y - 4x + 7 = 0$, $4y - 3x - 21 = 0$, $3y - 4x + 14 = 0$ form a figure which is — (S) a square

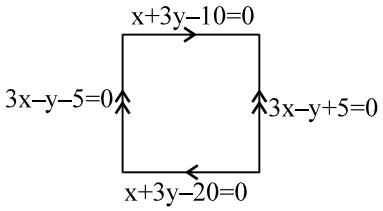
निम्न कॉलम को सुमेलित कीजिए :

Column-I	**Column-II**
(A) चार रेखाऐं $x + 3y - 10 = 0$, $x + 3y - 20 = 0$ $3x - y + 5 = 0$ तथा $3x - y - 5 = 0$ एक चतुर्भुज बनाती है जो कि	(P) एक चतुर्भुज जो ना तो समानान्तर है ना ही समलम्ब चतुर्भुज है, ना ही पतंग है
(B) बिन्दु A(1, 2), B(2, – 3), C(– 1, – 5) तथा D(–2, 4) क्रमवार किसके शीर्ष है	(Q) एक सामानान्तर चतुर्भुज
(C) रेखाऐं $7x + 3y - 33 = 0$, $3x - 7y + 19 = 0$, $3x - 7y - 10 = 0$ तथा $7x + 3y - 4 = 0$ चतुर्भुज बनाती है जो कि	(R) एक आयत जिसका क्षेत्रफल 10 वर्ग इकाई है
(D) चार रेखाऐं $4y - 3x - 7 = 0$, $3y - 4x + 7 = 0$, $4y - 3x - 21 = 0$, $3y - 4x + 14 = 0$ एक चतुर्भुज बनाती है जो कि	(S) एक वर्ग

Ans. (A) Q, R, S ; (B) P; (C) Q, S; (D) Q

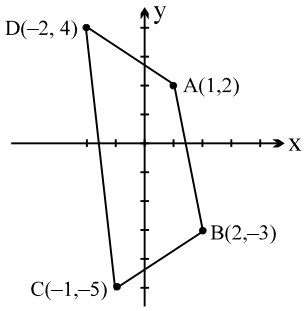
Sol. (A) $h_1 = \left|\frac{10}{\sqrt{10}}\right| = \sqrt{10}$

$h_2 = \left|\frac{10}{\sqrt{10}}\right| = \sqrt{10}$



$\Rightarrow$ a square of side $\sqrt{10}$

$\Rightarrow$ area = 10 $\Rightarrow$ Q, R, S



(B) $m_{AB} = \frac{5}{-1} = -5$

$m_{DC} = \frac{9}{-1} = -9$

not a parallelogram $\Rightarrow$ P

|||ly (C) and (D) can be checked

1. Let (x_1, y_1) ; (x_2, y_2) and (x_3, y_3) are the vertices of a triangle ABC respectively. D is a point on BC such that BC = 3BD. The equation of the line through A and D, is :

यदि (x_1, y_1) ; (x_2, y_2) तथा (x_3, y_3) त्रिभुज ABC का शीर्ष है। यदि रेखा BC पर एक बिन्दु D इस प्रकार से है कि BC = 3BD तो रेखा AD का समीकरण होगा :

(A) $\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} + 2\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0$ (B) $3\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} + \begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0$

(C) $\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} + 3\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0$ (D) $2\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} + \begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0$

Ans. D

Sol. BC = 3BD

$$\frac{BC}{BD} = \frac{3}{1}$$

D Point :

$$\left[\frac{x_3 + 2x_2}{3}, \frac{y_3 + 2y_2}{3}\right]$$

$$\Rightarrow AD = 3\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 2 \end{vmatrix} + 2\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_3 & y_3 & 2 \end{vmatrix} = 0$$

2. In a triangle ABC, side AB has the equation $2x + 3y = 29$ and the side AC has the equation, $x + 2y = 16$. If the mid – point of BC is $(5, 6)$ then the equation of BC is :

एक त्रिभुज ABC में, भुजा AB का समीकरण $2x + 3y = 29$ है तथा भुजा AC का समीकरण $x + 2y = 16$ है। यदि BC का मध्य बिन्दु $(5, 6)$ है तो BC का समीकरण होगा :

(A) $x - y = -1$ (B) $5x - 2y = 13$ (C) $x + y = 11$ (D) $3x - 4y = -9$

Ans. C

Sol. Let co-ordinates of B be (x_1, y_1) & C be (x_2, y_2)

$\therefore$ (5,6) is the mid point,

so, $\frac{x_1 + x_2}{2} = 5$, $\frac{y_1 + y_2}{2} = 6$

$\Rightarrow x_1 + x_2 = 10, y_1 + y_2 = 12$

B (x_1, y_1) lies on the line $2x + 3y = 29$

$\therefore 2x_1 + 3y_1 = 29$ ...(1)

$C(x_2, y_2)$ lies on the line $x + 2y = 16$,

$\therefore x_2 + 2y_2 = 16$ ...(2)

$\therefore$ putting x_1, y_1 in the form of x_2, y_2 in (1)

$2(10 - x_2) + 3(12 - y_2) = 29$ $\{x_1 = 10 - x_2, y_1 = 12 - y_2\}$

$\Rightarrow 20 - 2x_2 + 36 - 3y_2 = 29$

$\Rightarrow 2x_2 + 3y_2 = 27$ ...(3)

on subtracting (3) and (2) $\times 2$

$-y_2 = -5$

$y_2 = 5$

Putting y_2 in (2)

$x_2 + 2(5) = 16$

$x_2 = 6$

$x_1 = 10 - x_2$

$= 4$

$y_1 = 12 - 5 = 7$

Equation :

$$\frac{x - x_1}{x_2 - x_1} = \frac{y - y_1}{y_2 - y_1}$$

$$\Rightarrow \frac{x - 4}{2} = \frac{y - 7}{-2}$$

$\Rightarrow -x + 4 = y - 7$

$\Rightarrow x + y = 11$

3. Let $A \equiv (3, 2)$ and $B \equiv (5, 1)$. ABP is an equilateral triangle is constructed on the side of AB remote from the origin then the orthocentre of triangle ABP is :

माना $A \equiv (3, 2)$ एवं $B \equiv (5, 1)$ यदि समबाहु त्रिभुज ABP इस प्रकार बनाया जाता है कि P तथा मूल बिन्दु AB के विपरीत तरफ स्थित हो। त्रिभुज ABP का लम्ब केन्द्र होगा :

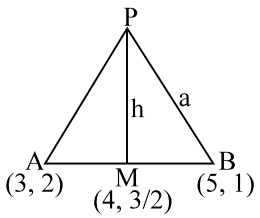
(A) $\left(4 - \frac{1}{2}\sqrt{3}, \frac{3}{2} - \sqrt{3}\right)$ (B) $\left(4 + \frac{1}{2}\sqrt{3}, \frac{3}{2} + \sqrt{3}\right)$

(C) $\left(4 - \frac{1}{6}\sqrt{3}, \frac{3}{2} - \frac{1}{3}\sqrt{3}\right)$ (D) $\left(4 + \frac{1}{6}\sqrt{3}, \frac{3}{2} + \frac{1}{3}\sqrt{3}\right)$

Ans. D

Sol. $m_{AB} = -1/2$

$m_{PM} = 2$



$h = \frac{a\sqrt{3}}{2}$ now $a = \sqrt{5} = AB$; $h = \frac{\sqrt{15}}{2}$

so point P $\frac{x - y}{\cos\theta} = \frac{y - 3/2}{\sin\theta} = h$

$P : x = 4 + \frac{\sqrt{3}}{2}, \quad y = \frac{3}{2} + \sqrt{3}$

so orthocentre/centroid is $\frac{\sum x_1}{3}, \frac{\sum y_1}{3}$

$= 4 + \frac{\sqrt{3}}{6}, \quad \frac{3}{2} + \frac{\sqrt{3}}{3}$

4. A variable line L = 0 is drawn through O(0, 0) to meet the lines L_1: x + 2y – 3 = 0 and L_2 : x + 2y + 4 = 0 at points M and N respectively. A point P is taken on L = 0 such that $\frac{1}{OP^2} = \frac{1}{OM^2} + \frac{1}{ON^2}$.

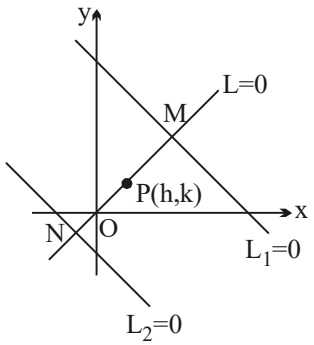
Locus of P is :

एक चर रेखा L = 0 जो कि O(0, 0) से जाती है तथा रेखा L_1: x + 2y – 3 = 0 एवं L_2 : x + 2y + 4 = 0 को M एवं N पर मिलती है। रेखा L पर एक बिन्दु P इस प्रकार से लिया जाता है कि $\frac{1}{OP^2} = \frac{1}{OM^2} + \frac{1}{ON^2}$ तो P का बिन्दु पथ होगा :

(A) $x^2 + 4y^2 = \frac{144}{25}$ (B) $(x + 2y)^2 = \frac{144}{25}$ (C) $4x^2 + y^2 = \frac{144}{25}$ (D) $(x - 2y)^2 = \frac{144}{25}$

Ans. B

Sol. Let the parametric equation of the variable line is



$$\frac{x-0}{\cos\theta} = \frac{y-0}{\sin\theta} = r \quad \Rightarrow \quad x = r\cos\theta;\ y = r\sin\theta$$

$\therefore$ putting $(x = r\cos\theta;\ y = r\sin\theta)$ in $L_1 = 0$, we get

$$\frac{1}{OM} = \frac{(\cos\theta + 2\sin\theta)}{3} \qquad ....(1)$$

|||ly putting the general point in $L_2 = 0$, we get

$$\frac{1}{ON} = \frac{-(\cos\theta + 2\sin\theta)}{4} \qquad ....(2)$$

[11th, 26-10-2009, st.line + cir]

Let $P = (h, k)$ and $OP = r \Rightarrow$ $r\cos\theta = h,\ r\sin\theta = k$

$$\therefore \quad \frac{1}{OP^2} = \frac{1}{OM^2} + \frac{1}{ON^2} \Rightarrow \quad \frac{1}{r^2} = \frac{(\cos\theta + 2\sin\theta)^2}{9} + \frac{(\cos\theta + 2\sin\theta)^2}{16}$$

$\Rightarrow 144 = 16(r\cos\theta + 2r\sin\theta)^2 + 9(r\cos\theta + 2r\sin\theta)^2 \Rightarrow 144 = 16(h + 2k)^2 + 9(h + 2k)^2$

$\therefore$ Locus of P(h, k) is

$$(x + 2y)^2 = \frac{144}{25} \text{ Ans.}$$

5. Two mutually perpendicular straight lines through the origin from an isosceles triangle with the line $2x + y = 5$. Then the area of the triangle is :

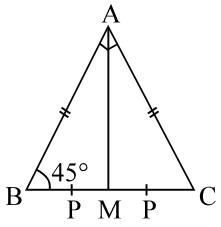
दो परस्पर लम्बवत रेखाऐं जो कि मूल बिन्दु से जाती है तथा रेखा $2x + y = 5$ के साथ समद्विबाहु त्रिभुज बनाती है, तो त्रिभुज का क्षेत्रफल होगा :

(A) 5 (B) 3 (C) 5/2 (D) 1

Ans. A

Sol. Area of right isosceles triangle = p^2

Area of equilateral triangle in terms of



$$P = \frac{5}{\sqrt{4+1}} = \sqrt{5}$$

6. The vertex of a right angle of a right angled triangle lies on the straight line $2x + y - 10 = 0$ and the two other vertices, at points $(2, -3)$ and $(4, 1)$ then the area of triangle in sq. units is :

यदि किसी समकोणीय त्रिभुज का समकोण रेखा $2x + y - 10 = 0$ पर है तथा बाकी दोनों शीर्ष क्रमशः $(2, -3)$ एवं $(4, 1)$ हो, तो त्रिभुज का क्षेत्रफल होगा :

(A) $\sqrt{10}$ (B) 3 (C) 33/5 (D) 11

Ans. B

Sol. $M_1M_2 = -1$

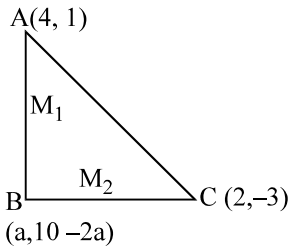
$$\frac{9-2a}{a-4} \times \frac{13-2a}{a-2} = -1$$

$$117 - 26a - 18a + 4a^2 = -(a^2 - 6a + 8)$$

$$5a^2 - 50a + 125 = 0$$

$$a = 5$$

so B is (5, 0)



so area $= \frac{1}{2} AB \times AC = \frac{1}{2} \sqrt{2} \times 3\sqrt{2} = 3$

7. Point 'P' lies on the line l { $(x, y) \mid 3x + 5y = 15$}. If 'P' is also equidistant from the coordinate axes, then P can be located in which of the four quadrants :

(A) I only (B) II only (C) I or II only (D) IV only

यदि बिन्दु 'P' रेखा l { $(x, y) \mid 3x + 5y = 15$} पर स्थित हो तथा निर्देशांक अक्ष से बराबर दूरी पर हो, तो 'P' इनमें से किस–किस चतुर्थांश में हो सकता है :

(A) I केवल (B) II केवल (C) I या II केवल (D) IV केवल

Ans. C

Sol. Let (x, y) lies in the 1st quadrant.

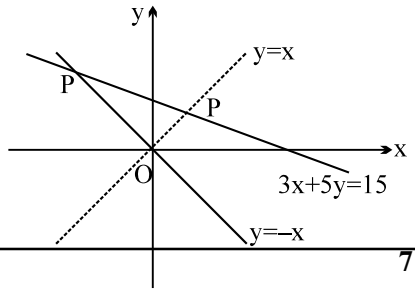
Hence $x = y = a > 0$ (say)

$$3a + 5a = 15$$

$$a = \frac{15}{8}.$$

Hence $P\left(\frac{15}{8}, \frac{15}{8}\right)$

note that P can lie either on $y = x$ or $y = -x$ in the 4th quadrant only.



Solving $y = -x$ and the line $x = -\frac{15}{2}$. Hence $P = \left(-\frac{15}{2}, \frac{15}{2}\right)$

8. The area of the parallelogram formed by the lines $3x + 4y = 7a$; $3x + 4y = 7b$; $4x + 3y = 7c$ and $4x + 3y = 7d$ is :

समान्तर चतुर्भुज की भुजाऐं यदि $3x + 4y = 7a$; $3x + 4y = 7b$; $4x + 3y = 7c$ एवं $4x + 3y = 7d$ तो उसका क्षेत्रफल होगा :

(A) $\frac{|(a-b))(c-d)|}{7}$ (B) $49\,|(a-b)\,(c-d)|$ (C) $\frac{|(a-b))(c-d)|}{49}$ (D) $7\,|(a-b)\,(c-d)|$

Ans. D

Sol. Equation of lines given are :

$3x + 4y = 7a$ $\qquad$ $4x + 3y = 7c$

$3x + 4y = 7b$ $\qquad$ $4x + 3y = 7d$

Here $m_1 = -\frac{3}{4}$ $\qquad$ $m_2 = -\frac{4}{3}$

$$\therefore \text{Area of prallelogram} = \left|\frac{(c_1 - c_2)(d_1 - d_2)}{m_1 - m_2}\right|$$

$$\Rightarrow \text{Area} = \left|\frac{\left(\frac{7a}{4} - \frac{7b}{4}\right)\left(\frac{7c}{3} - \frac{7d}{3}\right)}{\left(\frac{4}{3} - \frac{3}{4}\right)}\right|$$

$$\Rightarrow \text{Area} = \left|\frac{7(a-b) \times 7(c-d)}{4 \times 3} \times \frac{4 \times 3}{(16-9)}\right|$$

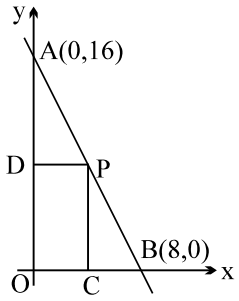
$\Rightarrow$ Area = $|7(a - b)\,(c - d)|$

Answer – Option 'D'

Paragraph for Question Nos. 9 to 11

In the diagram, a line is drawn through the points A(0, 16) and B(8, 0). Point P is chosen in the first quadrant on the line through A and B. Points C and D are chosen on the x and y-axis respectively, so that PDOC is a rectangle.

दिखाये गये चित्रानुसार बिन्दु A(0, 16) तथा B(8, 0) से रेखा खींची गयी है तथा इस रेखा पर एक बिन्दु P तथा x-अक्ष एवं y-अक्ष पर बिन्दु C एवं D इस प्रकार से चुना जाता है कि PDOC एक आयत है।



9. Perpendicular distance of the line AB from the point (2, 2) is :

बिन्दु (2, 2) से रेखा AB की लम्बवत् दूरी होगी :

(A) $\sqrt{4}$ (B) $\sqrt{10}$ (C) $\sqrt{20}$ (D) $\sqrt{50}$

Ans. C

Sol. Equation of line $2x + y = 16$ ....(1)

perpendicular from (2, 2) is equal to $\left|\frac{6-16}{\sqrt{5}}\right| = \left|2\cdot\sqrt{5}\right| = \sqrt{20}$

10. Sum of the coordinates of the point P if PDOC is a square is :

यदि PDOC एक वर्ग हो, तो P के निर्देशांको का योग क्या होगा :

(A) 32/3 (B) 16/3 (C) 16 (D) 11

Ans. A

Sol. If PDOC is a square then let P(a, a). Substituting the co-ordinates of P in (1)

$\therefore \quad 2a + a = 16 \quad \Rightarrow \quad a = \frac{16}{3}; \quad \text{sum} = \frac{32}{3}$

11. Number of possible ordered pair(s) of all positions of the point P on AB so that the area of the rectangle PDOC is 30 sq. units, is :

(A) three (B) two (C) one (D) zero

रेखा AB पर बिन्दु P की कुल कितनी स्थितिया होगी जब आयत PDOC का क्षेत्रफल 30 वर्ग मात्रक होगा :

(A) तीन (B) दो (C) एक (D) शून्य

Ans. B

Sol. If PDOC is a rectangle of area 30 then let P(a, b)

$ab = 30$ and $2a + b = 16$

solving $a = 3$ and $b = 10$

or $a = 5$ and $b = 6$

hence two possible positions of P Ans.

12. Let (x_r, y_r) $r = 1, 2, 3$ are the coordinates of the vertices of a triangle ABC. If D is the point on BC dividing it in the ratio of 1 : 2 reckoning from the vertex B, prove that the equation of the line AD is

$$2\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} + \begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0$$

Also find the equation of the line AE in the similar form where E is the harmonic conjugate of D w.r.t. the points B and C.

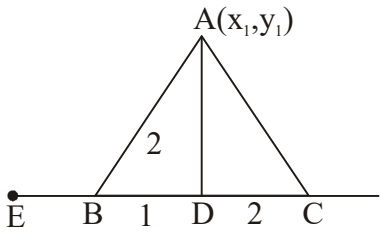
माना कि (x_r, y_r) $r = 1, 2, 3$ त्रिभुज ABC के शीर्ष का निर्देशांक है। यदि रेखा BC पर एक बिन्दु D ऐसा है। कि जो BC की 1 : 2 के अनुपात में काटता हो, तो सिद्ध करों कि रेखा AD का समीकरण

$$2\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} + \begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0 \text{ होगा।}$$

रेखा AE का भी समीकरण ज्ञात करो जो E बिन्दु D का B एवं C के सापेक्ष हरात्मक संयुग्मी (harmonic conjugate) है।

Ans. $2\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} - \begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0$

Sol.



$$\left(\frac{2x_2 + x_3}{3}, \frac{2y_2 + y_3}{3}\right)$$

$$\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ \frac{2x_2 + x_3}{3} & \frac{2y_2 + y_3}{3} & 1 \end{vmatrix} = 0$$

$$\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ 2x_2 + x_3 & 2y_2 + y_3 & 3 \end{vmatrix} = 0$$

$$2\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} + \begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0$$

13. A variable line passing through the origin intersects the lines $2x + 5y - 3 = 0$ and $4x + 7y - 3 = 0$ at P and Q respectively. If a point R is taken on the variable line such that OP, OR and OQ are in harmonical progression then find the locus of R.

मूल बिन्दु से जाने वाली चर रेखा, रेखाओं $2x + 5y - 3 = 0$ एवं $4x + 7y - 3 = 0$ को P तथा Q पर काटती है। यदि इस चर रेखा पर एक बिन्दु R इस प्रकार से है कि OP, OR एवं OQ हरात्मक श्रेणी में हो, तो R का बिन्दुपथ बताओ।

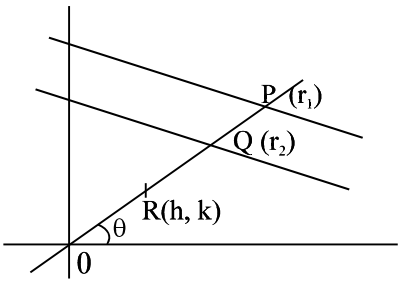
Ans. $x + 2y = 1$

Sol. $\frac{x}{\cos\theta} = \frac{y}{\sin\theta} = r$

$(r_1 \cos\theta , r_1 \sin\theta)$ lies on $2x + 5y - 3 = 0$

$$\frac{2}{3}\cos\theta + \frac{5}{3}\sin\theta = \frac{1}{r_1} \quad ....(1)$$

|||ly $\frac{4}{3}\cos\theta + \frac{7}{3}\sin\theta = \frac{1}{r_2}$ ...(2)



$$2\cos\theta + 4\sin\theta = \frac{1}{r_1} + \frac{1}{r_2} = \frac{2}{OR} \qquad \left[\text{Given } \frac{2}{OR} = \frac{1}{OP} + \frac{1}{OQ}\right]$$

$2(OR)\cos\theta + 4(OR)\sin\theta = 2$

$2h + 4k = 2 \quad \Rightarrow$ Locus is $x + 2y = 1$

14. A variable line passing through the origin intersects two given straight lines $2x + y = 4$ and $x + 3y = 6$ at R and S respectively. A point P is taken on this variable line. Find the equation to the locus of the point P if:

(a) OP is the arithmetic mean of OR and OS.

(b) OP is the geometric mean of OR and OS.

मूल बिन्दु से जाने वाली चर रेखा यदि सरल रेखाओं $2x + y = 4$ एवं $x + 3y = 6$ को R तथा S पर काटती है। इस चर रेखा पर बिन्दु

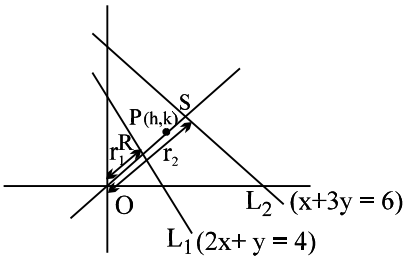
P इस प्रकार से लिया गया है कि तो P का बिन्दु पथ बताओ :

(a) यदि OR एवं OS का समान्तर माध्य OP है।

(b) यदि OR एवं OS एवं गुणोतर माध्य OP है।

Ans. (a) $2x^2+ 7xy + 3y^2 – 8x – 9y = 0$; (b) $2x^2+ 7xy + 3y^2 – 24 = 0$

Sol. (a)



Given $OP = \frac{r_1 + r_2}{2}$ ....(1)

equation of the line passes through O is $\frac{x}{\cos\theta} = \frac{y}{\sin\theta} = r$

For $r = r_2$ put x, y in $L_2 = 0 \Rightarrow r_2(\cos\theta + 3\sin\theta) = 6$ ....(2)

|||ly for $r = r_1$ put x, y in $L_1 = 0 \Rightarrow r_1(2\cos\theta + \sin\theta) = 4$ ....(3)

From (2) , (3) and (1) $\frac{6}{(\cos\theta + 3\sin\theta)} + \frac{4}{(2\cos\theta + \sin\theta)} = 2OP$

$$\frac{3}{(\cos\theta + 3\sin\theta)OP} + \frac{2}{(2\cos\theta + \sin\theta)OP} = 1 \quad ....(4)$$

Also $OP\cos\theta = h$ and $OP\sin\theta = k$

therefore from equation (4) $2x^2+ 7xy + 3y^2 – 8x – 9y = 0$ Ans:]

(b) Given $OP^2 = r_1 . r_2$ now proceed in the same way as done in part (a)

$2x^2+ 7xy + 3y^2 – 24 = 0$ Ans.

15. Find the sum of the abscissas of all the points on the line $x + y = 4$ that lie of a unit distance from the line $4x + 3y – 10 = 0$.

रेखा $x + y = 4$ पर स्थित सभी बिन्दुओं के भुजों का योग जो रेखा $4x + 3y – 10 = 0$ से इकाई दूरी पर स्थित है।

Ans. – 4

Sol. Any point on the line $x + y = 4$ can be taken as $(x_1, 4 – x_1)$. As it is at a unit distance from the line $4x + 3y – 10 = 0$, we get

$$\frac{|4x_1 + 3(4 - x_1) - 10|}{\sqrt{(4^2 + 3^2)}} = 1 \Rightarrow |x_1 + 2| = 5 \Rightarrow x_1 + 2 = \pm 5 \text{ Þ } x_1 = 3 \text{ or } -7$$

$\therefore$ required sum = 3 – 7 = – 4

16. Find the equation of lines parallel to $3x - 4y - 5 = 0$ at a unit distance from it.

रेखाओं का समीकरण जो $3x - 4y - 5 = 0$ के समान्तर एवं उससे इकाई दूरी पर स्थित है।

Ans. $3x - 4y = 0$ and $3x - 4y - 10 = 0$

Sol. Equation of line parallel to $3x - 4y - 5 = 0$ is given by, $3x - 4y + c = 0$

Also distance between given line is 1.

$$\left|\frac{c_1 - c_2}{\sqrt{a^2 + b^2}}\right|$$

$$\Rightarrow \left|\frac{c+5}{\sqrt{3^2 + 4^2}}\right| = 1$$

$\Rightarrow |c + 5| = 5 \Rightarrow c = 0, -10$

Hence line parallel to $3x - 4y - 5 = 0$ and unit distance from it is

$3x - 4y - 10 = 0$ or $3x - 4y = 0$

17. Prove that no line can be drawn through the point $(4, -5)$ so that its distance from $(-2, 3)$ will be equal to 12.

सिद्ध कीजिये की बिन्दु $(4, -5)$ से ऐसी कोई रेखा नही बनाई जा सकती है जो $(-2, 3)$ से 12 की दूरी पर स्थित है।

Ans.

Sol. The equation of line having slope m and passing through $(4, -5)$ is

$y + 5 = m(x - 4)$

$\Rightarrow mx - y - (4m + 5) = 0$

The distance of a point $(-2,3)$ from the line $mx - y - (4m + 5) = 0$ is 12.

So, $$\left|\frac{m(-2) - 3 - (4m + 5)}{\sqrt{m^2 + 1}}\right| = 12$$

$\Rightarrow 36m^2 + 64 + 96m = 144m^2 + 144$

$\Rightarrow 108m^2 - 96m + 80 = 0$

$\Rightarrow 27m^2 - 24m + 20 = 0$

Now, $b^2 - 4ac = (-24)^2 - 4 \times 27 \times 20 = -1584$

Thus $b^2 - 4ac < 0$

So slope cannot be imaginary number.

$\therefore$ No line can be drawn through the point (–2,3)

18. Find the equations of the sides of a square whose each side is of length 4 units and centre is (1, 1). Given that one pair of sides is parallel to $3x - 4y = 0$.

यदि किसी वर्ग का केन्द्र (1, 1) तथा वर्ग की भुजा की लम्बाई 4 हो, तो वर्ग की चारों भुजाओं की समीकरण बताओं (दिया है कि वर्ग की एक भुजा रेखा $3x - 4y = 0$ के समान्तर है।

Ans. $3x - 4y + 11 = 0$; $3x - 4y - 9 = 0$; $4x + 3y + 3 = 0$; $4x + 3y - 17 = 0$

Sol. Suppose, if possible.

Equation of line through (4, – 5) with slope of m is

$y + 5 = m(x - 4) \Rightarrow mx - y - 4m - 5 = 0$

Then $\dfrac{|m(-2) - 3 - 4m - 5|}{\sqrt{m^2 + 1}} = 12 \Rightarrow |-6m - 8| = 12\sqrt{(m^2 + 1)}$

On Squaring $(6m + 8)^2 = 144(m^2 + 1)$

$\Rightarrow 4(3m + 4)^2 = 144(m^2 + 1) \Rightarrow (3m + 4) = 36(m^2 + 1) \Rightarrow 27m^2 - 24m + 20 = 0$

Since the discriminant of (1) is $(-24)^2 - 4 \cdot 27 \cdot 20 = -1584$ which is negative, there is no real value of m. Hence no such line is possible.]

1. Area of the quadrilateral formed by the lines $|x| + |y| = 2$ is :

(A) 8 (B) 6 (C) 4 (D) none

रेखा $|x| + |y| = 2$ से बनने वाले चतुर्भुज का क्षेत्रफल होगा :

(A) 8 (B) 6 (C) 4 (D) कोई नहीं

Ans. A

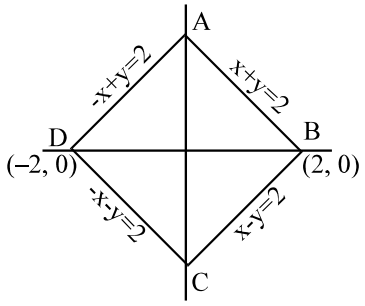
Sol. given lines are

$x + y = 2$

$-x + y = 2$

$x - y = 2$

$-x - y = 2$



ABCD is a square as AC = BD and $AC \perp BD$

also AB = BC = CD = AD now are ABCD $= 4$ area Δ AOB

$= 4 \times \frac{1}{2} \times 2 \times 2 = 8$ sq.units

2. A is a point on either of two lines $y + \sqrt{3}|x| = 2$ at a distance of $\frac{4}{\sqrt{3}}$ units from their point of intersection. The co-ordinates of the foot of perpendicular from A on the bisector of the angle between them are :

रेखाओं $y + \sqrt{3}|x| = 2$ के प्रतिच्छेद बिन्दु से $\frac{4}{\sqrt{3}}$ की दूरी पर इनमें से किसी एक रेखा पर बिन्दु A है। यदि बिन्दु A से दोनों रेखाओं के कोणों के अर्द्धक पर लम्ब डाला जाता है तो लम्ब के पाद का निर्देशांक होगा :

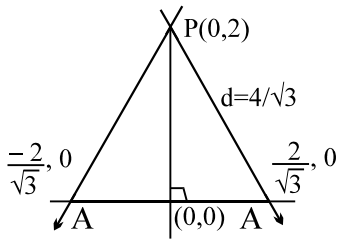
(A) $\left(-\frac{2}{\sqrt{3}}, 2\right)$ (B) (0, 0) (C) $\left(\frac{2}{\sqrt{3}}, 2\right)$ (D) (0, 4)

Ans. B

Sol. Draw figure

$y + \sqrt{3}x = 2$ for $x > 0$

$y = -\sqrt{3}x = 2$ for $x < 0$



3. If the straight lines, $ax + amy + 1 = 0$, $bx + (m+1)by + 1 = 0$ and $cx + (m+2)cy + 1 = 0$, $m \neq 0$ are concurrent then a, b, c are in :

(A) A.P. only for $m = 1$ (B) A.P. for all m

(C) G.P. for all m (D) H.P. for all m.

यदि सरल रेखा $ax + amy + 1 = 0$, $bx + (m+1)by + 1 = 0$ तथा $cx + (m+2)cy + 1 = 0$, $m \neq 0$ संगामी हो, तो a, b, c होगा :

(A) यदि $m = 1$ तो समान्तर श्रेणी में (B) m के प्रत्येक मान के लिये समान्तर श्रेणी में

(C) m के प्रत्येक मान के लिये गुणोत्तर श्रेणी में (D) m के प्रत्येक मान के लिये हरात्मक श्रेणी में

Ans. D

Sol. $D = \begin{vmatrix} a & am & 1 \\ b & b(m+1) & 1 \\ c & c(m+2) & 1 \end{vmatrix} = 0$; $\begin{vmatrix} a & 0 & 1 \\ b & b & 1 \\ c & 2c & 1 \end{vmatrix} = 0$

$a(b - 2c) + (2bc - bc) = 0$

$ab - 2ac + bc = 0$

$b(a + c) = 2ac$

$b = \dfrac{2ac}{a+c}$ $\Rightarrow$ a, b, c are in H.P.

4. Family of lines represented by the equation $(\cos\theta + \sin\theta)x + (\cos\theta - \sin\theta)y - 3(3\cos\theta + \sin\theta) = 0$ passes through a fixed point M for all real values of θ. The reflection of M in the line $x - y = 0$, is :

यदि चर रेखा $(\cos\theta + \sin\theta)x + (\cos\theta - \sin\theta)y - 3(3\cos\theta + \sin\theta) = 0$, θ के प्रत्येक मान के लिये स्थिर बिन्दु M से गुजरेगी तो बिन्दु M का रेखा $x - y = 0$ में प्रतिबिम्ब होगा :

(A) (6, 3) (B) (3, 6) (C) (– 6, 3) (D) (3, – 6)

Ans. B

Sol. Given equation is $\cos\theta(x + y - 9) + \sin\theta(x - y - 3) = 0$

$\Rightarrow$ $(x + y - 9) + \tan\theta(x - y - 3) = 0$, which is of the form $L_1 + \lambda L_2 = 0$

where $L_1 : x + y - 9 = 0$ and $L_2 : x - y - 3 = 0$

Hence the lines will always pass through the point of intersection of $L_1 = 0$ and $L_2 = 0$ i.e. M (6, 3).

So its reflection in the line $y = x$ will be (3, 6) Ans.

Alternatively: For objective put $\theta = 0$ and $\dfrac{\pi}{2}$, we get $x + y = 9$, $x - y = 3$

$\therefore \quad x = 6, \; y = 3 \qquad \Rightarrow \quad M(6, 3)]$

5. The line $(k + 1)^2x + ky - 2k^2 - 2 = 0$ passes through a point regardless of the value k. Which of the following is the line with slope 2 passing through the point?

यदि रेखा $(k + 1)^2x + ky - 2k^2 - 2 = 0$, k के प्रत्येक मान के लिये किसी बिन्दु से जाती है तो उस बिन्दु से गुजरने वाली इनमें से कौनसी रेखा होगी?

(A) $y = 2x - 8$ (B) $y = 2x - 5$ (C) $y = 2x - 4$ (D) $y = 2x + 8$

Ans. A

Sol. $(k + 1)^2x + ky = 2k^2 + 2$

$(k^2 + 2k + 1)x + ky = 2k^2 + 2$

$k^2(x - 2) + k(2x + y) + (x - 2) = 0$

$(k^2 + 1)(x - 2) + k(2x + y) = 0$

$(2x + y) + (k + 1/k)(x - 2) = 0$

Hence line passes through fixed point, $x = 2$ and $y = -4$

6. Consider a parallelogram whose sides are represented by the lines $2x + 3y = 0$; $2x + 3y - 5 = 0$; $3x - 4y = 0$ and $3x - 4y = 3$. The equation of the diagonal not passing through the origin, is :

एक समान्तर चतुर्भुज की भुजाऐं यदि $2x + 3y = 0$; $2x + 3y - 5 = 0$; $3x - 4y = 0$ एवं $3x - 4y = 3$ हैं, तो उस विकर्ण का समीकरण क्या होगा जो मूल बिन्दु से नहीं जाता :

(A) $21x - 11y + 15 = 0$ (B) $9x - 11y + 15 = 0$ (C) $21x - 29y - 15 = 0$ (D) $21x - 11y - 15 = 0$

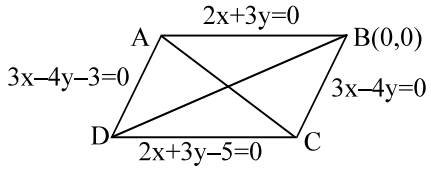
Ans. D

Sol. Equation of AC

$(2x + 3y)(3x - 4y) = (3x - 4y - 3)(2x + 3y - 5)$

or $5(3x - 4y) + 3(2x + 3y) - 15 = 0$

$21x - 11y - 15 = 0$ Ans.



equation of BD is

$(2x + 3y)(3x - 4y - 3) = (3x - 4y)(2x + 3y - 5)$

$5(3x - 4y) - 3(2x + 3y) = 0$

$9x - 29y = 0$

7. Statement-1: Let u, v, w satisfy the equations $uvw = -6$, $uv + vw + wu = -5$, $u + v + w = 2$ where $u > v > w$, then the set of value(s) of 'a' for which the points $P(u, -w)$ and $Q(v, a^2)$ lies on the same side

of the line $4x - y + 5 = 0$ are given by $(-3, 3)$.

Statement-2 : If two points $M(x_1, y_1)$ and $N(x_2, y_2)$ lies on the same side of the line $ax + by + c = 0$, then $(ax_1 + by_1 + c)(ax_2 + by_2 + c) > 0$.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

कथन-1: माना कि u, v तथा w समीकरण $uvw = -6$, $uv + vw + wu = -5$, $u + v + w = 2$ को सन्तुष्ट करती है। जहाँ $u > v > w$ तो 'a' को वह सारे मान, जबकि $P(u, -w)$ और $Q(v, a^2)$ रेखा $4x - y + 5 = 0$ के एक ही तरफ होगा, $(-3, 3)$ अन्तराल में होगा।

कथन-2: यदि $M(x_1, y_1)$ तथा $N(x_2, y_2)$ रेखा $ax + by + c = 0$ के एक ही तरफ हो, तो $(ax_1 + by_1 + c)(ax_2 + by_2 + c) > 0$ होगा।

(A) कथन–1 सत्य है, कथन–2 सत्य है, कथन–2 कथन–1 का सही स्पष्टीकरण है।

(B) कथन–1 सत्य है, कथन–2 सत्य है, कथन–2 कथन–1 का सही स्पष्टीकरण नही है।

(C) कथन–1 सत्य है, कथन–2 असत्य है।

(D) कथन–1 असत्य है, कथन– 2 सत्य है।

Ans. A

Sol. The equations whose roots are u, v, w is $x^3 - (u + v + w)x^2 + (uv + vw + wu)x - uvw = 0$

or $x^3 - 2x^2 - 5x + 6 = 0$

By trial method, $x = 1$ satisfies it.

$\therefore$ $x^3 - 2x^2 - 5x + 6 = (x - 1)(x^2 - x - 6) = (x - 1)(x - 3)(x + 2)$

$\therefore$ $x = 1, 3, -2$

$\therefore$ $u = 3, v = 1, w = -2$ (As $u > v > w$)

$\Rightarrow$ $P(u, -w) = (3, 2)$ and $Q(v, a^2) = (1, a^2)$

Now the given line is $4x - y + 5 = 0$

$\therefore$ $(4(3) - 2 + 5)\ (4(1) - a^2 + 5) > 0$

$\Rightarrow$ $15(9 - a^2) > 0 \Rightarrow a^2 - 9 < 0$

$\therefore$ $a \in (-3, 3)$

Hence option (A) is correct.

Paragraph for Question Nos. 8 to 10

Consider a family of lines $(4a + 3)x - (a + 1)y - (2a + 1) = 0$ where $a \in R$

रेखाओं के निकाय (family of lines), $(4a + 3)x - (a + 1)y - (2a + 1) = 0$ पर विचार करो जहाँ $a \in R$

8. The locus of the foot of the perpendicular from the origin on each member of this family, is :

इस समूह के प्रत्येक सदस्य पर (0, 0) से डाले गये लम्ब के पाद का बिन्दुपथ होगा :

(A) $(2x-1)^2+4(y+1)^2=5$ (B) $(2x-1)^2+(y+1)^2=5$

(C) $(2x+1)^2+4(y-1)^2=5$ (D) $(2x-1)^2+4(y-1)^2=5$

Ans. D

Sol. Given $(4a+3)x-(a+1)y-(2a+1)=0$

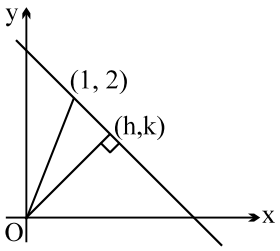
$(3x-y-1)+a(4x-y-2)=0$

family of lines passes through the fixed point P which is the intersection of

$3x-y=1$ and $4x-y=2$

solving P(1, 2),

now



$\frac{k}{h}\cdot\frac{k-2}{h-1}=-1$

$\therefore$ locus is $x(x-1)+y(y-2)=0$

$x^2+y^2-2y-x=0$

$\left(x-\frac{1}{2}\right)^2+(y-1)^2=\frac{5}{4}$

$(2x-1)^2+4(y-1)^2=5$

9. A member of this family with positive gradient making an angle of $\pi/4$ with the line $3x-4y=2$, is :

इस निकाय के उस सदस्य का समीकरण क्या होगा जिसकी प्रवणता धनात्मक है तथा रेखा $3x-4y=2$ से $\pi/4$ का कोण बनाती है :

(A) $7x-y-5=0$ (B) $4x-3y+2=0$ (C) $x+7y=15$ (D) $5x-3y-4=0$

Ans. A

Sol. Given $(4a+3)x-(a+1)y-(2a+1)=0$

$(3x-y-1)+a(4x-y-2)=0$

family of lines passes through the fixed point P which is the intersection of

$3x-y=1$ and $4x-y=2$

solving P(1, 2),

now

We have $y-2=m(x-1)$ ....(1)

this makes an angle of $\pi/4$ with $3x-4y=2$ with slope 3/4

$\therefore \left|\frac{m-(3/4)}{1+(3m/4)}\right|=1; \quad \frac{4m-3}{4+3m}=\pm 1 \quad \Rightarrow 4m-3=4+4m$ (with + ve sign)

$m=7$

with – ve sign

$4m - 3 = -4 - 3m$

$7m = -1 \quad \Rightarrow \quad m = -\frac{1}{7}$ (rejected)

hence the line is

$y - 2 = 7(x - 1)$

$7x - y - 5 = 0$

10. Minimum area of the triangle which a member of this family with negative gradient can make with the positive semi axes, is :

यदि ऋणात्मक प्रवणता वाली तथा इसी समूह की सदस्य यदि अक्षों से न्यूनतम क्षेत्रफल बनाये तो उसका क्षेत्रफल होगा :

(A) 8 (B) 6 (C) 4 (D) 2

Ans. C

Sol. Given $(4a + 3)x - (a + 1)y - (2a + 1) = 0$

$(3x - y - 1) + a(4x - y - 2) = 0$

family of lines passes through the fixed point P which is the intersection of

$3x - y = 1$ and $4x - y = 2$

solving P(1, 2),

now

Again $y - 2 = m(x - 1)$

$x = 0; \quad y = 2 - m; \quad y = 0, \; x = 1 - \frac{2}{m}$

$$\therefore \quad 2A = (2 - m)\left(1 - \frac{2}{m}\right) \quad (m < 0)$$

$$2A = 2 - m - \frac{4}{m} + 2 = 4 + \left(-m - \frac{4}{m}\right)$$

let $-m = M \qquad (M > 0)$

$$2A = 4 + M + \frac{4}{M}$$

$$= 4 + \left(\sqrt{M} - \frac{2}{\sqrt{M}}\right)^2 + 4$$

$$= 8 + \left(\sqrt{M} - \frac{2}{\sqrt{M}}\right)^2$$

area is minimum if $M = 2 \quad \Rightarrow \quad m = -2$

$2A|_{min} = 8 \quad \Rightarrow \quad A|_{min} = 4$

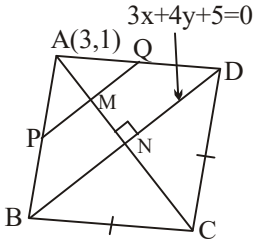
11. The equation of a line through the mid point of the sides AB and AD of rhombus ABCD, whose one diagonal is $3x - 4y + 5 = 0$ and one vertex is A(3, 1) is $ax + by + c = 0$. Find the absolute value of $(a + b + c)$ where a, b, c are integers expressed in lowest form.

एक समचतुर्भुज ABCD जिसका एक शीर्ष A (3, 1) तथा एक विकर्ण $3x - 4y + 5 = 0$ है यदि भुजा AB एवं AD के मध्य बिन्दु से खीचीं गयी रेखा का समीकरण यदि $ax + by + c = 0$ है तो $(a + b + c)$ का मापांक का मान क्या होगा जहाँ a, b, c पूर्णांक है तथा जिनका महत्तम समापवर्तक 1 है।

Ans. 1

Sol. Because line PQ passes through the mid point of AB & AD then PQ || BD and AN ⊥ BD, so M is also the mid point of AN (by similarity)

$$AN = \left|\frac{9-4+5}{5}\right| = 2$$



AM = 1

Suppose equation of PQ $\Rightarrow 3x - 4y + c = 0$.

Perpendicular from A on PQ isSt.line]

$$AM = \left|\frac{5+C}{5}\right| = 1 \Rightarrow C = 0 \text{ or } -10$$ then equations of PQ can be

$3x - 4y = 0$ or $3x - 4y - 10 = 0$

but PQ has unit distance from BD

Hence equation of PQ must be $3x - 4y = 0$

$\Rightarrow ax + by + c = 0 \qquad \Rightarrow \qquad |a + b + c| = |3 - 4 + 0| = 1$ Ans.

12. If the equations of the diagonals of the parallelogram formed by the lines, $2x - y + 7 = 0$; $2x - y - 5 = 0$; $3x + 2y - 5 = 0$ and $3x + 2y + 4 = 0$ are $ax + by - 5 = 0$ and $px + qy + 1 = 0$. Find the value of $(a + b + p + q)$.

एक समान्तर चतुर्भुज की भुजाऐं $2x - y + 7 = 0$; $2x - y - 5 = 0$; $3x + 2y - 5 = 0$ एवं $3x + 2y + 4 = 0$ है यदि उसका विकर्ण $ax + by - 5 = 0$ तथा $px + qy + 1 = 0$ है तो $(a + b + p + q)$ का मान होगा।

Ans. 40

Sol. Equation of the diagonal AC

$u_1 u_3 = u_2 u_4$

$(2x - y + 7)(3x + 2y - 5) = (3x + 2y + 4)(2x - y - 5)$

$-5(2x - y) + 7(3x + 2y) = -5(3x + 2y) + 4(2x - y) - 20$

$11x + 19y - 35 = -7x - 14y - 20$

$18x + 33y - 15 = 0$

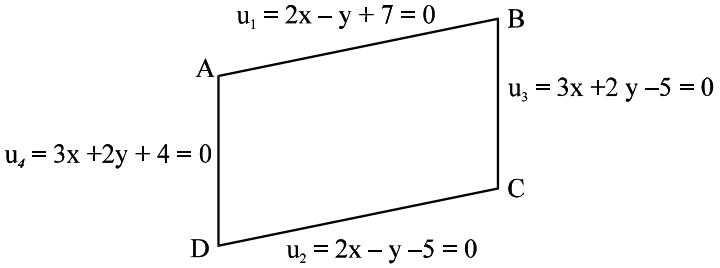
$6x + 11y - 5 = 0 \quad \Rightarrow$ a = 6 and b = 11

equation of the other diagonal ie. BD is

$u_1 u_4 = u_2 u_3$

$(2x - y + 7)(3x + 2y + 4)$

$= (2x - y - 5)(3x + 2y - 5)$



$4(2x - y) + 7(3x + 2y)$

$= -5(2x - y) - 5(3x + 2y) + 25$

$29x + 10y + 28 = -25x - 5y + 25$

$54x + 15y + 3 = 0$

$18x + 5y + 1 = 0 \quad \Rightarrow$ p = 18 and q = 5

$\Rightarrow$ (a + b + p + q) = 6 + 11 + 18 + 5 = 40 Ans.

13. The parallelogram is bounded by the lines $y = ax + c$; $y = ax + d$; $y = bx + c$ and $y = bx + d$ and has the area equal to 18. The parallelogram bounded by the lines $y = ax + c$; $y = ax - d$; $y = bx + c$ and $y = bx - d$ has area 72. Given that a, b, c and d are positive integers, find the smallest possible value of (a + b + c + d).

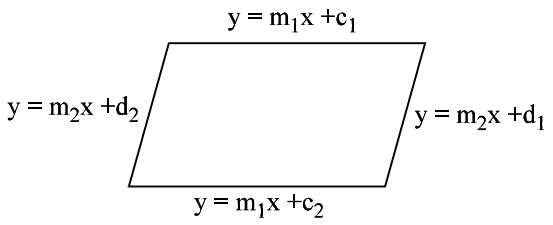
भुजाओं $y = ax + c$; $y = ax + d$; $y = bx + c$ एवं $y = bx + d$ से घिरे समान्तर चतुर्भुज का क्षेत्रफल 18 है तथा भुजाओं $y = ax + c$; $y = ax - d$; $y = bx + c$ एवं $y = bx - d$ से घिरे क्षेत्र का क्षेत्रफल 72 है यदि a, b, c तथा d धनात्मक पूर्णांक है तो (a + b + c + d) का न्यूनतम मान होगा।

Ans. 16

Sol. Area of the parallelogram $= \dfrac{|c_1 - c_2||d_1 - d_2|}{|m_1 - m_2|}$ (for the figure as shown)

hence Area of the parallelogram formed by the 1st set of 4 lines

$$= \frac{|c - d|^2}{|a - b|} = 18 \qquad ....(1)$$



and Area of the parallelogram formed by the 2nd set of 4 lines

$$= \frac{|c+d|^2}{|a-b|} = 72 \qquad ....(2)$$

from (1) $|c-d|^2 = |a-b| \cdot 18$

and from (2) $|c+d|^2 = |a-b| \cdot 72$. Since a, b, c, d $\in$ N,

Hence RHS must be a perfect square for some least value of $|a-b|$ which obviously can be equal to 2 (think ?)

$\Rightarrow$ $|a-b| = 2$

and $\boxed{a=1 \text{ and } b=3}$

hence $c-d=6$
and $c+d=12$ $\Rightarrow$ $\boxed{c=9 \text{ and } d=3}$

$\therefore$ $(a+b+c+d) = 16$ Ans.

14. Match the following column :

Column-I	**Column-II**
(A) If the line $x+2ay+a=0$, $x+3by+b=0$ & $x+4cy+c=0$ are concurrent, then a, b, c are in	(P) A.P.
(B) The lines, $ax+by+(a\alpha+b)=0$; $bx+cy+(b\alpha+c)=0$ and $(a\alpha+b)x+(b\alpha+c)y=0$ are concurrent if :	(Q) G.P.
(C) If the lines, $ax+2y+1=0$; $bx+3y+1=0$ & $cx+4y+1=0$ passes through the same point then a, b, c are in	(R) H.P.
(D) Let a, b, c be distinct non–negative numbers. If the lines $ax+ay+c=0$, $x+1=0$ & $cx+cy+b=0$ pass through the same point then a, c, b are in	(S) $(x-\alpha)$ is a factor of $ax^2+2bx+c$

निम्न कॉलम को सुमेलित कीजिए :

Column-I	**Column-II**
(A) यदि रेखा $x+2ay+a=0$, $x+3by+b=0$ एवं $x+4cy+c=0$ संगामी है तो a, b, c होगा	(P) समान्तर श्रेणी
(B) यदि रेखा $ax+by+(a\alpha+b)=0$; $bx+cy+(b\alpha+c)=0$ तथा $(a\alpha+b)x+(b\alpha+c)y=0$ संगामी हो, तो	(Q) गुणोत्तर
(C) यदि रेखा $ax+2y+1=0$; $bx+3y+1=0$ एवं $cx+4y+1=0$ संगामी है तो a, b, c होगा	(R) हरात्मक

(D) यदि रेखा $ax + ay + c = 0$, $x + 1 = 0$ एवं $cx + cy + b=0$ संगामी है जहां a, b, c भिन्न भिन्न तथा अऋणात्मक संख्याऐं है तो a, c, b होगा

(S) $(x-\alpha)\, ax^2 + 2bx + c$ का भाजक है

Ans. (A) R; (B) S; (C) P; (D) Q

Sol. (A) $\rightarrow$ R

$$\begin{vmatrix} 1 & 2a & a \\ 1 & 3b & b \\ 1 & 4c & c \end{vmatrix} \Rightarrow 0 \Rightarrow \begin{vmatrix} 1 & 2a & a \\ 0 & (3b-2a) & (b-a) \\ 0 & (4c-2a) & (c-a) \end{vmatrix} = 0$$

$\Rightarrow (3b - 2a)(c - a) - (b - a)(4c - 2a) = 0$

$\Rightarrow 3bc - 3ab - 2ca + 2a^2 - 4bc + 2ab + 4ca - 2a^2 = 0$

$\Rightarrow -ab - bc + 2ca = 0$

$\Rightarrow 2ca = ab + bc$

$\Rightarrow \dfrac{2}{b} = \dfrac{1}{a} + \dfrac{1}{c} \rightarrow$ H.P.

(B) $\rightarrow$ S

$$\begin{vmatrix} a & b & a\alpha+b \\ b & c & b\alpha+c \\ a\alpha+b & b\alpha+c & 0 \end{vmatrix} = 0 \Rightarrow \begin{vmatrix} a & b & 0 \\ b & c & 0 \\ a\alpha+b & b\alpha+c & -b\alpha-c-a\alpha^2-b\alpha \end{vmatrix}$$

$\Rightarrow -\left(a\alpha^2 + 2b\alpha + c\right)\left(ac - b^2\right) = 0$

$\Rightarrow a\alpha^2 + 2b\alpha + c = 0$ or $b^2 = ac \Rightarrow$ a,b,c are in G.P. Q is not answer, as it is not asking for a,b,c are in?

(C) $\rightarrow$ P

$$\begin{vmatrix} a & 2 & 1 \\ b & 3 & 1 \\ c & 4 & 1 \end{vmatrix} = 0 \qquad \Rightarrow -a + 2b - c = 0$$

$2b = a + c$

(D) $\rightarrow$ Q

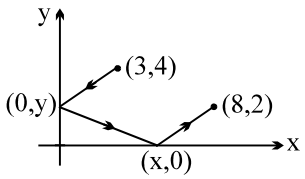
$$\begin{vmatrix} a & a & c \\ 1 & 0 & 1 \\ c & c & b \end{vmatrix} = 0 \Rightarrow -ab + c^2 - ac + ac$$

$c^2 = ab$

a, c, b $\rightarrow$ G.P.0

1. Suppose that a ray of light leaves the point (3, 4), reflects off the y-axis towards the x-axis, reflects off the x-axis, and finally arrives at the point (8, 2). The value of x, is :

माना कि एक रेखा किरण बिन्दु (3, 4) से अपनी यात्र शुरू करती है तथा चित्रनुसार परावर्तित होकर बिन्दु (8, 2) पर पहुँचती हो, तो x का मान होगा :



(A) $x = 4\frac{1}{2}$ (B) $x = 4\frac{1}{3}$ (C) $x = 4\frac{2}{3}$ (D) $x = 5\frac{1}{3}$

Ans. B

Sol. $\tan\theta = \frac{y}{x} = \frac{2}{8-x} = \frac{4-y}{3} \Rightarrow y = \frac{26}{11}$

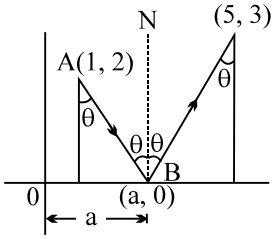
2. A ray of light passing through the point A (1, 2) is reflected at a point B on the x – axis and then passes through (5, 3). Then the equation of AB is :

एक किरण जो बिन्दु A (1, 2) से गुजरती है तथा x अक्ष पर बिन्दु B पर परावर्तित होकर (5, 3) से गुजरती है तो AB का समीकरण है:

(A) $5x + 4y = 13$ (B) $5x - 4y = -3$ (C) $4x + 5y = 14$ (D) $4x - 5y = -6$

Ans. A

Sol. $\frac{a-1}{2} = \frac{5-a}{3} \Rightarrow a = \frac{13}{5}$



3. The co–ordinates of the point of reflection of the origin (0, 0) in the line $4x - 2y - 5 = 0$ is :

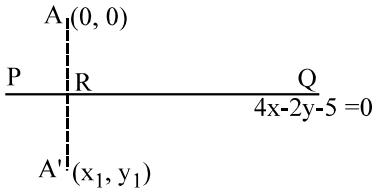
मूल बिन्दु का रेखा $4x - 2y - 5 = 0$ से परावर्तन के पश्चात् बिन्दु होगा :

(A) $(1, -2)$ (B) $(2, -1)$ (C) $\left(\frac{4}{5}, -\frac{2}{5}\right)$ (D) $(2, 5)$

Ans. B

Sol. $m_{AA'} \times m_{PQ} = -1 \Rightarrow \frac{y_1}{x_1} \cdot 2 = -1 \Rightarrow 2y_1 = -x_1$

$R\left(\frac{x_1}{2}, \frac{y_1}{2}\right)$ lies on $4x - 2y - 5 = 0 \Rightarrow 2x_1 - y_1 = 5$ $\Rightarrow$ (B)



4. m, n are integer with $0 < n < m$. A is the point (m, n) on the cartesian plane. B is the reflection of A in the line $y = x$. C is the reflection of B in the y-axis, D is the reflection of C in the x-axis and E is the reflection of D in the y-axis. The area of the pentagon ABCDE is :

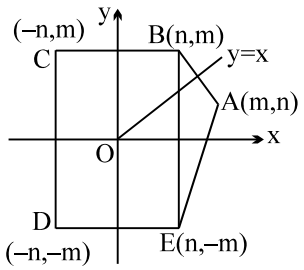
यदि $0 < n < m$ जहां $m, n \in I$ एवं A (m, n) कार्तीय तल पर एक बिन्दु है यदि A का रेखा $y = x$ के सापेक्ष परावर्तन बिन्दु B है तथा B का y अक्ष के सापेक्ष परावर्तन बिन्दु C है तथा C का x अक्ष के सापेक्ष परावर्तन D हो तथा D का y-अक्ष के सापेक्ष परावर्तन E है तो पंचभुज ABCDE का क्षेत्रफल होगा :

(A) $2m(m+n)$ (B) $m(m+3n)$ (C) $m(2m+3n)$ (D) $2m(m+3n)$

Ans. B

Sol. Area of rectangle BCDE = 4mn

Area of $\Delta ABC = \frac{2m(m-n)}{2}$



$= m^2 - mn$

$\therefore$ area of pentagon $= 4mn + m^2 - mn$

$= m^2 + 3mn$ Ans.

5. In a triangle ABC, if A (2, – 1) and $7x - 10y + 1 = 0$ and $3x - 2y + 5 = 0$ are equations of an altitude and an angle

bisector respectively drawn from B, then equation of BC is :

त्रिभुज ABC में यदि A (2, – 1) तथा 7x – 10y + 1 = 0 एवं 3x – 2y + 5 = 0 शीर्ष B से डाले गये ऊँचाई तथा कोण B का अर्द्धक है, तो रेखा BC का समीकरण होगा :

(A) x + y + 1 = 0 (B) 5x + y + 17 = 0 (C) 4x + 9y + 30 = 0 (D) x – 5y – 7 = 0

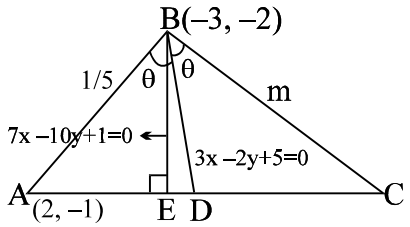
Ans. B

Sol. BD and BE are intresect at B

Coordinates of B are (–3, –2)

$m_{AB} = 1/5$

$$\tan\theta = \left|\frac{\frac{3}{2}-\frac{1}{5}}{1+\frac{3}{10}}\right| = \left|\frac{\frac{3}{2}-m}{1+\frac{3m}{2}}\right|$$



$$\Rightarrow \quad 1 = \left|\frac{3-2m}{2+3m}\right| \quad \text{or} \quad \pm 1 = \frac{3-2m}{2+3m}$$

$\Rightarrow$ m = 1/5 (rejected) or – 5

equation of BC = $y + 2 = -5(x + 3) \Rightarrow 5x + y + 17 = 0$

Alternatively : Take image of (2, –1) in the line BD to get a point on BC

6. Number of straight line(s) passing through the point P(4, 3) whose x-intercept is a prime number and whose y-intercept is a positive integer, is equal to :

(A) 0 (B) 1 (C) 2 (D) more than 2 but finite

P(4, 3) से गुजरने वाली सारी रेखाओं की संख्या जिसका x-अन्तःखण्ड अभाज्य संख्या है तथा y-अन्तःखण्ड धनात्मक पूर्णांक है :

(A) 0 (B) 1 (C) 2 (D) 2 अधिक लेकिन निश्चित

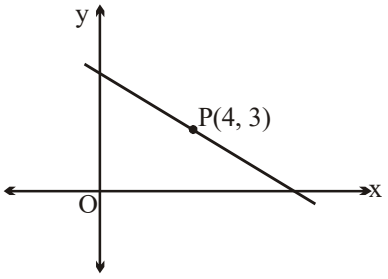
Ans. C

Sol. We must have $m < 0$.

(If $m > 0$ then either x or y intercept must be negative.)

Let $(y - 3) = m(x - 4)$

Now, x-intercept $= 4 - \frac{3}{m}$ and y-intercept $= 3 - 4m$



For x-intercept to be positive prime and y-intercept to be positive integer, possible values of m are

i.e. m = – 1 or m = – 3. Ans.

Paragraph for Question nos. 7 to 9

Consider two points $A \equiv (1, 2)$ and $B \equiv (3, -1)$. Let M be a point on the straight line $L \equiv x + y = 0$.

यदि $A \equiv (1, 2)$ तथा $B \equiv (3, -1)$ एवं रेखा $L \equiv x + y = 0$ पर एक बिन्दु M है।

7. If M be a point on the line $L = 0$ such that $AM + BM$ is minimum, then the reflection of M in the line $x = y$ is :

यदि M ऐसा बिन्दु है कि $AM + BM$ न्यूनतम है, तो $x = y$ रेखा पर बिन्दु M का परावर्तन होगा :

(A) $(1, -1)$ (B) $(-1, 1)$ (C) $(2, -2)$ (D) $(-2, 2)$

Ans. B

Sol. Let C be the reflection of A in $L \equiv x + y = 0$

$\Rightarrow C \equiv (-2, -1)$. Now $AM + BM = CM + BM$, which is minimum if B, M and C are collinear.

Now, equation of BC is

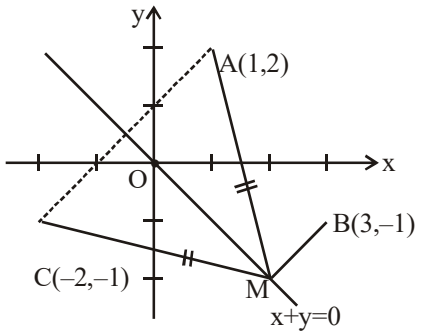
$(y + 1) = 0\ (x + 2)$

$\Rightarrow y + 1 = 0$ ... (1)

$\therefore$ On solving equation (1) with $x + y = 0$, we get

$M \equiv (1, -1)$

Hence the reflection of M in the line $x = y$ is $(-1, 1)$ Ans.



8. If M be a point on the line $L = 0$ such that $|AM - BM|$ is maximum, then the distance of M from $N \equiv (1, 1)$ is :

यदि M ऐसा बिन्दु है कि $|AM - BM|$ अधिकतम है, तो बिन्दु M की $N \equiv (1, 1)$ से दूरी :

(A) $5\sqrt{2}$ (B) $73\sqrt{5}$ (C) (D) 10

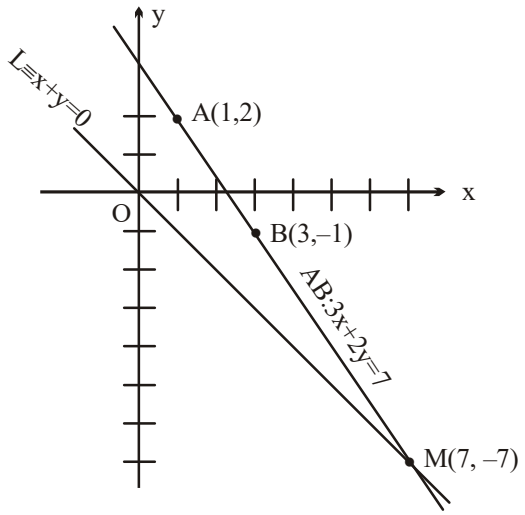
Ans. D

Sol. We have $|AM - BM| \le AB$

$\therefore$ For $|AM - BM|$ to be maximum, points A, B and M must be collinear.

Now, equation of AB is $(y - 2) = \dfrac{-3}{2}(x - 1)$

$\Rightarrow 3x + 2y = 7$ ... (1)



$\therefore$ On solving equation (1) with $x + y = 0$, we get

$M \equiv (7, -7)$

Hence distance of M (7, – 7) from

N (1, 1) = MN = $\sqrt{(7-1)^2 + (-7-1)^2}$ = $\sqrt{36+64}$ = 10 units

9. If M be a point on the line $L = 0$ such that $|AM - BM|$ is minimum, then the area of ΔAMB equals :

यदि M ऐसा बिन्दु है कि $|AM - BM|$ न्यूनतम है, तो त्रिभुज ΔAMB का क्षेत्रफल होगा :

(A) $\frac{13}{4}$ (B) $\frac{13}{2}\frac{13}{6}$ (C) (D) $\frac{13}{8}$

Ans. A

Sol. 0

Paragraph for Question Nos. 10 to 11

An equilateral triangle ABC has its centroid at the origin and the base BC lies along the line $x + y = 1$

एक समबाहु त्रिभुज का केन्द्रक $(0, 0)$ पर है तथा उसका आधार रेखा $x + y = 1$ पर है, तो

10. Area of the equilateral ΔABC is :

ΔABC का क्षेत्रफल होगा :

(A) $\frac{3\sqrt{3}}{2}$ (B) $\frac{3\sqrt{3}}{4}$ (C) $\frac{3\sqrt{2}}{2}$ (D) $\frac{2\sqrt{3}}{4}$

Ans. A

Sol. $GD = \frac{1}{\sqrt{2}}$; $AG = \sqrt{2}$

parametric through (0, 0)

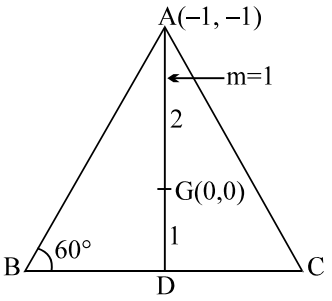
$$\frac{x}{\cos\theta} = \frac{y}{\sin\theta} = -\sqrt{2} \quad \text{(think !)}$$

$$\cos\theta = -\frac{1}{\sqrt{2}}; \quad \sin\theta = -\frac{1}{\sqrt{2}}$$

$x = -1; \quad y = -1$

hence coordinates of D are $(-1/2, -1/2)$

$$AD = p = \frac{3}{\sqrt{2}};$$



$$\text{Area of equilateral triangle} = \frac{p^2}{\sqrt{3}} = \frac{9}{2}\cdot\frac{1}{\sqrt{3}} = \frac{3\sqrt{3}}{2}$$

11. Gradient of the other two lines are :

बाकी दोनों भुजाओं की प्रवणता होगी :

(A) $\sqrt{3}, \sqrt{2}$ (B) $\sqrt{3}, \frac{1}{\sqrt{3}}$ (C) $\sqrt{2}+1, \sqrt{2}-1$ (D) $2+\sqrt{3}, 2-\sqrt{3}$

Ans. D

Sol. equation of line through A, $y + 1 = m(x + 1)$ ....(1)

(1) makes an angle of 60° with BC

$$\left|\frac{m+1}{1-m}\right| = \sqrt{3} \qquad \Rightarrow \qquad m = 2 - \sqrt{3} \quad \text{or} \quad 2+\sqrt{3} \quad \text{Ans.}$$

12. A ray of light is sent along the line $x - 2y - 3 = 0$. Upon reaching the line $3x - 2y - 5 = 0$, the ray is reflected from it. Find the equation of the line containing the reflected ray.

एक किरण रेखा $x - 2y - 3 = 0$ से जाती है तथा रेखा $3x - 2y - 5 = 0$ पर पहुँचने पर इससे परावर्तित हो जाती हैं, तो परावर्तित किरण का समीकरण बताओ।

Ans. $29x - 2y = 31$

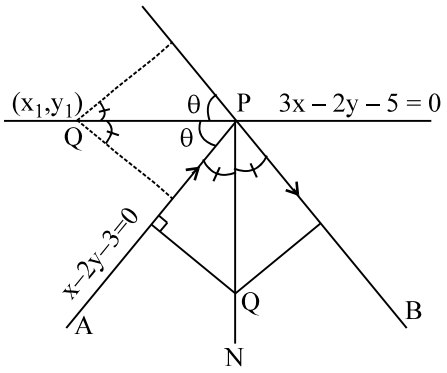
Sol. Equation of the reflected ray BP can be takens as

$x - 2y - 3 + \lambda(3x - 2y - 5) = 0$ ....(1)n

Now perpendicular from $Q(x_1, y_1)$ are (1)

and $x - 2y - 3 = 0$ should be equal

$$\left|\frac{x_1-2y_1-3}{\sqrt{5}}\right| = \left|\frac{x_1-2y_1-3+\lambda(3x_1-2y_1-5)}{\sqrt{(3\lambda+1)^2+4(\lambda+1)^2}}\right|$$



or $(3\lambda + 1)^2 + 4(\lambda + 1)^2 = 5]$

$9\lambda^2 + 1 + 6\lambda + 4\lambda^2 + 8\lambda + 4 = 5$

$13\lambda^2 + 14\lambda = 0 \Rightarrow \lambda = -\frac{14}{13}$ (as $\lambda \neq 0$)

$\therefore$ required equation is $x - 2y - 3 - \frac{14}{13}(3x - 2y - 5) = 0$

$13x - 26y - 39 = 42x - 28y - 70 \quad \Rightarrow 29x - 2y = 31$ Ans.

1. Let PQR be a right angled isosceles triangle, right angled at P (2, 1). If the equation of the line QR is $2x + y = 3$, then the equation representing the pair of lines PQ and PR is :

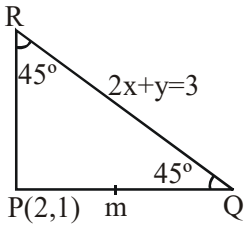
ΔPQR एक समकोणीय समद्विबाहु त्रिभुज है। जिसका समकोण P (2, 1) पर है। यदि QR का समीकरण $2x + y = 3$ है, तो रेखा PQ एवं PR के युग्म का समीकरण होगा :

(A) $3x^2 - 3y^2 + 8xy + 20x + 10y + 25 = 0$ (B) $3x^2 - 3y^2 + 8xy - 20x - 10y + 25 = 0$

(C) $3x^2 - 3y^2 + 8xy + 10x + 15y + 20 = 0$ (D) $3x^2 - 3y^2 - 8xy - 10x - 15y - 20 = 0$

Ans. B

Sol. Let m be the slope of PQ, then



$$\tan 45° = \left|\frac{m-(-2)}{1+m(-2)}\right|$$

$$\Rightarrow 1 = \left|\frac{m+2}{1-2m}\right| \qquad \Rightarrow \pm 1 = \frac{m+2}{1-2m}$$

$\Rightarrow m = -1/3$ or $m = 3$

As PR also makes $\angle 45°$ with RQ . $\therefore$ The above two values of m are for PQ and PR.

$\therefore$ Equation of PQ $y - 1 = -\frac{1}{3}(x-2)$

$\Rightarrow x + 3y - 5 = 0$

and equation of PR is $\Rightarrow 3x - y - 5 = 0$

$\therefore$ Combined equation of PQ and PR is

$(x - 3y - 5)(3x - y - 5) = 0$

$\Rightarrow 3x^2 - 3y^2 + 8xy - 20x - 10y + 25 = 0$

2. A pair of perpendicular straight lines is drawn through the origin forming with the line $2x + 3y = 6$ an isosceles triangle right angled at the origin . The equation to the line pair is :

यदि रेखा युग्म जो कि लम्बवत् हो तथा (0, 0) से खींचा जाता है तथा रेखा $2x + 3y = 6$ से समद्विबाहु त्रिभुज बनाता हो, तो रेखा युग्म का समीकरण होगा :

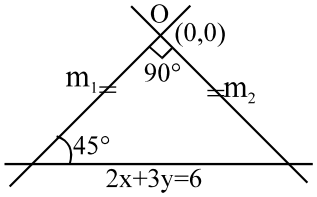
(A) $5x^2 - 24xy - 5y^2 = 0$ (B) $5x^2 - 26xy - 5y^2 = 0$

(C) $5x^2 + 24xy - 5y^2 = 0$ (D) $5x^2 + 26xy - 5y^2 = 0$

Ans. A

Sol. $2x + 3y = 6$

$$\tan 45^\circ = \left|\frac{m-(\frac{-2}{3})}{1+m(\frac{-2}{3})}\right|$$



get two value of m $\quad m_1 = -5, \quad m_2 = 1/5$

3. If the line $y = mx$ bisects the angle between the lines $ax^2 + 2h\,xy + by^2 = 0$ then m is a root of the quadratic equation :

यदि रेखा $y = mx$ रेखाओं $ax^2 + 2h\,xy + by^2 = 0$ के बीच कोण की अर्द्धक हो, तो m इनमें से किस द्विघातीय का मूल होगा :

(A) $hx^2 + (a-b)x - h = 0$ (B) $x^2 + h(a-b)x - 1 = 0$

(C) $(a-b)x^2 + hx - (a-b) = 0$ (D) $(a-b)x^2 - hx - (a-b) = 0$

Ans. A

Sol. Equation of the angle bisector $\frac{x^2 - y^2}{a-b} = \frac{xy}{h}$; Now put $y = mx$

4. If the equation $ax^2 - 6xy + y^2 + 2gx + 2fy + c = 0$ represents a pair of lines whose slopes are m and m^2, then sum of all possible values of a is :

यदि समीकरण $ax^2 - 6xy + y^2 + 2gx + 2fy + c = 0$ एक ऐसा रेखायुगम है। जिसकी प्रवणता m एवं m^2 हो, तो a के सम्भावित मानों का योग होगा :

(A) 17 (B) – 19 (C) 19 (D) – 17

Ans. B

Sol. The given lines will be parallel to lines $ax^2 - 6xy + y^2 = 0$, so

$$\left(\frac{y}{x}\right)^2 - 6\left(\frac{y}{x}\right) + a = 0 \begin{matrix} \langle \ m \\ \ \ m^2 \end{matrix}$$

$\therefore \quad m + m^2 = 6 \qquad ....(1)$

From (1), $m = -3$ or 2

and $m^3 = a \qquad ....(2)$

From (2), $a = -27$ or 8

Hence sum of all possible values of $a = -27 + 8 = -19$ Ans.

5. Through a point A on the x-axis a straight line is drawn parallel to y-axis so as to meet the pair of straight lines $ax^2 + 2hxy + by^2 = 0$ in B and C. If AB = BC then :

x-अक्ष पर बिन्दु A से y-अक्ष के समान्तर रेखा खींची जाती है तथा रेखा युग्म $ax^2 + 2hxy + by^2 = 0$ को क्रमशः B एवं C पर काटती

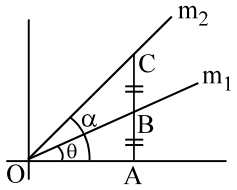
है यदि AB = BC तो :

(A) $h^2 = 4ab$ (B) $8h^2 = 9ab$ (C) $9h^2 = 8ab$ (D) $4h^2 = ab$

Ans. B

Sol. Given AB = BC; also $m_1 + m_2 = -\frac{2h}{b}$ and $m_1 m_2 = \frac{a}{b}$

$\tan\theta = \frac{AB}{OA} = m_1$; $m_2 = \tan\alpha = \frac{2AB}{OA} = 2m_1$



$\frac{m_2}{m_1} = 2$; $\frac{m_2 + m_1}{m_2 - m_1} = \frac{2+1}{2-1} = 3$

$\Rightarrow -\frac{\frac{2h}{b}}{\sqrt{\frac{4h^2}{b^2} - \frac{4a}{b}}} = 3$ hence, $\frac{4h^2}{b^2} \cdot \frac{b^2}{4(h^2 - ab)} = 9$ or $9h^2 - 9ab = h^2$

$\Rightarrow 8h^2 = 9ab$

6. Let $S = \{(x, y) \mid x^2 + 2xy + y^2 - 3x - 3y + 2 = 0\}$, then S :

(A) consists of two coincident lines. (B) consists of two parallel lines which are not coincident.

(C) consists of two intersecting lines. (D) is a parabola.

माना $S = \{(x, y) \mid x^2 + 2xy + y^2 - 3x - 3y + 2 = 0\}$, तो समुच्चय S में :

(A) दो सम्पाती रेखायें होगी (B) दो समानान्तर रेखायें होगी जो कि सम्पाती नही होगी

(C) दो प्रतिच्छेदी रेखायें होगी (D) एक परवलय होगा

Ans. B

Sol. $(x + y)^2 - 3(x + y) + 2 = 0$

$t^2 - 3t + 2 = 0$ $\Rightarrow (t - 2)(t - 1) = 0$ $\Rightarrow (x + y - 2)(x + y - 1) = 0$

$\Rightarrow$ two parallel lines which are non coincident

7. The greatest slope along the graph represented by the equation $4x^2 - y^2 + 2y - 1 = 0$, is :

समीकरण $4x^2 - y^2 + 2y - 1 = 0$ द्वारा निरूपित वक्र पर अधिकतम ढाल है :

(A) – 3 (B) – 2 (C) 2 (D) 3

Ans. C

Sol. $y^2 - 2y + 1 = 4x^2$

$(y - 1) = 2x$ or $-2x$

$y = 2x + 1$ or $y = 1 - 2x$

greatest slope = 2 Ans.

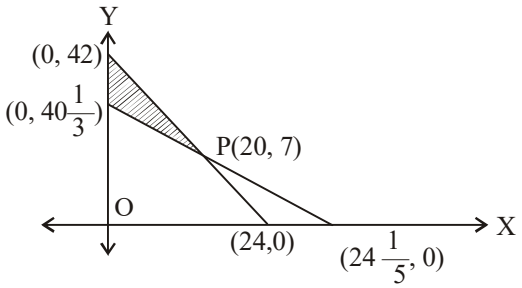
8. The area of the triangular region in the first quadrant bounded on the left by the y-axis, bounded above by the line $7x + 4y = 168$ and bounded below by the line $5x + 3y = 121$, is :

प्रथम चतुर्थांथ में y-अक्ष के दांयी तरफ, रेखा $7x + 4y = 168$ के नीचे तथा रेखा $5x + 3y = 121$ के ऊपर घिरे त्रिभुजाकार क्षेत्र का क्षेत्रफल होगा :

(A) $\frac{50}{3}$ (B) $\frac{52}{3}$ (C) $\frac{53}{3}$ (D) 17

Ans. A

Sol. Clearly lines intersect at P(20, 7)



$\therefore$ Area of shaded region $= \frac{1}{2}\left(42 - 40\frac{1}{3}\right)20 = \frac{1}{2}\left(\frac{5}{3}\right)20 = \frac{50}{3}$ (square units)

9. The equation $9x^3 + 9x^2y - 45x^2 = 4y^3 + 4xy^2 - 20y^2$ represents 3 straight lines, two of which pass through the origin. Find the area of the triangle formed by these lines (in sq. units).

समीकरण $9x^3 + 9x^2y - 45x^2 = 4y^3 + 4xy^2 - 20y^2$, 3 सरल रेखाओं को निरूपित कर रहा है। जहाँ उनमें से दो रेखाऐं (0, 0) से जा रही हो, तो उन तीनों रेखाओं से बनने वाले त्रिभुज का क्षेत्रफल बताओ।

Ans. 30

Sol. $D = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$ and $D = 0$ is condition of concurrency

$D = -(a^3 + b^3 + c^3 - 3abc) = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$

(A) if $a + b + c = 0$ but $\sum a^2 \neq \sum ab$ i.e. a, b, c are not all equal, then $D = 0$

hence lines are concurrent $\Rightarrow$ (Q)

(B) if $a + b + c = 0$ and $\sum a^2 = \sum ab$ $\Rightarrow$ $a = b = c$

$\therefore$ $a = 0$; $b = 0$; $c = 0$

$\Rightarrow$ lines becomes identical and of the form $0x + 0y + 0 = 0$

any ordered pair (x, y) will satisfy $\Rightarrow$ complete xy plane $\Rightarrow$ (P)

(C) if $a + b + c \neq 0$ and $\sum a^2 \neq \sum ab$ $\Rightarrow$ a, b, c are not all equal $\Rightarrow D \neq 0$

In this case equations represents set of lines which are neither concident nor concurrent $\Rightarrow$ (S)

(D) if $a + b + c \neq 0$ and $\sum a^2 = \sum ab$ $\Rightarrow$ $a = b = c$

10. Find the value of K for which the equation $2x^2 - xy + Ky^2 + 8x + 7y - 10 = 0$ may represent a pair of lines. For this value of K show that this equation can be transformed into a homogeneous equation of second degree by translating the origin to a properly chosen point. Also find the acute angle between the line pair represented by the given general equation.

K का वह मान ज्ञात करो जब समीकरण $2x^2 - xy + Ky^2 + 8x + 7y - 10 = 0$ रेखा युग्म निरूपित करे। दिखाओं कि K के इस मान के लिये उपरोक्त समीकरण को विस्थापित (shift) करके 2 घातीय समघात समीकरण बनाया जा सकता है दी हुयी रेखा युग्म के बीच का न्यूनकोण भी ज्ञात करो।

Ans. $K = -1$; $(-1, 4)$; $2x^2 - xy - y^2 = 0$; $\theta = \tan^{-1}(3)$

Sol. $2x^2 - xy + Ky^2 + 8x + 7y - 10 = 0$

$2(x^2 + 2x + 1) + 4x - 2 - xy - (y^2 - 8y + 16) - 10 - y + 16 = 0$

$2(x + 1)^2 - (y - 4)^2 - x(y - 4) - (y - 4) = 0$

$2(x + 1)^2 - (y - 4)^2 - (x + 1)(y - 4) = 0$

$2X^2 - Y^2 - XY = 0$]

Alternative: Compute the lines and shift the origin at there point of intersection.

11. Find the area enclosed by the graph of $x^2y^2 - 9x^2 - 25y^2 + 225 = 0$.

वक्र $x^2y^2 - 9x^2 - 25y^2 + 225 = 0$ से घिरे क्षेत्र का क्षेत्रफल बताओं।

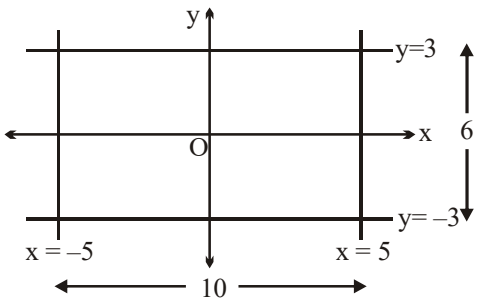
Ans. 60

Sol. Factorising, we get

$x^2(y^2 - 9) - 25(y^2 - 9) = 0$

$\Rightarrow (y^2 - 9)(x^2 - 25) = 0$

$\therefore$ Area $= 10 \times 6 = 60$ sq. units.



12. Set of family of lines are described in column-I and their mathematical equation are given in column-II. Match the entry of column-I with suitable entry of column-II : (*m* and *a* are parameters)

Column-I **Column-II**

(A) having gradient 3	(P) $mx - y + 3 - 2m = 0$
(B) having y intercept three times the x-intercept	(Q) $mx - y + 3m = 0$
(C) having x intercept (–3)	(R) $3x + y = 3a$
(D) concurrent at (2, 3)	(S) $3x - y + a = 0$

रेखाओं के समूह कॉलम-I में दिया गया हैं तथा उनका समीकरण कॉलम-II में दिया गया है : (जहाँ m एवं a प्राचल है।) तो कॉलम को सुमेलित कीजिए।

Column-I	**Column-II**
(A) प्रवणता 3 है	(P) $mx - y + 3 - 2m = 0$
(B) y अन्तःखण्ड़ = 3. (x-अन्तःखण्ड)	(Q) $mx - y + 3m = 0$
(C) x अन्तःखण्ड (–3) है	(R) $3x + y = 3a$
(D) (2, 3) पर संगामी है	(S) $3x - y + a = 0$

Ans. (A) S; (B) R; (C) Q; (D) P

Sol.